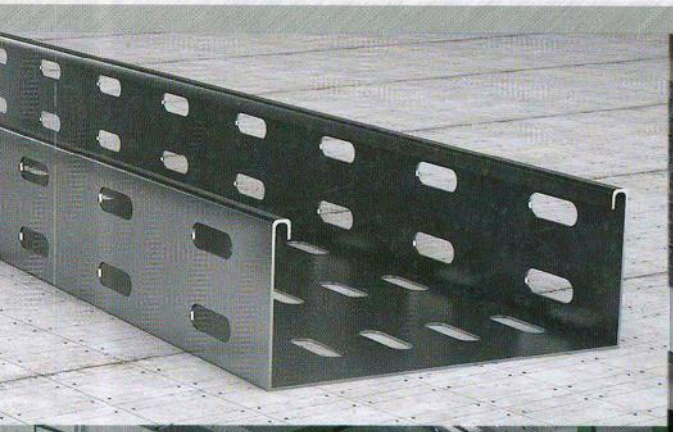
MODERN METAL
INDUSTRIES COMPANY







Cable Management System

Cable Tray - Cable Ladder - Cable Tray Basket - Cable Duct (Trunking) - Channel & Supports

Registered ISO 9001:2008

IAF MEMBER OF MULTILATERAL RECOGNITION ARRANGEMENT

Modern Metal Industries Co.

المصنع الحديث للمنتجات الحديدية
MODERN METAL INDUSTRIES

المصنع الحديث للمنتجات الحديدية
MODERN METAL INDUSTRIES

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Madinah Metal Industries Co.

Introduction Letter

Cable Support Management System

Modern Metal as the ability to take up the gigantic projects in kingdom as well as overseas in GCC Countries and Middle East since 1999. We have well qualified and well equipped technical staffs which we can offer to assist customers in the correct selection of standard products.

Modern Metal manufacture and supply a full range of Cable Support System including Cable Tray, Cable Trunks, Cable Ladder, Support Channels complying with the most well known International Standards such as NEMA, IEC, for HDG as ASTM 123 or BS EN 1461.

Our Strategy for the purchase of raw material from many sourees i.e. Sabic Unicoil in kingdom of Saudi Arabia, Al Ghuriar in U.A.E, G.I. coil material as per standard BS 2989, Hot Rolled coil material as per standard BS 1449, Cold Rolled coil material as per standard BS 2994, and Stainless Steel coils complying with BS 1449: Part 2 (304S15) AISI 304, A2. DIN, Material No. 1.430.

We have our own specification to provide the customer the quality products and excellent service according according to their requirements with the following materials specification for special applications.

Modern Matal is fully equipped with the state of the art tools and CNC and CNC roll forming machine line for fater production, also specialized labor and staff with long experience in theis product, also specialized Engineering for QA/QC and Safety department since 10 years.

Modern Metal also offers full engineering services related to support system products such as load caluclations, drawings and specifications.

Modern Metal Cable Support System is approved by the most of the Engineering design & Consulting Offices in the Kingdom and is beign used for all kinds of Projects including mega size and land mark projects as indicated in Modern metal reference list. Our commitment is to satisfy our clients with high standard products, prompt delivery and efficient services.



Selection Process

A number of basic decisions must be made before a cable tray system be specified MMICO has developed a simple six step process to gude you in the process:

1. Select Material and Finish
2. Select the Tray Load Class
3. Select the Tray Type
4. Select the Tray Size
5. Select the Fittings
6. Consider Deflection

Each step is described in detial below:

1-Select Material & Finish

METALS
MMICO. a pioneer in the development of cable trays. first made a complete study of the metals available in order to create those systems that would be unmatched in economy and strength. Inherent are the problems connected with the environment and its effect on the life of metals, including weathering and corrosion by electro - chemical and chemical attack.

Steel - Hot Dip Galvanized After Fabrication
Commercial quality steel is used for this type of coating, which is described below. This steel is either hot rolled, pickled and oiled, or cold rolled, mild carbon steel.

Steel - Mill Galvanized (Pre-Galvanized)
Where heavy zinc coating is not necessary, steel with a Continuous hot rolled zinc coating conforming to ASTM specification A653 designation G90 is used. Products made with mill galvanized steel are generally lower in cost than the same products that are hot dip galvanized after fabrication. However, see the limitations under “Finishes” below.

Stainless Steel
Stainless steel type 304 and 316 is available for most MMICO products. Complete cable tray systems can be furnished in stainless steel.

Aluminum Alloys
The principal aluminum used for Ladder, Tray side rails, rungs, and channels is extruded, copper - free alloy 6063 - T6. This alloys is used for structural members where extreme stiffness and high tensile strength are mechanically required to resist high unit stresses. This is called a “magnesium alloy” Alloy 5052-H32 is used for components fabricated from sheet (except covers) such as connectors, blind ends, dropouts, etc. This is a “magnesium alloy”, copper free, and very compatible with alloy 6063 -T6. Alloys 3003 and 3105 are used for aluminum covers. These alloys are very compatible with alloys 6063-T6 and 5052-H32.

Finishes

Steel-Hot Dip Galvanized After Fabrication
This high-quality coating is created in a multi- step process and applied to the completed product after all fabrication and welding. First the steel is cleaned by immersing it in a caustic dip tank, then pickled, and finally washed. The article is then dipped into a molten zinc bath. All surfaces are thoroughly coated. Hot dip galvanizing after all fabrication and welding is done on MMICO products in accordence with ASTM Specification A123. or BS EN ISO 1461: 1999 This coating, applied to steel trays, provides greatest protection at lowest cost.

Mill Galvanized (Pre Galvanized)
This coating is applied onto steel sheets at the mll in accordance with ASTM Specification A653 and results in a coating thickness of 0.90 ounces per square foot total for both sides. This coating adheres tightly to the base metal at bends and compound curves. Being a hot dip coating prior to tray fabrication, and of a controlled thickness about one - third that of hot dip galvanized after fabrication, it is ant generally used outdoors. It is not recommended for areas where there is a concentration of moisture and/or chemical deposits. This material gives excellent service in dry, normal, or controlled atmospheres.

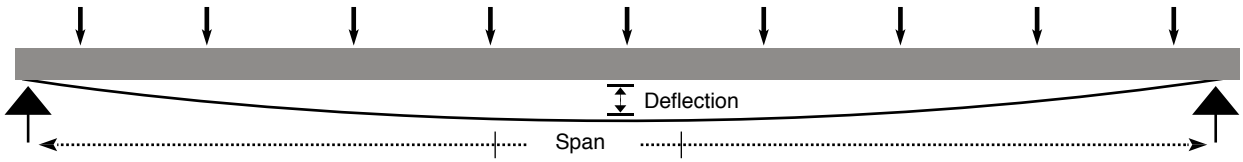
Other Coatings
Occasionally, other highly special coatings are required such as powder coating. epoxy painting. standard painting, primer paint, and others MMICO engineers are available or consultation on theses coatings.

2- Select the Tray Class/ Load Capacity (Loading)

The standard classes of cable trays, as related to their maximum design loads and to the associated design support spacing based on a simple beam span requirement, shall be designated in accordance with Table 1. Please note the load ratings in Table 1 are those most commonly used. other load ratings are acceptable. (according to NEMA VE-1) Costs Very between different load classes. Since labor and coupling costs are similar for a given length of tray, the heavier classes are more cost-effective on a load length basis. The designer should therefore specify the lightest class of tray compatible with the weight requirements of the cable tray.

TABLE1 **Load / Span Class Esignation**

LOAD km/m (lb/ft)		SPAN m (ft)				
		2.4 (8)	3.0 (10)	3.7 (12)	4.9 (16)	6.0 (20)
37	(25)	-	A	-	-	-
67	(45)	-	-	-	-	D
74	(50)	8A	-	12A	16A	20A
97	(65)	-	C	-	-	-
112	(75)	8B	-	12B	16B	E or 20B
149	(100)	8C	-	12C	16C	20C
179	(120)	-	D	-	-	-
299	(200)	-	E	-	-	-



Note: 8 A/B/C, 12 A/B/C, 16 A/B/C, and 20 A/B/C, are the traditional NEMA designations.
A, C, D, and E are the conventional C S A designations.

2- Select the Tray Class / Load Capacity

Cable Loads : The cable load is the total wenght, expressed in (Kg/m). of all the cables that will be placed in the cable tray.

Snow Loads : The additional design load from snowfall should be determined by the following formule:

Ice Loads : The Additional load design due to the ice is determined by the following fromula:

Wi = W × Tix Di/ 144

Where: WI = ice load (b/lin ear foot) W= width of the tray (inches)

Ti = maximum ic thickness (inches). Di = 75 lb/ft - ice density

ice thickness will vary depending on installation location.

Wind Loads : The additional loading to be considered is the effect of the impact pressure normal to the side rail.

This loading is Determined by the following formula:

Wp = 0.00256 × V xH/12

Where: Wp = loading due to the wind (li/lin ear foot) V = wind velocity (mph)

H = Height of the side rail (inches).

Concentrated Loads:

A concenttrated static load is not included in the Teble 1. Some user applications my require that a given concentrated

static load be imposed over and above the working load.

Such a concentrated static load represents a s static weight applied on the centerline of the tray at mid span. When so

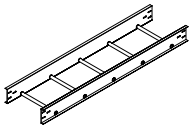
specified, the concentrated static load may be converted to an equivalent uniform load (We) in kilograms/ metre (pouds/linear

foot) using the following formula, and added to the static weight of cable in the tray:

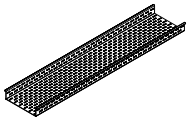
$$We = \frac{2 \times (\text{concentrated static load, kg (lb)})}{\text{Span length, m (ft)}}$$

3- Select the Tray Type

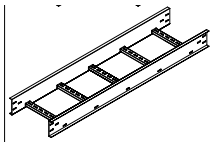
Tray Type : The type of cables to be installed in the cable tray or channel will dictate the tray type needed.



Swage Ladder :
Is a structure consisting of two side rails, connected by individual rungs and is manufactured in accordance with NEMA Standard # VE-1. rungs are fastened to the side members by swaging process.



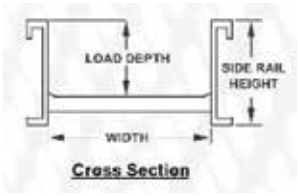
Welded Type Ladder :
Is a prefabricated metal structure consisting of rungs, NEMA Standerd VE-1. MMICO rungs are fastened to the side rails with MIG-arc-welding system.



Perforated Type Cable Tray :
System is ventilated with inside & outside return flange. This Perforation makes the structure to have less weight and Provides good ventilation, thus, protecting the cable from extreme heat. This kind of system provide high quality, cost effective and reliable solution to cable distribution and installation work.

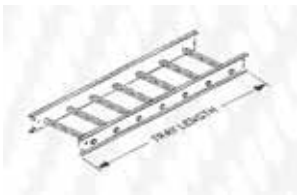
4- Select the Tray Size

Tray Size : Is essential to consider the type and quantity of cables to be installed in the cable tray or channel.



Load Depth :
Interior depth of the cable tray that is avilable for cable fill. Specifications for cable trays should include a specific requirement for the tray load depth. Due to design variations between different manufacturers, it is imprtant to specify an exact load depth to ensure equal performance. Standard MMICO load depths are 3",4",5", and 6", Custom depths are available, please consult factory.

Side Rail Height :
The overall height of the cable tray side rail. In general, side rail heights for MMICO cable trays
available, please consult factory for more information.



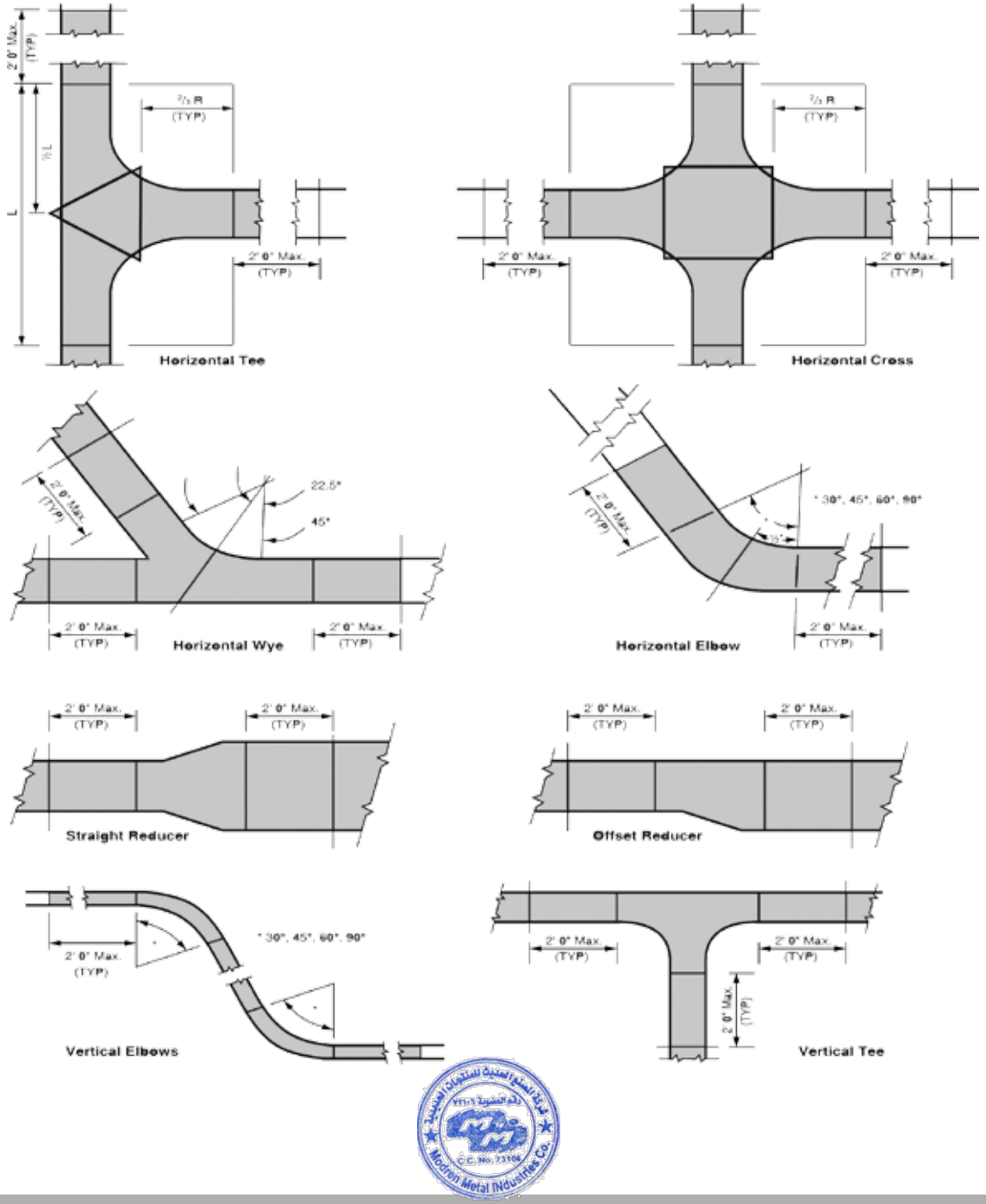
Tray Length :
The overall length of each cable tray straight section. The spacing of the cable tray supports will determine the length requirements for straight trays. The support span should not be greater than the straight section length to ensure that no more than one splice plate connection is located between supports. Standard MMICO tray lengths are 10;12;20; and 24; Custom lengths are available, please consult factory.

Width :
Interior width of the tray or channel that is available for cable fill. Standard widths are 6",9",12",18", 24",30", and 36" for MMICO tray and 3",4", and 6" for MMICO Channel. Non standard widths are also available.

5- Select the Fittings

Fittings are used to change the size or direction of the cable tray. The most imprtant decision to be made in fitting design concerns radius. The radius of the bend, whether horizontal or vertical. can be 12, 24, 36 in., of even greater on a custom basis. THe selection reqiures a compromise with the considerations being available space, minimum bending radius of cables, ease of cable pulling, and cost. The typical radius is 24 in. Fittings are also available for 30 , 45 , 60 , and 90 angles. It may be necessary to add supports to the tray at these points. Refer to NEMA VE2 Installation Guidelines for suggested support locations.

Support Locations for Fittings



6- Consider Deflection

The NEMA load test is a simple beam, uniformly distributed load test. (see Figure 1 . 2)
This type of test was initially selected because :

* It was easiest to test.
* It represents the worst case beam condition compared to contnuous or fixed cenfigurations. When consulting the manufacturer's catalog for deflaction information, the designer must verify whether the data shown represents simple or contiunous beam deflection. If continuous beam deflection is show, the calculation facto shuld be given.

NEMA has one criterion for acceptance under their load test : the ability to support 150% of the rated load.

Test Load = 1.5x rated load x length

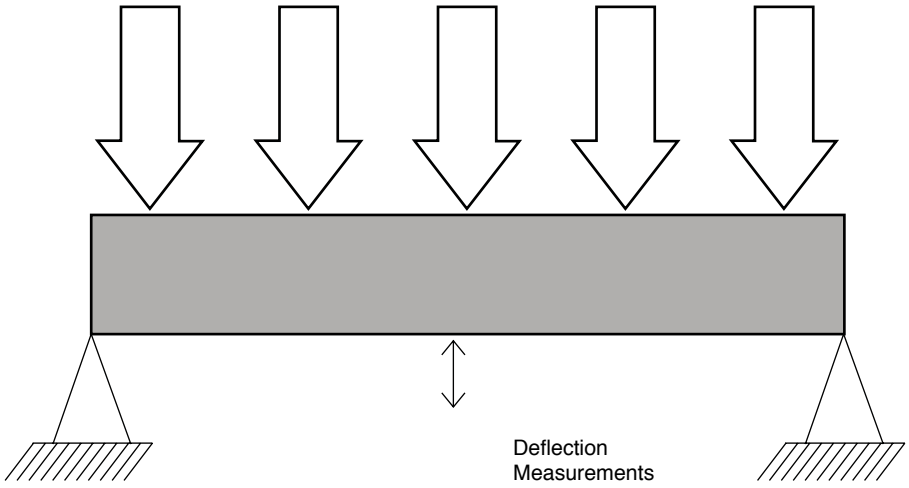


Figure 1.2

Theoretical maximum deflection for a simple beam, uniformly distributed load may be calculated as :

$$0130 \frac{WL}{EI}$$

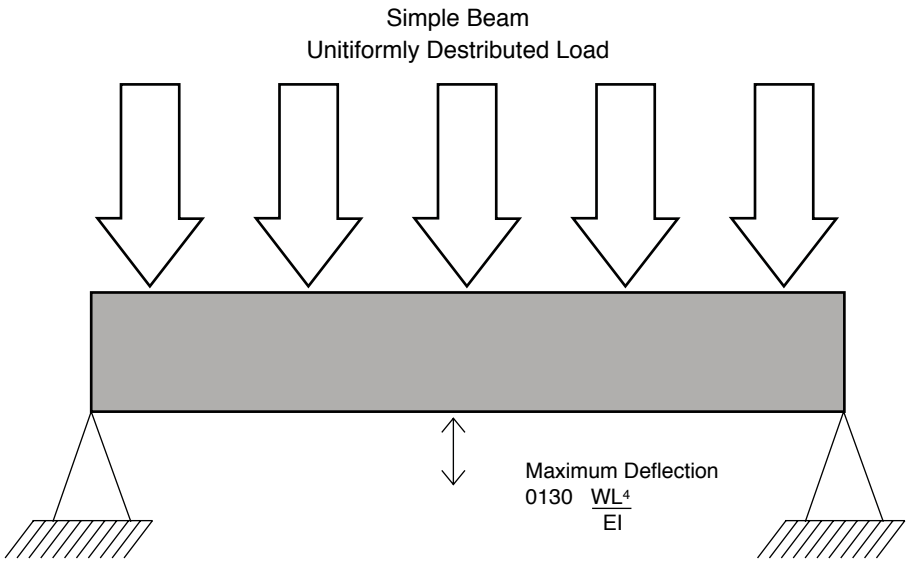
WHERE : W = Load in lb/ft
L = Length in inches
E Modulus of Elasticity
I = Moment of Inertia

The maximum deflection calculation for a continuous beam of two spans with a uniformly distributed load is :

$$00541 \frac{WL}{EI}$$

A continuous beam of two spans therefore has a theoretical maximum deflection of only 42% of its simple beam deflection. As the number of spans increases, the beam behaves increasingly like a fixed beam. and the maxximum deflection continues to decrease As this occurs, the system's load carrying capability increases. sse Figure 1.3

Simple vs. Continuous Beam Deflection



Continuous Beam - Two Spans
Uniformly Distributed Load

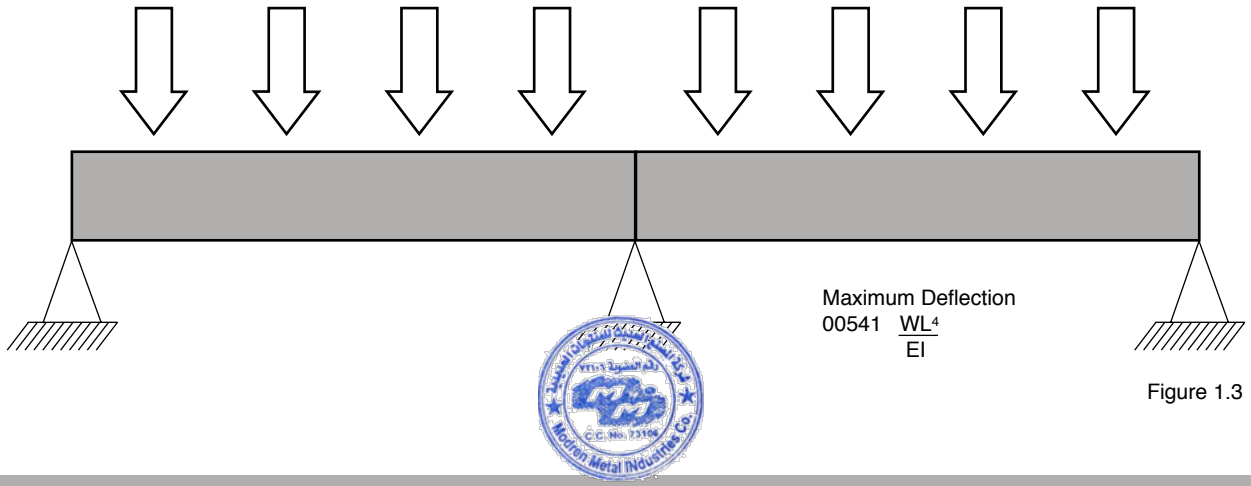
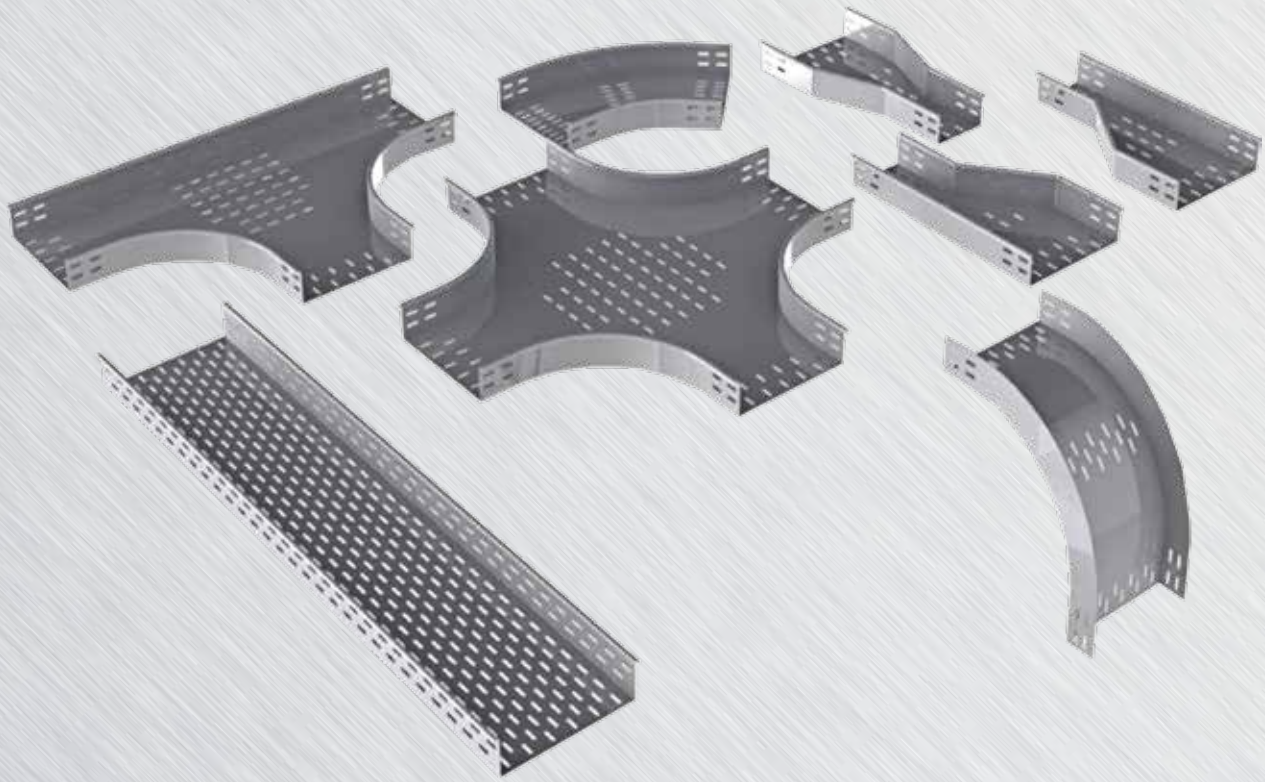


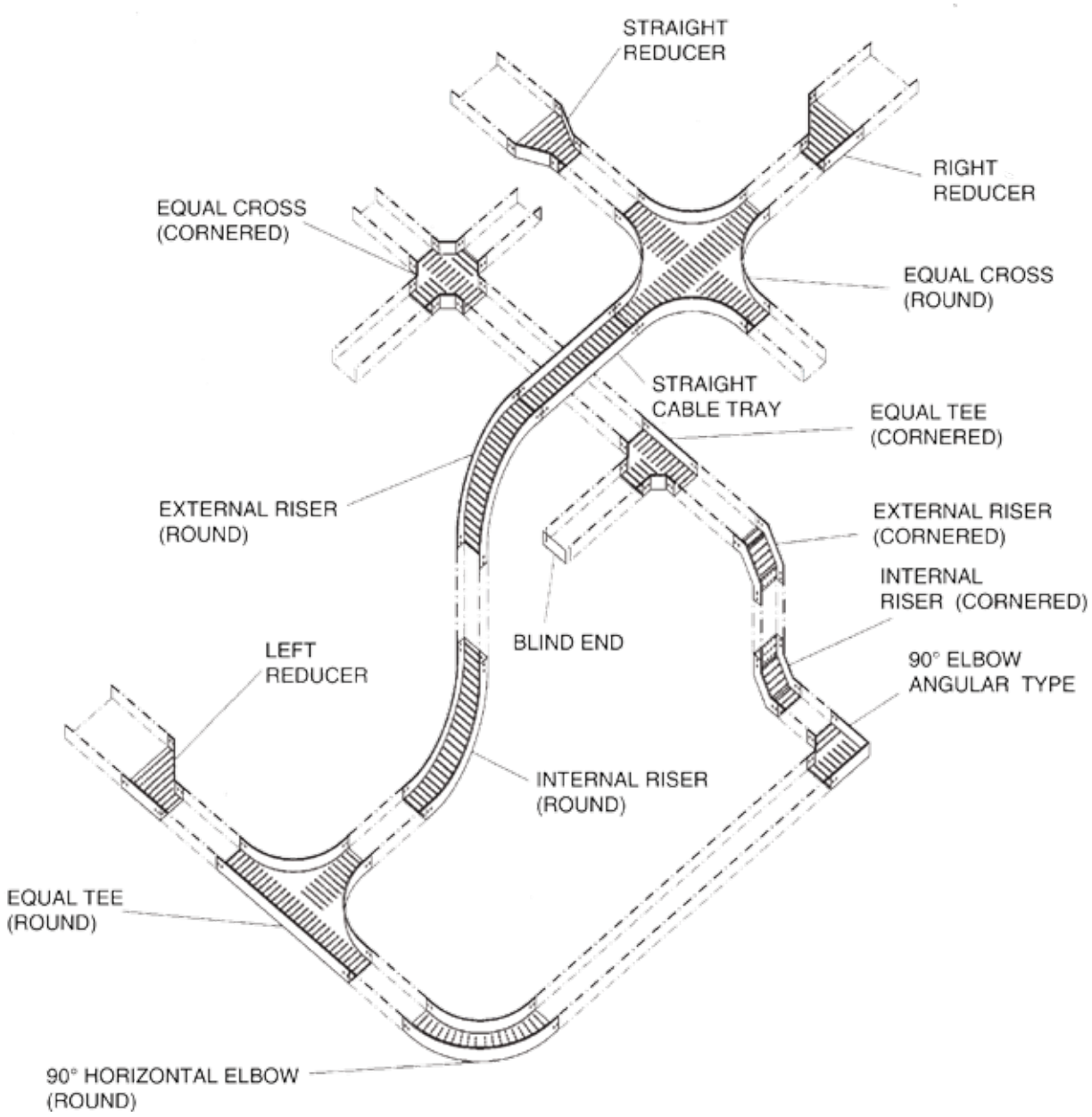
Figure 1.3



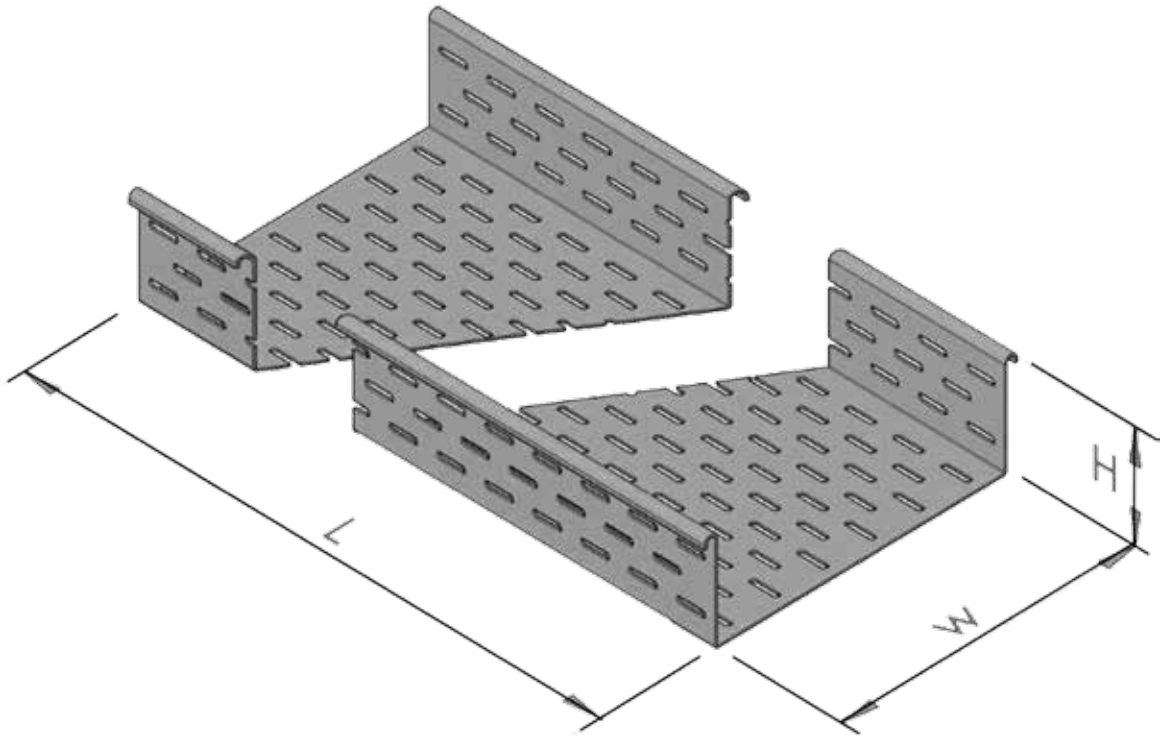
CABLE TRAY

MODERN METAL INDUSTRIES

Cable Tray Typical Layout



General Technical Information



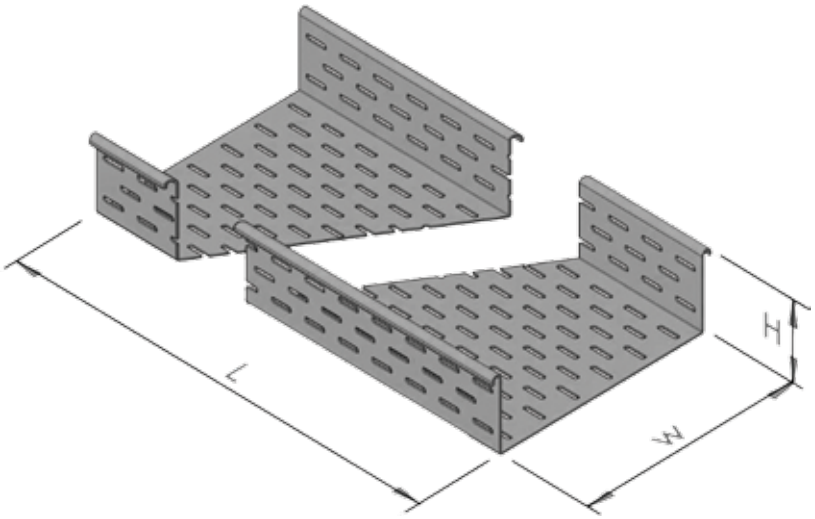
MMICO Classic preforated steel cable trays comply with the most well known standards name - VE-1, G.I to BS 2989, cold rolled to BS 2994 1976 galvanizing to BS EN ISO 1462 or ASTM 123. MMICO cable trays are provided with continuous returned flanges which helps sustain the desired cable loding.

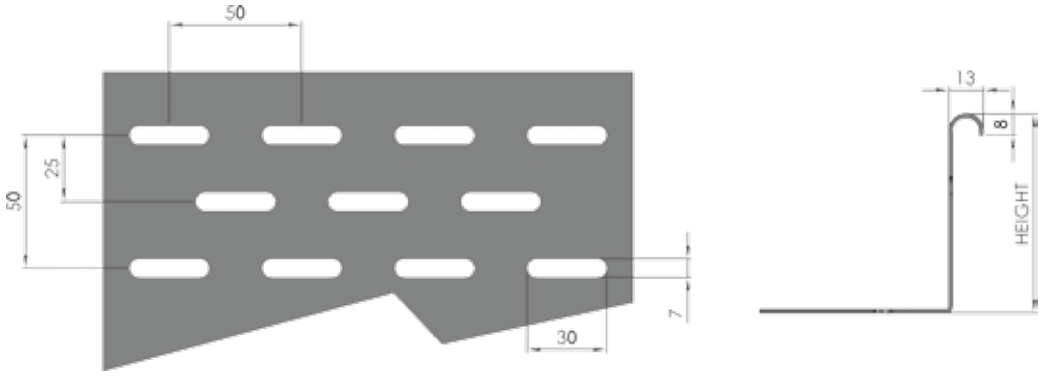
Manufactured from a series of welded C-Channel System and Tubular pipe swaged system with standard Height of 110, 135, 160 185mm. (4.25”, 5.25”, 6.25”, 7.25”). Side rail flanges could be inward or outward. Standard length 2.44 to 6.0 M.

COMPLIANCE STATEMENT

Modern Metal Industries Company never makes any design changes or new products. All products are being manufactured and in compliance with the SAUDI ARAMCO Engineering Standard SAES-P-104 and SAMSS and as well as other related International Standards

General Technical Information

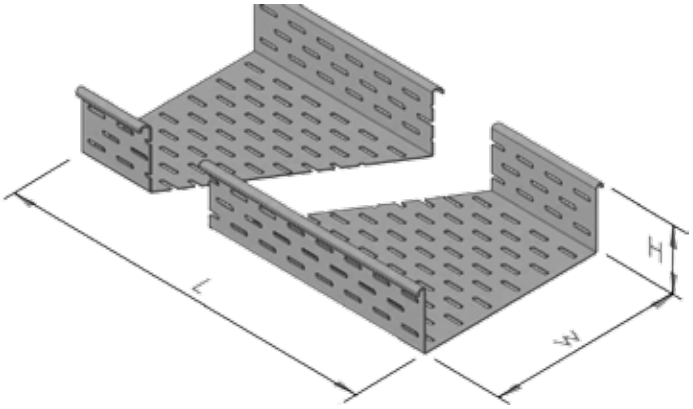


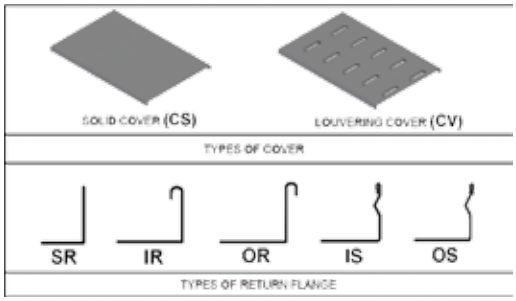


LIGHT DUTY CABLE TRAY
‹›U›› Shaped with returned flange, manufactured from 1.2 to 1.5 mm thickness perforated shets, flanges could be inward or woutward. Tray height is normally 35 & 50 standard length of 2.44 to 3.0 M.

HEAVY DUTY CABLE TRAY
‹›U›› Shaped with returned flange, manufactured form 1.5 to 2.0 mm thickness perforated sheets, flanges could be inward or outward. tray height is normally 50 to 100 mm standard length 2.44 to 3.0 M.

Cable Tray Straight





Ordering Code	CT					/		/		/	

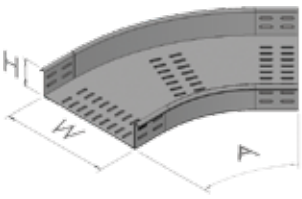
Cable Tray	CT									
With Cover	CS	CV								
Without Cover	-									
Straight	SR									
Round Inside Return	IR									
Round Outside Return	OR									
Inside Return Flonge Snap On	IS									
Oulside Return Flange Snap On	OS									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	50	100	150	200	300	450	600	900	1000	
Helght (H) mm	50	75	85	100						
Thickness (T) mm	1.0	1.25	1.5	2.0						
Length (L) mm	2000	2440	3000							

Ordering Code Example : CTCSORHDG600/50/2.0/3000

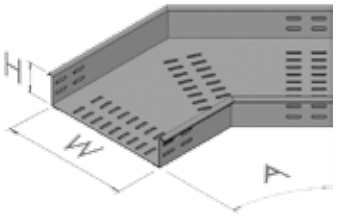
NOTES:

- Cover fixation as per required. (For types of Fixations, see page160)
- Standard cover thickness is 1.5 mm for below 300 mm width and 2 mm for 300 mm width above.
- Special orders can be manufactured upon request.

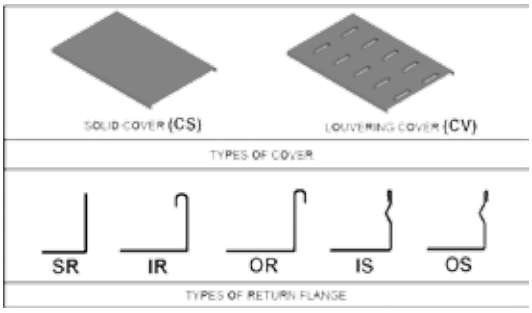
Cable Tray Horizontal Elbow



Round Type



Cornered Type



Ordering Code	CTLH							/		/		/	

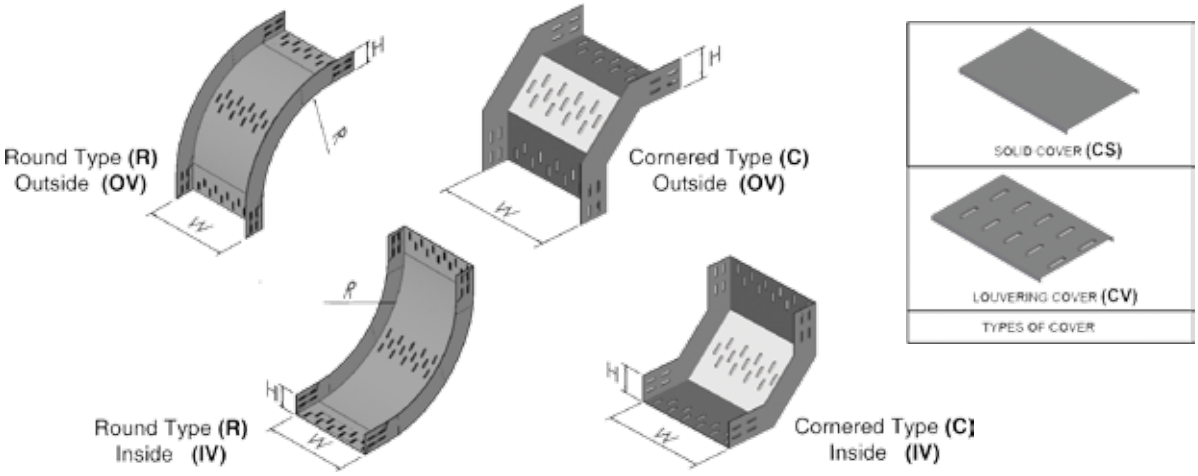
Cable Tray Horizontal Elbow	CTLH						
With Cover	CS	CV					
Without Cover	-						
Angle	45°	90°					
Round Type	D						
Cornered Type 45	R						
Radius	300	600	900				
Straight	SR						
Round Inside Return	IR						
Round Outside Return	OR						
Inside Return Flonge Snap On	IS						
Oulside Return Flange Snap On	OS						
Black Steel	BST						
Hot Dip Galvanized after Fabrication	HDG						
Hot Dip Galvanized and powder Coating	HDP						
Mill Galvanized	MG						
Steel Epoxy Coating	EPC						
Aluminum	AL						
Stainless Steel 304	S4						
Stainless Steel 316	S6						
Width (W) mm	50	100	150	200	300	1100	
Helght (H) mm	50	75	85	100			
Thickness (T) mm	1.0	1.25	1.5	2.0			

Ordering Code Example : CTLHCS45D300ORHDG200/75/1.5

NOTES:

- Cover fixation as per required.
- Standard cover thickness is 1.5 mm for below 3[illegible]mm width and 2 mm for 300 mm width above.
- Special orders can be manufactured upon req[illegible]

Cable Tray Vertical Elbow



Ordering Code	CTLV							/		/		/	

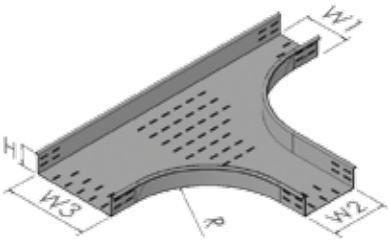
Cable Tray Horizontal Elbow	CTLH									
With Cover	CS	CV								
Without Cover	-									
Angle	45°	90°								
Outside Vertical Elbow	OV									
Inside Vertical Elbow	IV									
Round Type	D									
Cornered Type 45	R									
Radius	300	600	900							
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	50	100	150	200	300	450	600	900	1000	
Helght (H) mm	50	75	85	100						
Thickness (T) mm	1.0	1.25	1.5	2.0						

Ordering Code Example : CTLVCS45OVD300HDG200/75/1.5

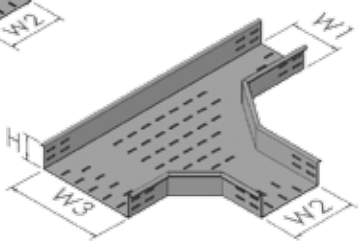
NOTES:

- Cover fixation as per required.
- Only straight type of return flange for this fittings.
- Standard cover thickness is 1.5 mm for below 300 mm width and 2 mm for 300 mm width above.
- Special orders can be manufactured upon request.

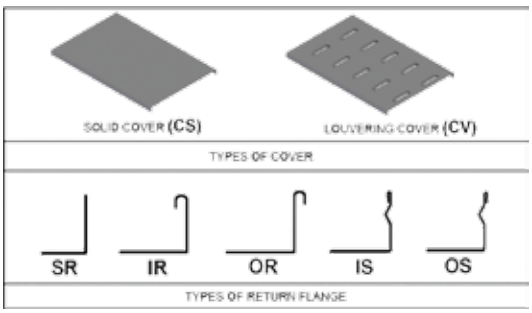
Cable Tray Horizontal Tee



Round Type **R**



Cornered Type **C**



Ordering Code	CTT							/		/		/	

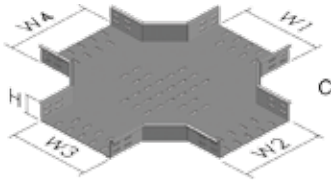
Cable Tray Horizontal Elbow	CTT									
With Cover	CS	CV								
Without Cover	-									
Round Type	D									
Cornered Type 45	R									
Radius	300	600	900							
Straight	SR									
Round Inside Return	IR									
Round Outside Return	OR									
Inside Return Flonge Snap On	IS									
Oulside Return Flange Snap On	OS									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W1/W2/W3) mm	50	100	150	200	300	450	600	900	1000	
Helght (H) mm	50	75	85	100						
Thickness (T) mm	1.0	1.25	1.5	2.0						

Ordering Code Example : CTTCSD300ORHDG450/300/200/100/2.0

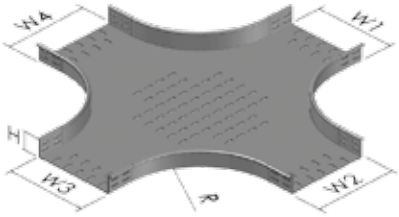
NOTES:

- Cover fixation as per required.
- Standard cover thickness is 1.5 mm for below 3[illegible] mm [illegible] and 2 mm for 300 mm width above.
- Special orders can be manufactured upon req[illegible]

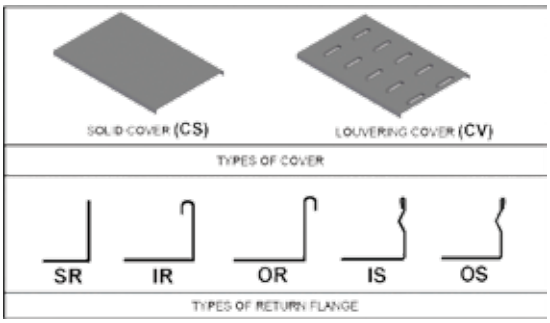
Cable Tray Horizontal Cross



Cornered Type **C**



Round Type **R**



Ordering Code	CTX							/		/		/	

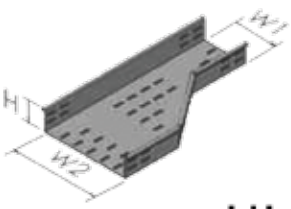
Cable Tray Horizontal Elbow	CTX									
With Cover	CS	CV								
Without Cover	-									
Round Type	D									
Cornered Type 45	R									
Radius	300	600	900							
Straight	SR									
Round Inside Return	IR									
Round Outside Return	OR									
Inside Return Flonge Snap On	IS									
Oulside Return Flange Snap On	OS									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W1/W2/W3/W4) mm	50	100	150	200	300	450	600	900	1000	
Helght (H) mm	50	75	85	100						
Thickness (T) mm	1.0	1.25	1.5	2.0						

Ordering Code Example : CTXCSD300ORHDG450/200/300/600/100/2.0

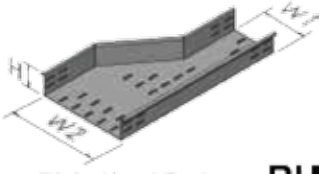
NOTES:

- Cover fixation as per required.
- Standard cover thickness is 1.5 mm for below 2[illegible] mm width and 2 mm for 300 mm width above.
- Special orders can be manufactured upon req[illegible]

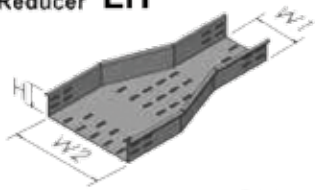
Cable Tray Reducers



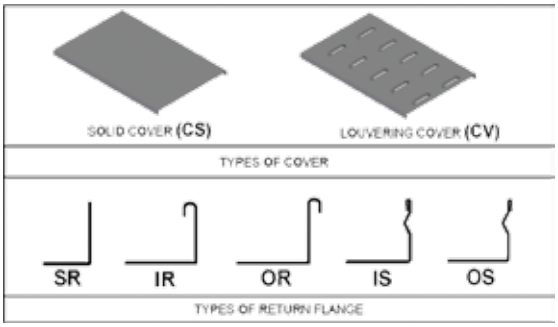
Left Hand Reducer **LH**



Right Hand Reducer **RH**



Straight Reducer **ST**



Ordering Code	CTX						/		/		/	

Cable Tray Horizontal Elbow	CTX	
With Cover	CS	CV
Without Cover	-	
Starigth	ST	
Right Hand	RH	
Left hand	LH	
Straight	SR	
Round Inside Return	IR	
Round Outside Return	OR	
Inside Return Flonge Snap On	IS	
Oulside Return Flange Snap On	OS	
Black Steel	BST	
Hot Dip Galvanized after Fabrication	HDG	
Hot Dip Galvanized and powder Coating	HDP	
Mill Galvanized	MG	
Steel Epoxy Coating	EPC	
Aluminum	AL	
Stainless Steel 304	S4	
Stainless Steel 316	S6	

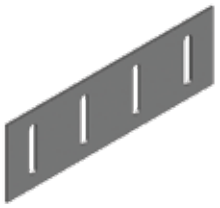
Width (W1/W2/W3/W4) mm	50	100	150	200	300	450	600	900	1000
Helght (H) mm	50	75	85	100					
Thickness (T) mm	1.0	1.25	1.5	2.0					

Ordering Code Example : CTYCSRHORHDG600/450/100/2.0

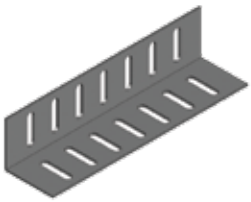
NOTES:

- Cover fixation as per required.
- Standard cover thickness is 1.5 mm for below 300 mm width and 2 mm for 300 mm width above.
- Round type bend is not available for this fittings.
- Special orders can be manufactured upon request.

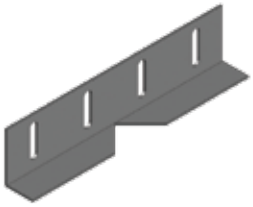
Cable Tray Accessories



Straight Joint
(CT01)



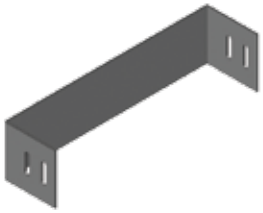
Angle Straight Connector
(CT02)



Angle Joint
(CT03)



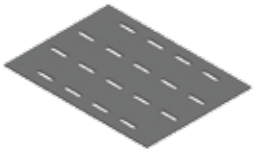
VerticalJoint
(CT04)



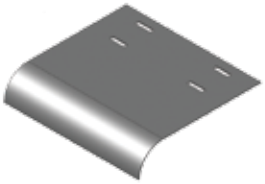
End PlateJoint
(CT05)



Reducing Connector
(CT06)



Fish Plate
(CT07)



Drop Out
(CT08)

Cable Tray Accessories



Separator Strip
(CT09)



Cover Side Clamp
(CT10)



Cover Double Clamp Connector
(CT11)



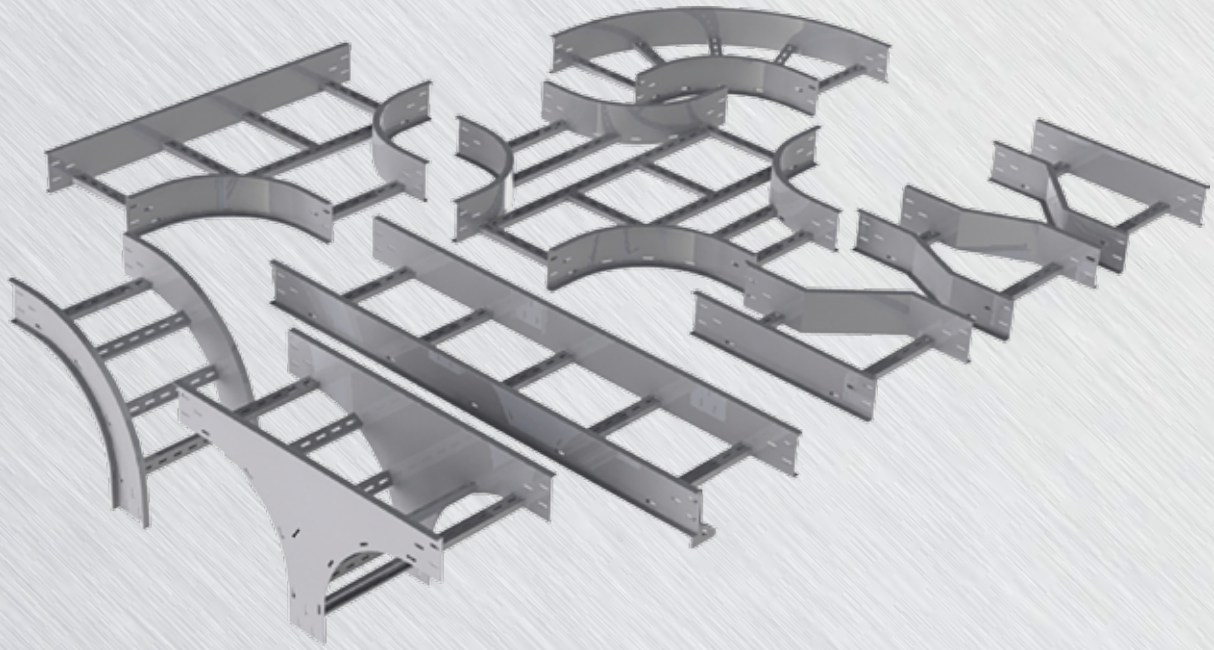
Cover Bracket
(CT12)



Cover Lock Device
(CT13)



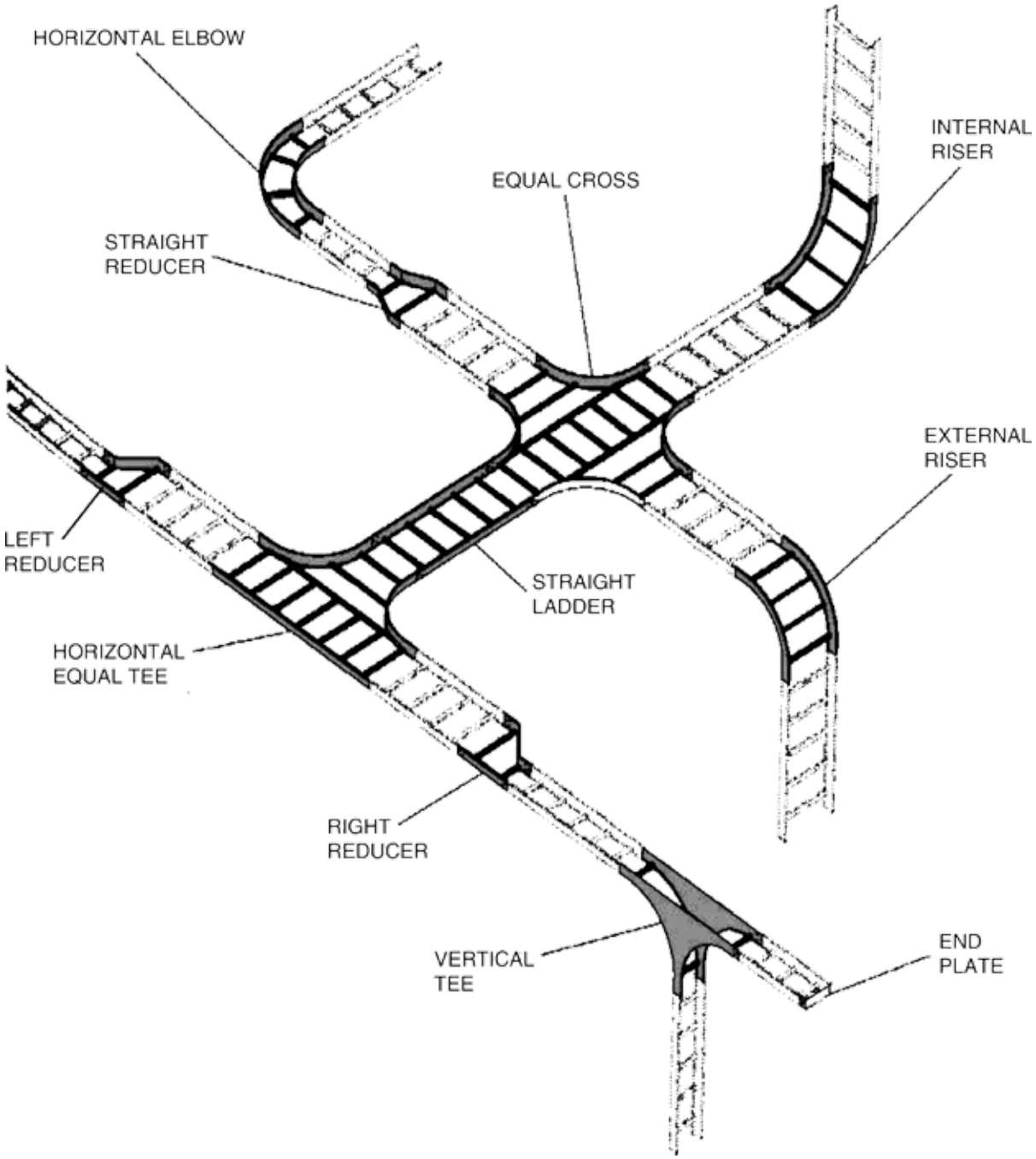
Wrap Around Cover Connector
(CT14)



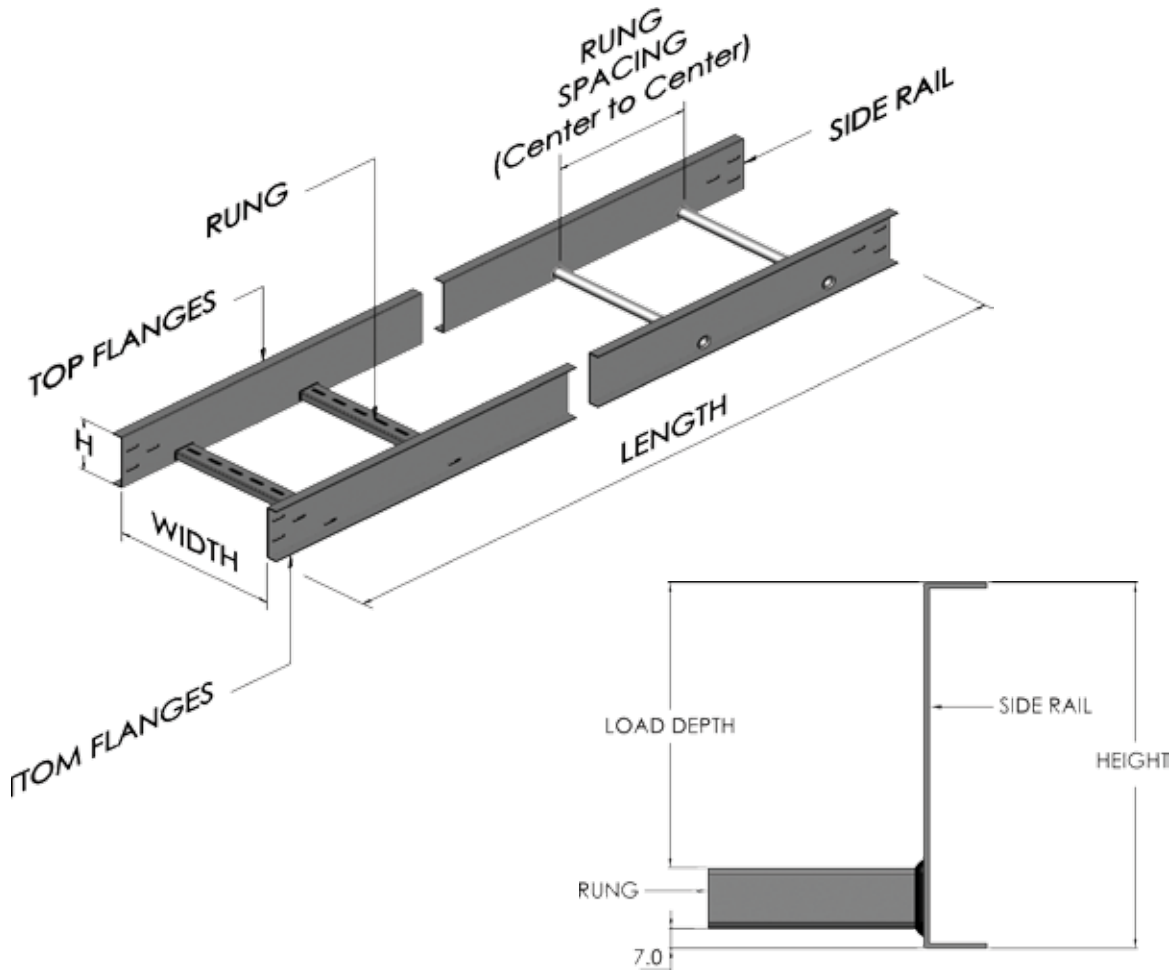
CABLE LADDER

MODERN METAL INDUSTRIES

Cable Ladder Typical Layout



General Technical Information



MMICO has a comprehensive system of cable ladders. MMICO Ladders come in two categories Tubular Swaged Type & C - Channel welded system. MMICO Cable Ladders are comply NEMA VE-1, BS EN ISO 1 461 or ASTM 123 for HDG and for Cold Rolled BS 2994 1976.

Manufactured from a series of welded C-Channel System and Tubular pipe swaged system with standard Height of 110, 135, 160 185mm. (4.25”, 5.25”, 6.25”, 7.25”). Side rail flanges could be inward or outward. Standard length 2.44 to 6.0 M.

COMPLIANCE STATEMENT

Modern Metal Industries Company never makes any design changes or new products. All products are being manufactured and in compliance with the SAUDI ARAMCO Engineering Standard SAES-P-104 and SAMSS and as well as other related International Standards.

Technical Specification - Steel Cable Ladder

Physical Properties, Load and Deflection Data for Steel Ladder							Working (Allowable) Load Capacity-Evenly Distributed Span (ft.) - Simple Beam - Safety Factor : 1.5																	
NEMA Class Load per Span		MMICO System Number	Load Depth	Height	Flange	Thickness	6Ft. (1.8M)			8Ft. (2.4M)			10Ft. (3.1M)			12Ft. (3.7M)			16Ft. (4.9M)			20Ft. (6.1M)		
			mm	mm	mm	mm	w	d	k	w	d	k	w	d	k	w	d	k	w	d	k	w	d	k
8A	8= 8' - 0"	STSS 08A3	76	108	21	1.2	292	0.23	+	164	0.42	0.003	103	0.63	0.006	69	0.89	0.013						
		STSS 08A4	102	133	21	1.2	383	0.18	+	210	0.31	0.001	127	0.46	0.004	83	0.62	0.007						
		STSS 08A5	127	159	21	1.2	357	0.11	+	194	0.18	+	119	0.28	0.002	79	0.38	0.005						
	A=50 lb./ft.	STSS 08A6	152	184	21	1.2	420	0.09	+	227	0.16	+	139	0.22	0.002	92	0.30	0.003						
8B	(74.4 kg/M)	STSS 08B3	76	108	21	1.2	292	0.23	+	164	0.42	0.003	103	0.63	0.006	69	0.89	0.013						
		STSS 08B4	102	133	21	1.2	383	0.18	+	210	0.31	0.001	127	0.46	0.004	83	0.62	0.007						
	B=75 lb./ft.	STSS 08B5	127	159	21	1.2	357	0.11	+	194	0.18	+	119	0.28	0.002	79	0.38	0.005						
	(111.6 kg/M)	STSS 08B6	152	184	21	1.2	420	0.09	+	227	0.16	+	139	0.22	0.002	92	0.30	0.003						
8C	C=100 lb./ft.	STSS 08C3	76	108	21	1.2	292	0.23	+	164	0.42	0.003	103	0.63	0.006	69	0.89	0.013						
		STSS 08C4	102	133	21	1.2	383	0.18	+	210	0.31	0.001	127	0.46	0.004	83	0.62	0.007						
	(148.8 kg/M)	STSS 08C5	127	159	21	1.2	357	0.11	+	194	0.18	+	119	0.28	0.002	79	0.38	0.005						
		STSS 08C6	152	184	21	1.2	420	0.09	+	227	0.16	+	139	0.22	0.002	92	0.30	0.003						
12A	50 lb./ft.	STSS 12A3	76	108	21	1.2	292	0.23	+	164	0.42	0.003	103	0.63	0.006	69	0.89	0.013						
		STSS 12A4	102	133	21	1.2	383	0.18	+	210	0.31	0.001	127	0.46	0.004	83	0.62	0.007						
	(74.4 kg/M)	STSS 12A5	127	159	21	1.2	357	0.11	+	194	0.18	+	119	0.28	0.002	79	0.38	0.005						
	12' span	STSS 12A6	152	184	21	1.2	420	0.09	+	227	0.16	+	139	0.22	0.002	92	0.30	0.003						
12B	75 lb./ft.	STSS 12B3	76	108	21	1.2	360	0.26	+	198	0.46	0.002	124	0.70	0.006	84	0.98	0.012						
		STSS 12B4	102	133	21	1.2	383	0.18	+	210	0.31	0.001	127	0.46	0.004	83	0.62	0.007						
	(111.6 kg/M)	STSS 12B5	127	159	21	1.2	357	0.11	+	194	0.18	+	119	0.28	0.002	79	0.38	0.005						
	12' span	STSS 12B6	152	184	21	1.2	680	0.12	+	328	0.19	+	182	0.26	0.001	107	0.31	0.003						
12C	100 lb./ft.	STSS 12C3	76	108	21	1.5	453	0.27	+	249	0.47	0.002	156	0.71	0.005	106	1.00	0.009						
		STSS 12C4	102	133	21	1.2	456	0.20	+	244	0.33	0.001	152	0.51	0.003	103	0.71	0.007						
	(148.8 kg/M)	STSS 12C5	127	159	21	1.5	460	0.11	+	250	0.19	+	154	0.29	0.002	103	0.40	0.004						
	12' span	STSS 12C6	152	184	21	1.2	680	0.12	+	328	0.19	+	182	0.26	0.001	107	0.31	0.003						
16A	50 lb./ft.	STSS 16A3	76	108	32	1.2				290	0.53	0.002	182	0.81	0.004	126	1.71	0.009	65	1.91	0.029			
		STSS 16A4	102	133	32	1.2				382	0.42	0.001	245	0.66	0.003	170	0.94	0.006	86	1.54	0.018			
	(74.4 kg/M)	STSS 16A5	127	159	32	1.2				268	0.20	+	166	0.31	0.002	111	0.43	0.004	58	0.71	0.012			
	16' span	STSS 16A6	152	184	32	1.2				322	0.17	+	199	0.26	0.001	133	0.36	0.003	69	0.58	0.008			
16B	75 lb./ft.	STSS 16B3	76	108	32	1.5				381	0.56	0.001	244	0.88	0.004	169	1.27	0.008	88	2.08	0.024			
		STSS 16B4	102	133	32	1.2				382	0.42	0.001	245	0.66	0.003	170	0.94	0.006	86	1.54	0.018			
	(111.6 kg/M)	STSS 16B5	127	159	32	1.2				426	0.31	+	259	0.46	0.002	171	0.62	0.004	86	0.99	0.012			
	16' span	STSS 16B6	152	184	32	1.2				322	0.17	+	199	0.26	0.001	133	0.36	0.003	75	0.63	0.008			
16C	100 lb./ft.	STSS 16C3	76	108	32	1.9				487	0.58	0.001	312	0.91	0.003	216	1.31	0.006	113	2.16	0.019			
		STSS 16C4	102	133	32	1.5				444	0.42	+	273	0.63	0.002	186	0.89	0.005	100	1.51	0.015			
	(148.8 kg/M)	STSS 16C5	127	159	32	1.5				574	0.35	+	367	0.55	0.001	245	0.76	0.003	123	1.22	0.010			
	16' span	STSS 16C6	152	184	32	1.2				504	0.25	+	306	0.37	0.001	202	0.51	0.003	101	0.80	0.008			
20A	50 lb./ft.	STSS 20A3	76	108	32	1.5										169	1.27	0.008	88	2.08	0.024	52	2.98	0.057
		STSS 20A4	102	133	32	1.5										186	0.89	0.005	100	1.51	0.015	53	1.94	0.037
	(74.4 kg/M)	STSS 20A5	127	159	32	1.5										245	0.76	0.003	123	1.22	0.010	66	1.59	0.024
	20' span	STSS 20A6	152	184	32	1.5										178	0.38	0.002	93	0.62	0.007	55	0.91	0.017
20B	75 lb./ft.	STSS 20B3	76	108	38	1.9										251	1.36	0.005	131	2.24	0.017	77	3.23	0.042
		STSS 20B4	102	133	32	1.9										280	1.02	0.004	145	1.67	0.012	85	2.4	0.028
	(111.6 kg/M)	STSS 20B5	127	159	32	1.5										269	0.80	0.003	139	1.28	0.009	81	1.82	0.022
	20' span	STSS 20B6	152	184	32	1.5										291	0.59	0.002	148	0.95	0.006	85	1.33	0.016
20C	100 lb./ft.	STSS 20C3	76	108	38	2.5										316	1.26	0.004	164	2.06	0.013	100	3.09	0.013
		STSS 20C4	102	133	38	1.9										304	1.00	0.003	157	1.00	0.010	100	2.5	0.025
	(148.8 kg/M)	STSS 20C5	127	159	32	1.9										296	0.79	0.003	152	0.79	0.008	105	1.91	0.018
	20' span	STSS 20C6	152	184	32	1.9										379	0.66	0.002	191	0.66	0.005	102	1.37	0.013

to Convert 1.5 safety factor to 2.0 - Multiply w,d,k by - 75

w = Allowable working load (lbs/ft) to convert to kg/M multiply by 148

d = Deflection at midspan (inches) to convert to mm multiply by 25.4.

k = Deflection for each pound of load (inches).

\+ = Deflection is less than 0001/ inch.

Technical Specification - Aluminum Cable Ladder

Physical Properties, Load and Deflection Data for Aluminum Ladder							Working (Allowable) Load Capacity-Evenly Distributed Span (ft.) - Simple Beam - Safety Factor : 1.5																	
NEMA Class Load per Span		MMICO System Number	Load Depth	Height	Flange	Thickness	6Ft. (1.8M)			8Ft. (2.4M)			10Ft. (3.1M)			12Ft. (3.7M)			16Ft. (4.9M)			20Ft. (6.1M)		
			mm	mm	mm	mm	w	d	k	w	d	k	w	d	k	w	d	k	w	d	k	w	d	k
8A	8= 8' - 0" A=50 lb./ft. (74.4 kg/M)	STSAL 08A3	76	108	20.6	1.5	181	0.34	0.002	102	0.60	0.006	65	0.94	0.014	45	1.36	0.030						
		STSAL 08A4	102	133	20.6	1.5	324	0.33	0.001	182	0.59	0.003	117	0.87	0.007	77	1.26	0.016						
		STSAL 08A5	127	159	20.6	2.0	359	0.22	+	191	0.37	0.002	115	0.55	0.005	75	0.74	0.010						
		STSAL 08A6	152	184	20.6	1.5	416	0.18	+	221	0.31	0.001	133	0.87	0.003	87	0.62	0.007						
8B	B=75 lb./ft. (111.6 kg/M)	STSAL 08B3	76	108	20.6	1.5	181	0.34	0.002	102	0.60	0.006	65	0.94	0.014	45	1.36	0.030						
		STSAL 08B4	102	133	20.6	1.5	324	0.33	0.001	182	0.59	0.003	117	0.87	0.007	77	1.26	0.016						
		STSAL 08B5	127	159	20.6	2.0	359	0.22	+	191	0.37	0.002	115	0.55	0.005	75	0.74	0.010						
		STSAL 08B6	152	184	20.6	1.5	416	0.18	+	221	0.31	0.001	133	0.87	0.003	87	0.62	0.007						
8C	C=100 lb./ft. (148.8 kg/M)	STSAL 08C3	76	108	20.6	1.5	181	0.34	0.002	102	0.60	0.006	65	0.94	0.014	45	1.36	0.030						
		STSAL 08C4	102	133	20.6	1.5	324	0.33	0.001	182	0.59	0.003	117	0.87	0.007	77	1.26	0.016						
		STSAL 08C5	127	159	20.6	2.0	359	0.22	+	191	0.37	0.002	115	0.55	0.005	75	0.74	0.010						
		STSAL 08C6	152	184	20.6	1.5	416	0.18	+	221	0.31	0.001	133	0.87	0.003	87	0.62	0.007						
12A	50 lb./ft. (74.4 kg/M) 12' span	STSAL 12A3	76	108	20.6	1.5	222	0.35	0.002	125	0.63	0.005	80	0.99	0.012	55	1.42	0.026						
		STSAL 12A4	102	133	20.6	1.5	324	0.33	0.001	182	0.59	0.003	117	0.87	0.007	77	1.26	0.016						
		STSAL 12A5	127	159	20.6	2.0	359	0.22	+	191	0.37	0.002	115	0.55	0.005	75	0.74	0.010						
		STSAL 12A6	152	184	20.6	1.5	416	0.18	+	221	0.31	0.001	133	0.46	0.003	87	0.62	0.007						
12B	75 lb./ft. (111.6 kg/M) 12' span	STSAL 12B3	76	108	20.6	2.0	354	0.45	0.001	191	0.76	0.004	117	1.14	0.010	78	1.57	0.020						
		STSAL 12B4	102	133	20.6	1.5	324	0.33	0.001	182	0.59	0.003	117	0.87	0.007	77	1.26	0.016						
		STSAL 12B5	127	159	20.6	2.0	359	0.22	+	191	0.37	0.002	115	0.55	0.005	75	0.74	0.010						
		STSAL 12B6	152	184	20.6	1.5	416	0.18	+	221	0.31	0.001	133	0.46	0.003	87	0.62	0.007						
12C	100 lb./ft. (148.8 kg/M) 12' span	STSAL 12C3	76	108	20.6	2.5	448	0.39	+	251	0.69	0.003	161	1.08	0.007	112	1.55	0.014						
		STSAL 12C4	102	133	20.6	2.0	394	0.31	+	222	0.55	0.002	142	0.86	0.006	100	1.24	0.012						
		STSAL 12C5	127	159	20.6	2.0	480	0.24	+	257	0.40	0.002	156	0.60	0.004	102	0.81	0.008						
		STSAL 12C6	152	184	20.6	2.5	540	0.17	+	286	0.29	0.001	172	0.43	0.003	111	0.57	0.005						
16A	50 lb./ft. (74.4 kg/M) 16' span	STSAL 16A3	76	108	20.6	2.5				251	0.69	0.003	161	1.08	0.007	112	1.55	0.014	51	2.26	0.044	25	2.75	0.110
		STSAL 16A4	102	133	31.75	2.5				260	0.44	0.002	166	0.69	0.004	96	0.82	0.009	51	1.36	0.027	28	1.83	0.065
		STSAL 16A5	127	159	31.75	2.0				273	0.37	0.001	155	0.51	0.003	101	0.69	0.007	51	1.10	0.022	24	1.28	0.053
		STSAL 16A6	152	184	31.75	2.0				284	0.26	+	182	0.41	0.002	110	0.52	0.005	52	0.78	0.015	30	1.1	0.037
16B	75 lb./ft. (111.6 kg/M) 16' span	STSAL 16B3	76	108	31.75	2.0				342	0.72	0.002	219	1.12	0.005	152	1.61	0.011	82	2.74	0.033	52	4.27	0.082
		STSAL 16B4	102	133	31.75	3.0				349	0.51	0.001	223	0.80	0.004	155	1.15	0.016	79	1.85	0.023	51	2.89	0.057
		STSAL 16B5	127	159	31.75	1.5				380	0.49	0.001	222	0.69	0.003	154	1.00	0.007	78	1.61	0.021	50	2.51	0.050
		STSAL 16B6	152	184	38.1	1.5				377	0.32	+	241	0.50	0.002	149	0.64	0.004	78	1.06	0.014	50	1.65	0.033
16C	100 lb./ft. (148.8 kg/M) 16' span	STSAL 16C3	76	108	31.75	2.5				444	0.71	0.002	284	0.70	0.002	188	1.53	0.008	106	2.73	0.026	65	4.07	0.063
		STSAL 16C4	102	133	31.75	2.0				525	0.55	0.001	336	0.86	0.003	233	1.24	0.005	119	2.01	0.017	76	3.14	0.041
		STSAL 16C5	127	159	44.45	2.0				481	0.41	+	308	0.64	0.002	215	0.92	0.004	121	1.63	0.013	77	2.55	0.033
		STSAL 16C6	152	184	38.1	1.5				542	0.44	+	318	0.62	0.002	221	0.90	0.004	119	1.53	0.013	76	2.39	0.032
20A	50 lb./ft. (74.4 kg/M) 20' span	STSAL 20A3	76	108	31.75	2.0										152	1.61	0.011	82	2.74	0.033	52	4.27	0.082
		STSAL 20A4	102	133	31.75	3.0										155	1.51	0.007	79	1.85	0.023	51	2.89	0.057
		STSAL 20A5	127	159	31.75	1.5										154	1.00	0.006	78	1.61	0.021	50	2.51	0.050
		STSAL 20A6	152	184	38.1	1.5										221	0.90	0.004	119	1.53	0.013	76	2.39	0.032
20B	75 lb./ft. (111.6 kg/M) 20' span	STSAL 20B3	76	108	38.1	2.0										224	1.53	0.007	120	2.59	0.022	77	4.05	0.053
		STSAL 20B4	102	133	31.75	2.0										233	1.24	0.005	119	2.01	0.017	76	3.14	0.041
		STSAL 20B5	127	159	44.45	2.0										215	0.92	0.004	121	1.63	0.013	77	2.55	0.033
		STSAL 20B6	152	184	38.1	1.5										221	0.90	0.004	119	1.53	0.013	76	2.39	0.032
20C	100 lb./ft. (148.8 kg/M) 20' span	STSAL 20C3	76	108	38.1	2.5										298	1.79	0.006	161	3.07	0.019	103	4.79	0.047
		STSAL 20C4	102	133	31.75	2.5										290	1.35	0.005	156	2.30	0.015	100	3.59	0.036
		STSAL 20C5	127	159	44.45	2.5										316	1.23	0.004	178	2.19	0.012	114	3.43	0.030
		STSAL 20C6	152	184	50	2.0										313	0.93	0.003	176	1.66	0.099	113	2.59	0.023

to Convert 1.5 safety factor to 2.0 - Multiply w,d,k by - 75

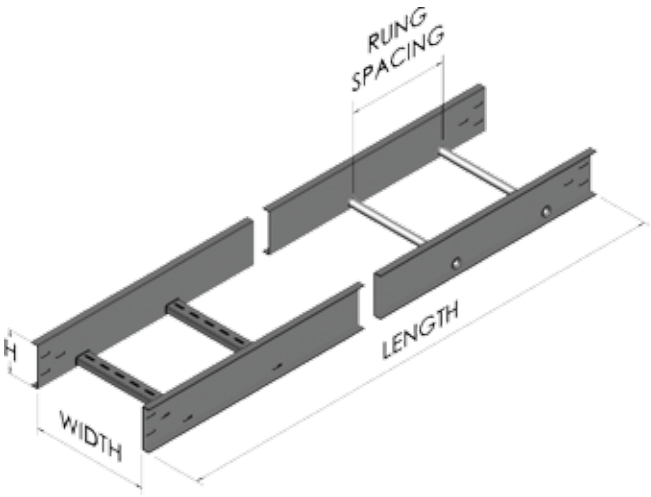
w = Allowable working load (lbs/ft) to convert to kg/M multiply by 148

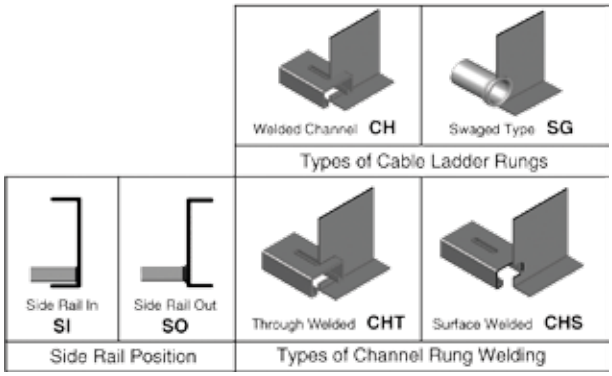
d = Deflection at midspan (inches) to convert to mm multiply by 25.4.

k = Deflection for each pound of load (inches).

\+ = Deflection is less than 0001/ inch.

Cable Ladder Straight





Ordering Code	CL					/		/		/		/		/	

Cable Ladder	CL										
Swaged Type	SG										
Welded Channel	CHS	CHT									
Side Rail In	SI										
Side Rail Out	SO										
Rung Spacing	mm	150	229	250	300						
	inches	6”	9”	10”	12”						
Black Steel	BST										
Hot Dip Galvanized after Fabrication	HDG										
Hot Dip Galvanized and powder Coating	HDP										
Mill Galvanized	MG										
Steel Epoxy Coating	EPC										
Aluminum	AL										
Stainless Steel 304	S4										
Stainless Steel 316	S6										
Width (W)	mm	100	150	200	300	450	600	750	900	1000	
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”	
Side Rail Height (H)	mm	110	125	137	150	184					
	inches	4”	5”	5.25”	6”	7.25”					
Flange Sizes	mm	21	32	38	44						
	inches	0.8”	1.25”	1.5”	1.75”						
Thickness (T)	mm	1.5	2.0	2.5							
	inches	16GA	14GA	12GA							
Length (L)	mm	2000	2440	3000							
	inches	6.5”	8”	10”							
NEMA CLASS (N)	8A	12A	16A	8B	12B	16B	8C	12C	16C		

Ordering Code Example : CLCHSSO300HDG6008/3000/2/32/125/C

NOTES:

- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs [illegible] for welded type cable lader.
- Special orders can be manufactured upon req[illegible]

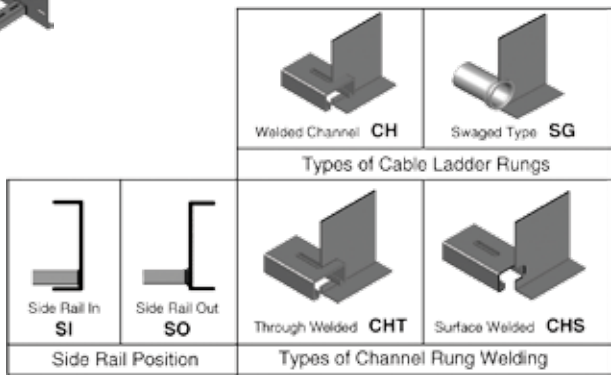
Cable Ladder Horizontal Elbow



Round Type (R)



Cornered Type (C)



Ordering Code	CLLH			/					/		/		/		/	

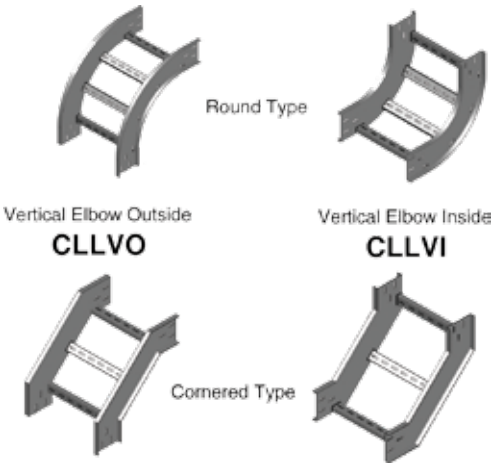
Cable Ladder Horizontal Elbow	CLLH									
Swaged Type	SG									
Welded Channel	CHS	CHT								
Radius (R)	300	600	900							
Angle (A)	30	45	60	90						
Side Rail In	SI									
Side Rail Out	SO									
Rung Spacing	mm	150	229	250	300					
	inches	6”	9”	10”	12”					
Black Steel					BST					
Hot Dip Galvanized after Fabri cation					HDG					
Hot Dip Galvanized and powder Coating					HDP					
Mill Galvanized					MG					
Steel Epoxy Coating					EPC					
Aluminum					AL					
Stainless Steel 304					S4					
Stainless Steel 316					S6					
Width (W)	mm	100	150	200	300	450	600	750	900	1000
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”
Side Rail Height (H)	mm	110	125	137	150	184				
	inches	4”	5”	5.25”	6”	7.25”				
Flange Sizes	mm	21	32	38	44					
	inches	0.8”	1.25”	1.5”	1.75”					
Thickness (T)	mm	1.5	2.0	2.5						
	inches	16GA	14GA	12GA						
Round Type	R									
Cornered Type	C									

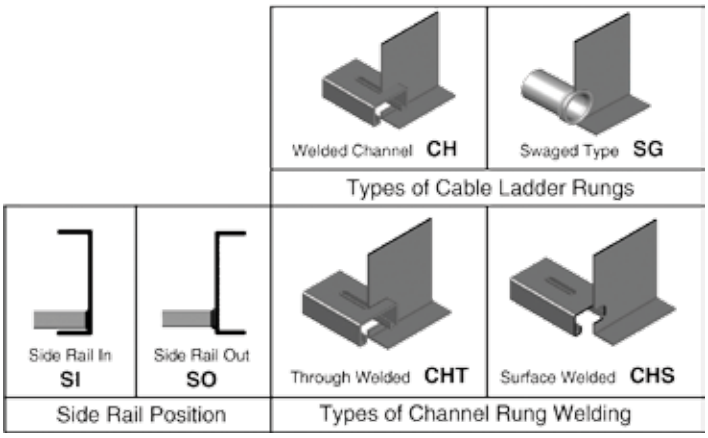
Ordering Code Example : CLLHCCHT300/90°SO250HD450/125/32/2/R

NOTES:

- Standard pipe size for swaged type cable ladd[illegible] steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs [illegible] for welded type cable lader.
- Special orders can be manufactured upon req[illegible]st.

Cable Ladder Vertical Elbow





Ordering Code	CLLV			/					/		/		/	

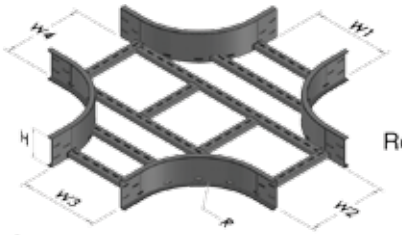
Cable Ladder Vertical Elbow	CLLV									
Swaged Type	SG									
Welded Channel	CHS	CHT								
Radius (R)	300	600	900							
Angle (A)	30	45	60	90						
Side Rail In	SI									
Side Rail Out	SO									
Rung Spacing mm	150	229	250	300						
Rung Spacing inches	6”	9”	10”	12”						
Black Steel	BST									
Hot Dip Galvanized after Fabri cation	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	100	150	200	300	450	600	750	900	1000	
Width (W) inches	4”	6”	8”	12”	18”	24”	30”	36”	40”	
Side Rail Height (H) mm	110	125	137	150	184					
Side Rail Height (H) inches	4”	5”	5.25”	6”	7.25”					
Flange Sizes mm	21	32	38	44						
Flange Sizes inches	0.8”	1.25”	1.5”	1.75”						
Thickness (T) mm	1.5	2.0	2.5							
Thickness (T) inches	16GA	14GA	12GA							
Round Type	R									

Ordering Code Example : CLLVOCHT300/90SO250HDG450/125/32/R

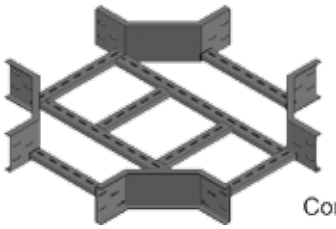
NOTES:

- Standard pipe size for swaged type cable ladder is 1¼“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

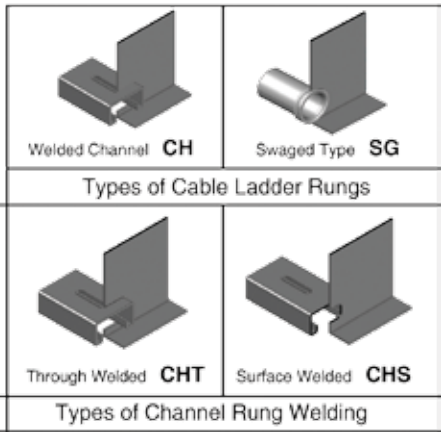
Cable Ladder Horizontal Cross



Round Type (F



Cornered Type (R)



Ordering Code	CLX					/		/		/		

Cable Ladder Horizontal Cross		CLX
Swaged Type	SG	
Welded Channel	CHS	CHT
Side Rail In	SI	
Side Rail Out	SO	

Black Steel	BST
Hot Dip Galvanized after Fabri cation	HDG
Hot Dip Galvanized and powder Coating	HDP
Mill Galvanized	MG
Steel Epoxy Coating	EPC
Aluminum	AL
Stainless Steel 304	S4
Stainless Steel 316	S6

Width (W)	mm	100	150	200	300	450	600	750	900	1000
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”
Side Rail Height (H)	mm	110	125	137	150	184				
	inches	4”	5”	5.25”	6”	7.25”				
Flange Sizes	mm	21	32	38	44					
	inches	0.8”	1.25”	1.5”	1.75”					
Thickness (T)	mm	1.5	2.0	2.5						
	inches	16GA	14GA	12GA						

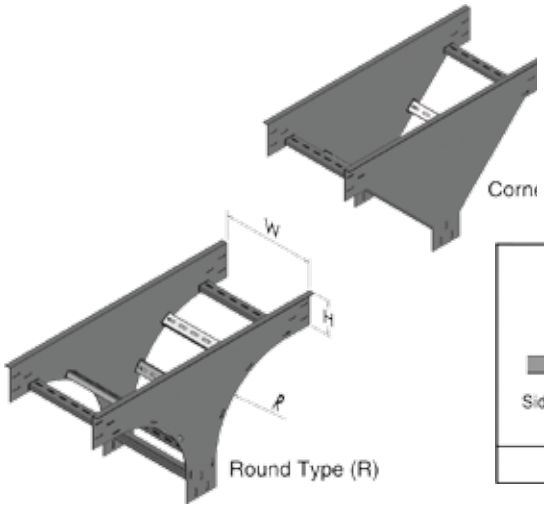
Round Type	R
Cornered Type	C

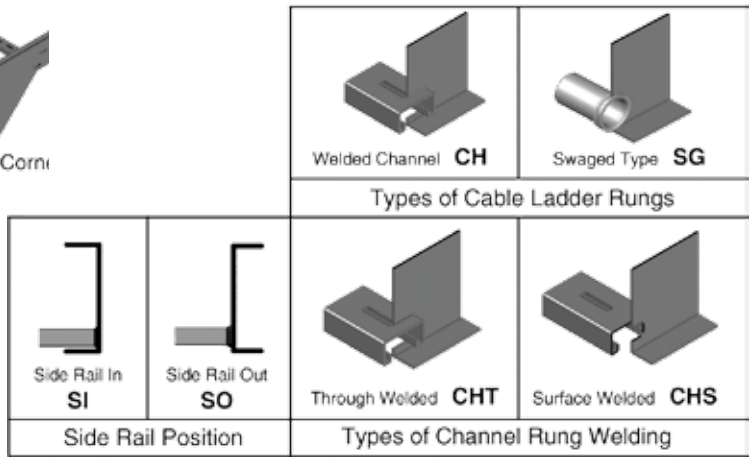
Ordering Code Example : CLTHSGSI600HD300/200/300/110/21/2/R

NOTES:

- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

Cable Ladder Vertical Tee





Ordering Code	CLTV					/		/	/	/	

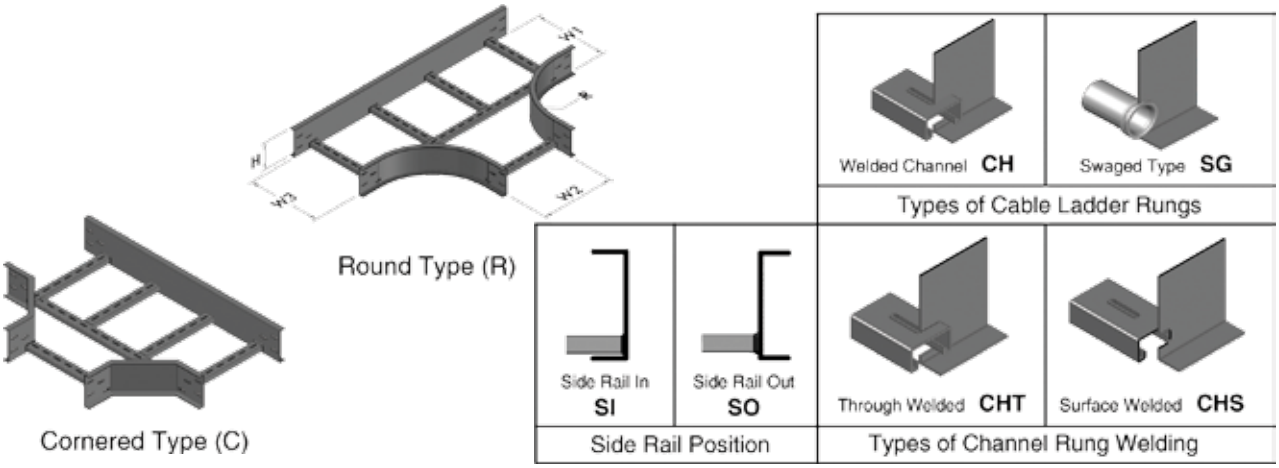
Cable Ladder Vertical Tee		CLTV									
Swaged Type	SG										
Welded Channel	CHS	CHT									
Side Rail In	SI										
Side Rail Out	SO										
Black Steel		BST									
Hot Dip Galvanized after Fabri cation		HDG									
Hot Dip Galvanized and powder Coating		HDP									
Mill Galvanized		MG									
Steel Epoxy Coating		EPC									
Aluminum		AL									
Stainless Steel 304		S4									
Stainless Steel 316		S6									
Width (W)	mm	100	150	200	300	450	600	750	900	1000	
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”	
Side Rail Height (H)	mm	110	125	137	150	184					
	inches	4”	5”	5.25”	6”	7.25”					
Flange Sizes	mm	21	32	38	44						
	inches	0.8”	1.25”	1.5”	1.75”						
Thickness (T)	mm	1.5	2.0	2.5							
	inches	16GA	14GA	12GA							
Round Type	R										
Cornered Type	C										

Ordering Code Example : CLTVSGSI600HDP300/200/300/110/21/2/R

NOTES:

- Standard pipe size for swaged type cable ladder is 1 1/2“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

Cable Ladder Horizontal Tee



Ordering Code	CLTH					/		/			/	

Cable Ladder Vertical Tee	CLTH	
Swaged Type	SG	
Welded Channel	CHS	CHT
Side Rail In	SI	
Side Rail Out	SO	
Black Steel	BST	
Hot Dip Galvanized after Fabri cation	HDG	
Hot Dip Galvanized and powder Coating	HDP	
Mill Galvanized	MG	
Steel Epoxy Coating	EPC	
Aluminum	AL	
Stainless Steel 304	S4	
Stainless Steel 316	S6	

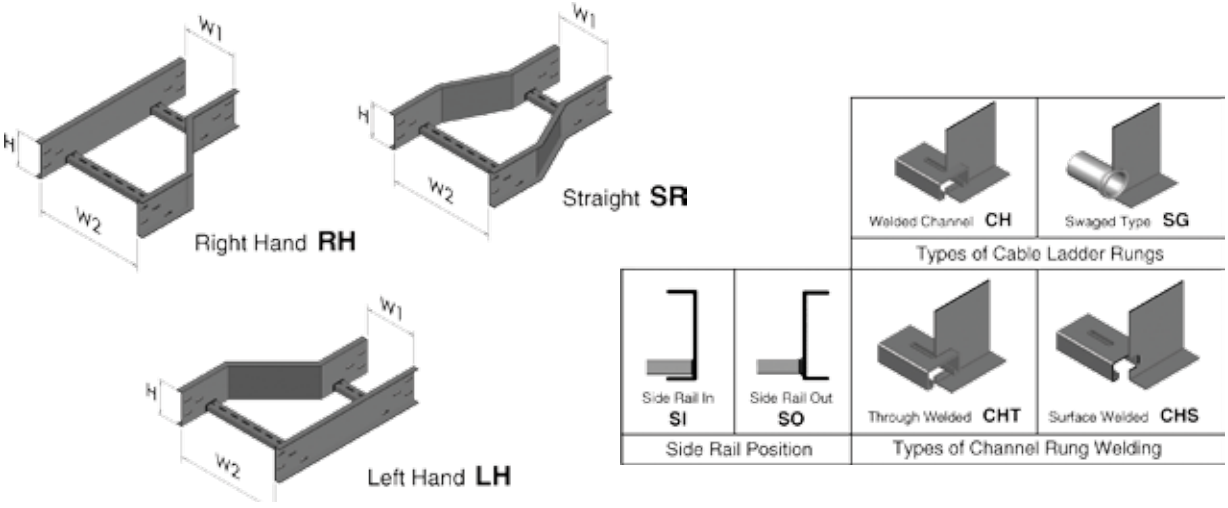
Width (W)	mm	100	150	200	300	450	600	750	900	1000
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”
Side Rail Height (H)	mm	110	125	137	150	184				
	inches	4”	5”	5.25”	6”	7.25”				
Flange Sizes	mm	21	32	38	44					
	inches	0.8”	1.25”	1.5”	1.75”					
Thickness (T)	mm	1.5	2.0	2.5						
	inches	16GA	14GA	12GA						

Round Type	R
Cornered Type	C

Ordering Code Example : CLXSGSO300HDG450/300/500/200/125/32/2/R

NOTES:
- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

Cable Ladder Reducers



Ordering Code	CLY						/		/		/		/	

Cable Ladder Reducer		CLY
Swaged Type	SG	
Welded Channel	CHS	CHT
Swaged Type	SG	
Welded Channel	CHS	
Straight	SR	
Right Hand	RH	
Left Hand	LH	

Black Steel	BST
Hot Dip Galvanized after Fabri cation	HDG
Hot Dip Galvanized and powder Coating	HDP
Mill Galvanized	MG
Steel Epoxy Coating	EPC
Aluminum	AL
Stainless Steel 304	S4
Stainless Steel 316	S6

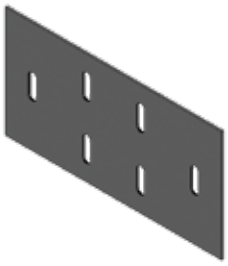
Width (W)	mm	100	150	200	300	450	600	750	900	1000
	inches	4”	6”	8”	12”	18”	24”	30”	36”	40”
Side Rail Height (H)	mm	110	125	137	150	184				
	inches	4”	5”	5.25”	6”	7.25”				
Flange Sizes	mm	21	32	38	44					
	inches	0.8”	1.25”	1.5”	1.75”					
Thickness (T)	mm	1.5	2.0	2.5						
	inches	16GA	14GA	12GA						

Ordering Code Example : CLYCHTSORHHDG450/200/125/38/2

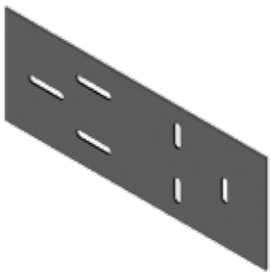
NOTES:

- Standard pipe size for swaged type cable ladder is 1 1/2“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

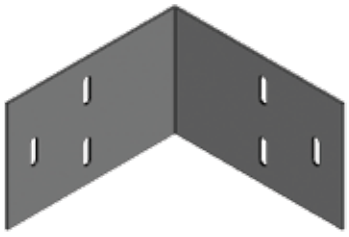
Cable Ladder Accessories



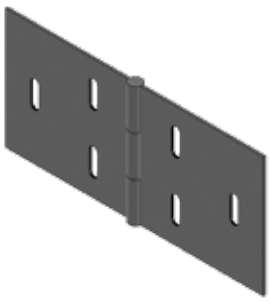
Straight Connector
(CL01)



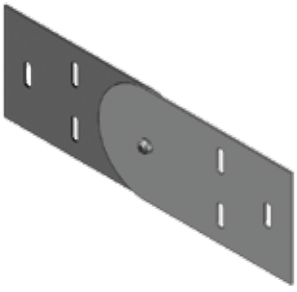
Expansion Connector
(CL02)



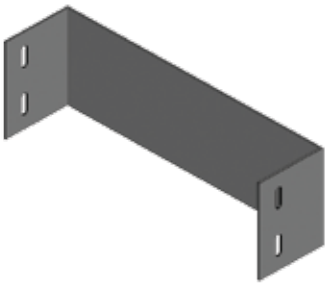
Angle Connector
(CL03)



Adjustable Horizontal Joint
(CL04)

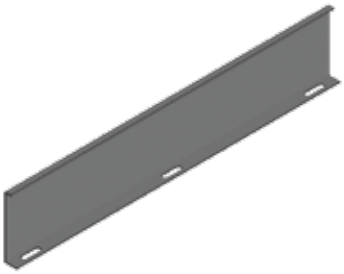


Adjustable Vertical Joint
(CL05)

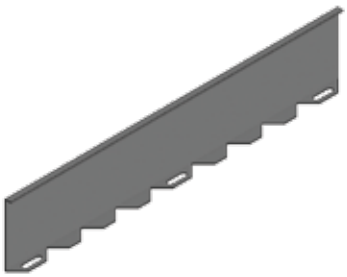


Blind End Plate
(CL06)

Cable Ladder Accessories



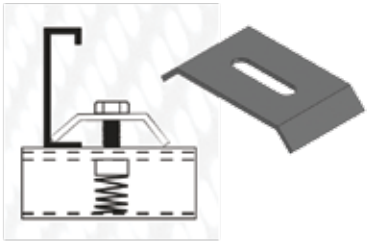
Barrier Strip Straight
(CL07)



Barrier Strip Horizontal Fitting
(CL08)



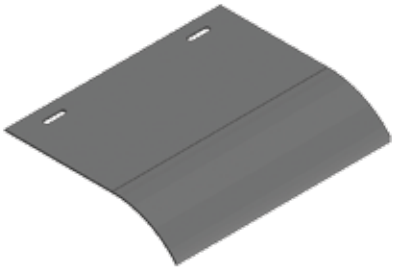
Side Clamp
(CL09)



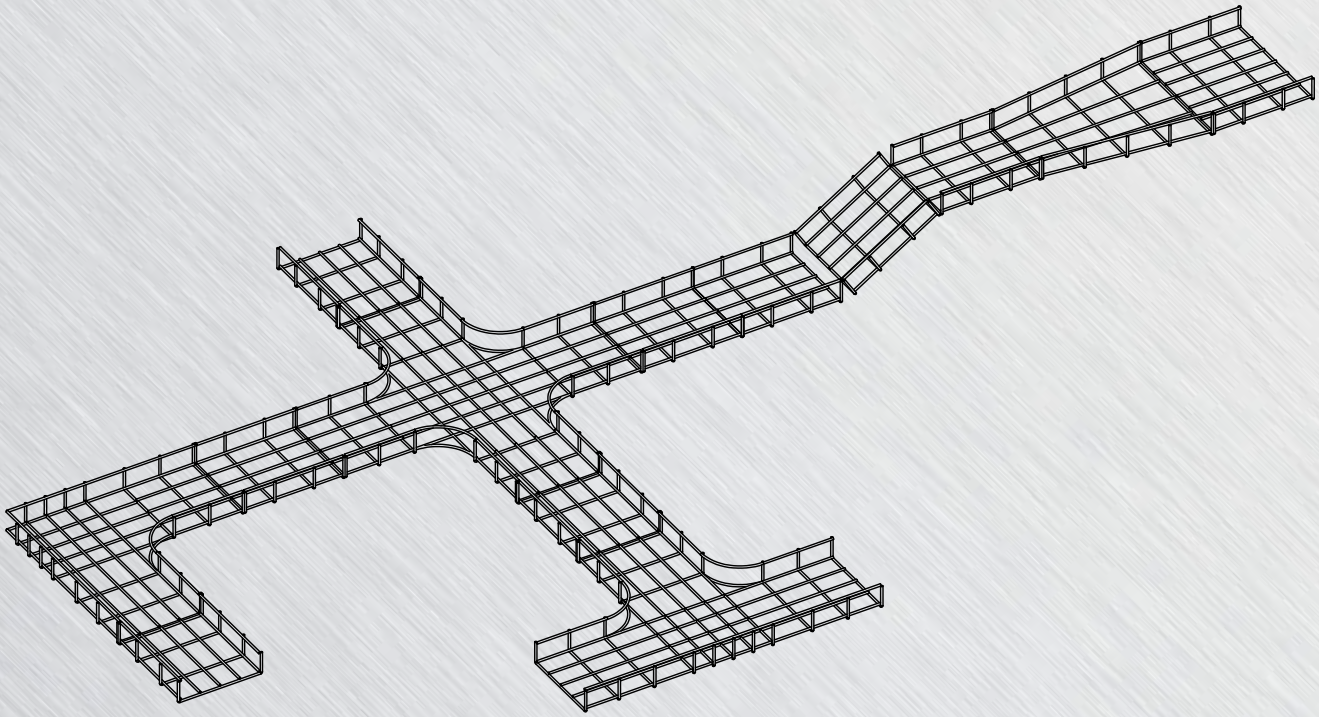
Hold Down Clamp
(CL10)



Bonding Jumper
(CL11)



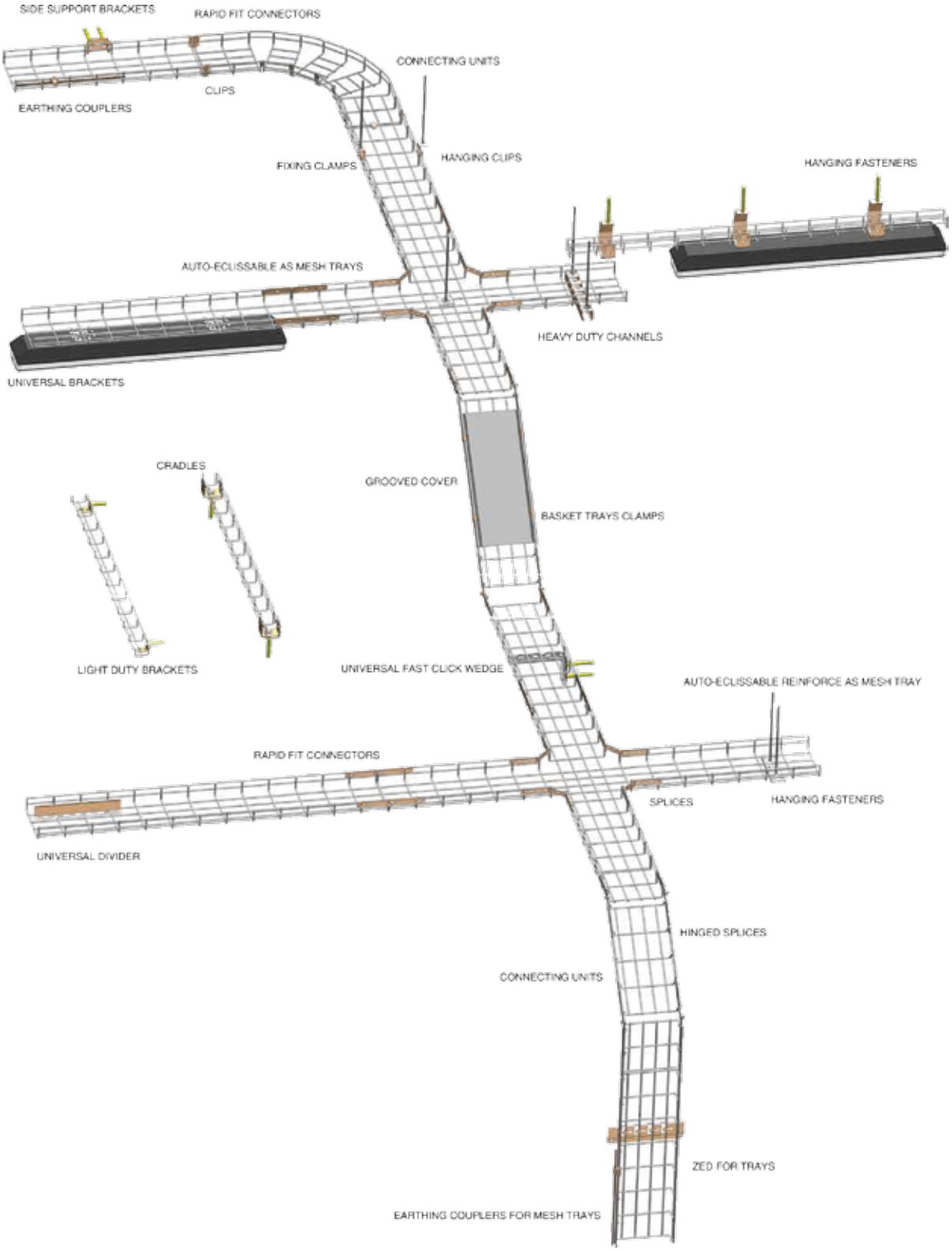
Drop Out Plate
(CL12)



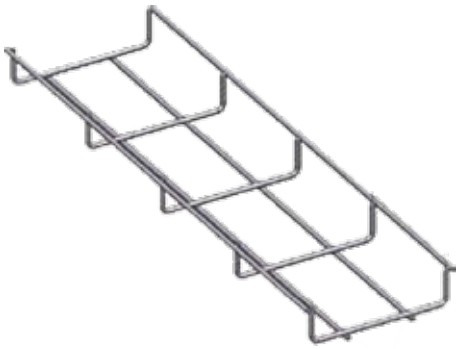
CABLE TRAY BASKET TYPE

MODERN METAL INDUSTRIES

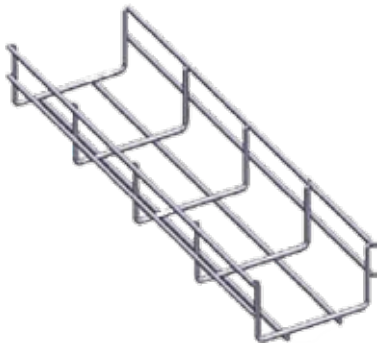
Layout


SIDE SUPPORT BRACKETS
RAPID FIT CONNECTORS
CONNECTING UNITS
CLIPS
EARTHING COUPLERS
FIXING CLAMPS
HANGING CLIPS
HANGING FASTENERS
AUTO-ECLISSABLE AS MESH TRAYS
HEAVY DUTY CHANNELS
UNIVERSAL BRACKETS
CRADLES
GROOVED COVER
BASKET TRAYS CLAMPS
LIGHT DUTY BRACKETS
UNIVERSAL FAST CLICK WEDGE
AUTO-ECLISSABLE REINFORCE AS MESH TRAY
RAPID FIT CONNECTORS
SPLICES
HANGING FASTENERS
UNIVERSAL DIVIDER
HINGED SPLICES
CONNECTING UNITS
ZED FOR TRAYS
EARTHING COUPLERS FOR MESH TRAYS

Basket Trays



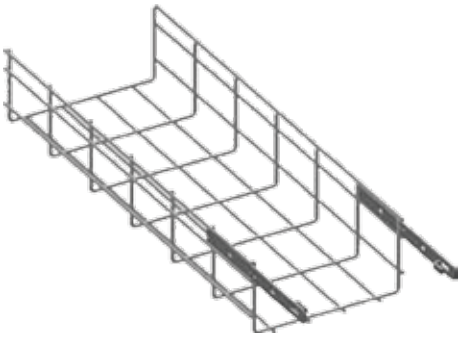
	Pre-Galvanized	HDG		SS304
MMB1 Height 30	Width mm	Code	Code	Code
MMB1-50	50	GI05	HG05	SS05
MMB1-100	100	GI10	HG10	SS10
MMB1-150	150	GI15	HG15	SS15
MMB1-200	200	GI20	HG20	SS20
MMB1-300	300	GI30	HG30	SS30



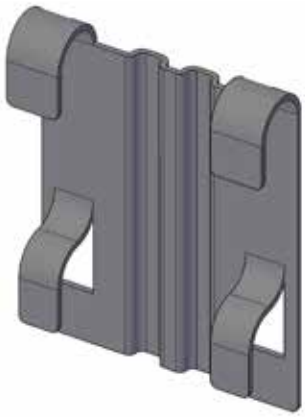
	Pre-Galvanized	HDG		SS304
MMB2 Height 50	Width mm	Code	Code	Code
MMB2-50	50	GI05	HG05	SS05
MMB2-100	100	GI10	HG10	SS10
MMB2-150	150	GI15	HG15	SS15
MMB2-200	200	GI20	HG20	SS20
MMB2-250	250	GI25	HG25	SS25
MMB2-300	300	GI30	HG30	SS30
MMB2-400	400	GI40	HG450	SS40
MMB2-500	500	GI50	HG50	SS50
MMB2-600	600	GI60	HG60	SS60

* **Note:** Stainless Steel 316 are available upon re

Basket Trays/ Accessories

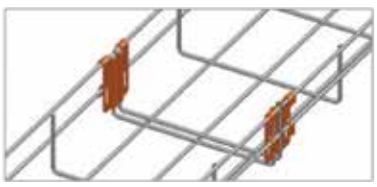


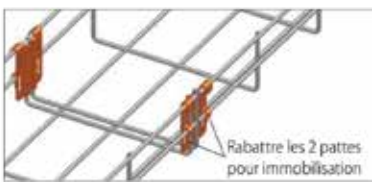
	Pre-Galvanized	HDG		SS304
MMB3 Height 110	Width mm	Code	Code	Code
MM3-100	100	GI10	HG10	SS10
MM3-100	200	GI20	HG20	SS20



	Pre-Galvanized	HDG	SS304
	Code	Code	Code
M-ER	GI850	HG250	SS650





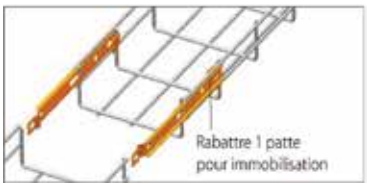


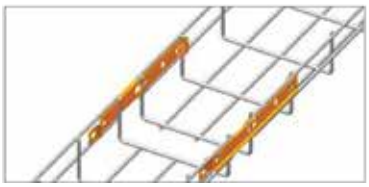
*** Note:** Stainless Steel 316 are available upon re[illegible]

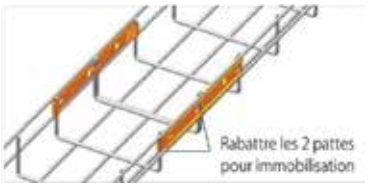
Basket Trays Accessories



	Pre-Galvanized	HDG	SS304
	Code	Code	Code
M-ECR	GI851	HG251	SS651

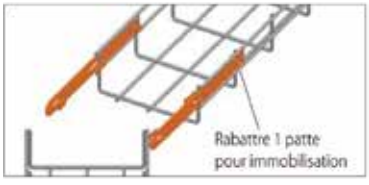


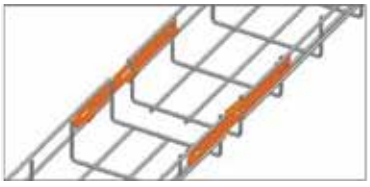


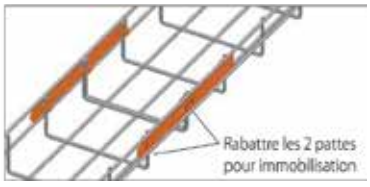




	Pre-Galvanized	HDG	SS304
	Code	Code	Code
M-ECR	GI852	HG252	SS652



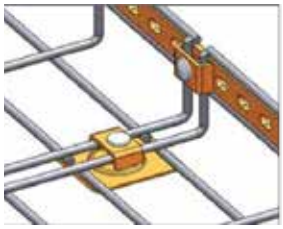




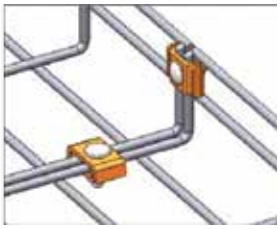
Basket Trays Accessories



	Pre-Galvanized	HDG	SS304
	Code	Code	Code
MM-CL20	GI850002	HG250002	SS650002
MM-CL30	GI850003	HG250003	SS650003
MM-CL40	GI850004	HG250004	SS650004
MM-B6X20	GI850004	HG250004	SS650004









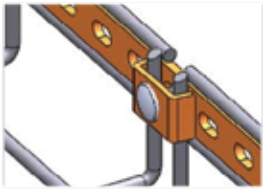
	Pre-Galvanized	HDG	SS304
	Code	Code	Code
MM-KIT-CL20	GI850010	HG250010	SS650010

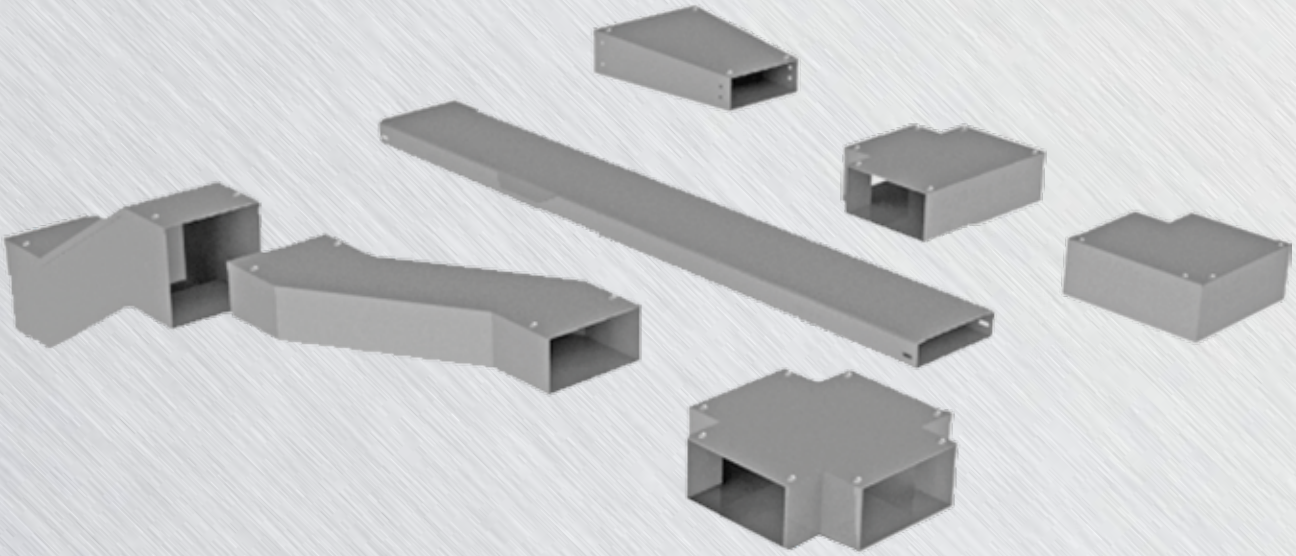


	Pre-Galvanized	HDG	SS304
	Code	Code	Code
MM-CL275	GI850005	HG250005	SS650005
MM-CL1100	GI850006	HG250006	SS650006





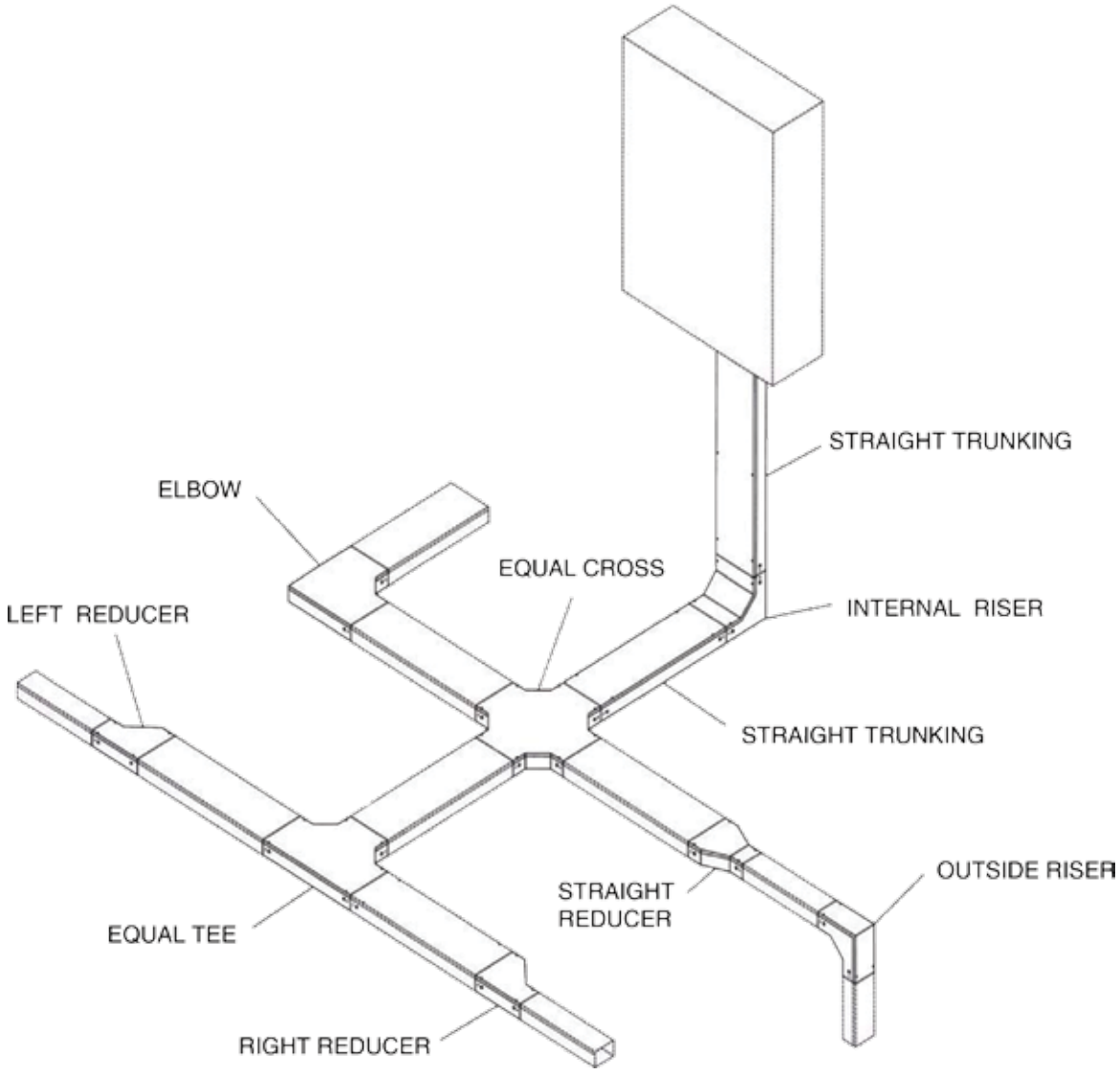




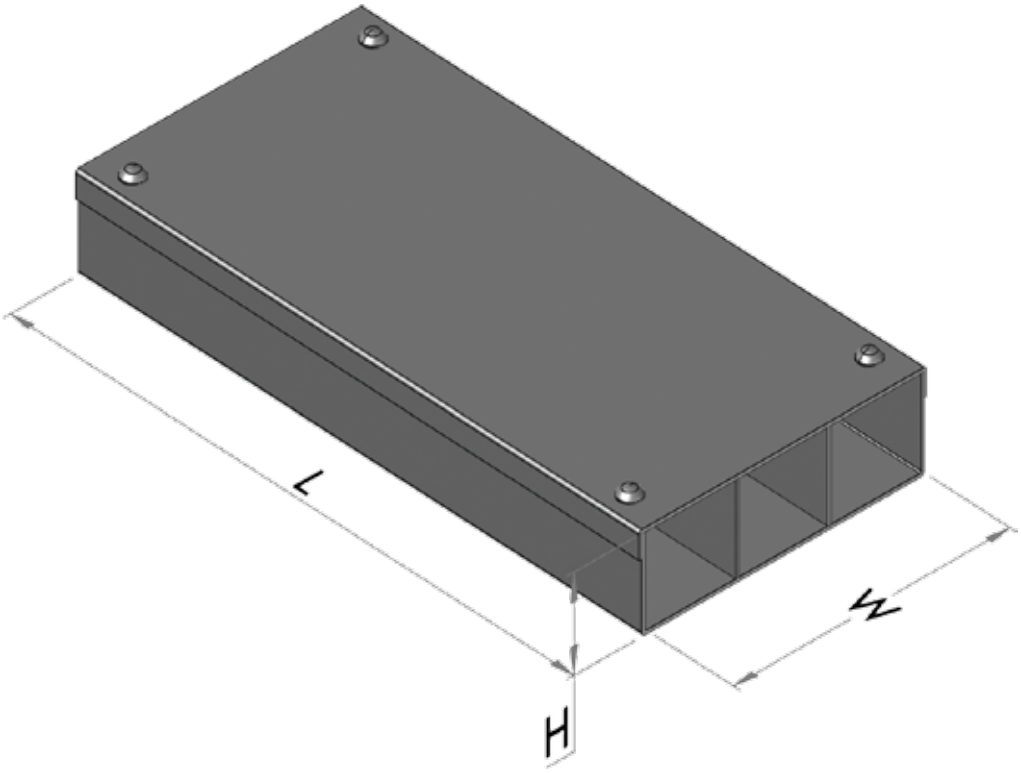
CABLE DUCT (TRUNKING)

MODERN METAL INDUSTRIES

Cable Duct Typical Layout



General Technical Information



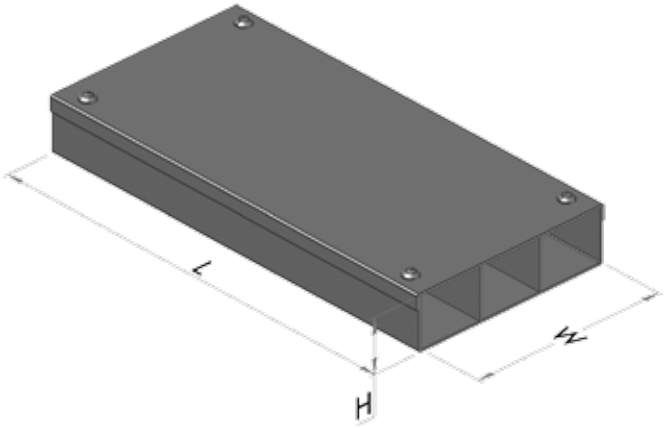
MMICO Cable Duct (Trunking) offers the widest range of measurements with or without dividing straps. A wide range of accessories and fittings complies with BS 4678.

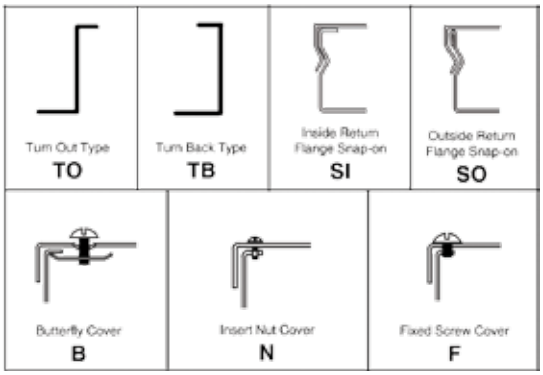
MMICo Standard accessories for Cable Duct (trunking) cover a wide consisting of elbows, tees, angles and others using manufactured steel G.I or HDG. MMICo accessories are supplied with removable covers. Complies with BS 4678 .

COMPLIANCE STATEMENT

Modern Metal Industries Company never makes any design changes or new products. All products are being manufactured and in compliance with the SAUDI ARAMCO Engineering Standard SAES-P-104 and SAMSS and as well as other related International Standards.

Cable Duct (Trunking)





Ordering Code	CD						/		/		/	

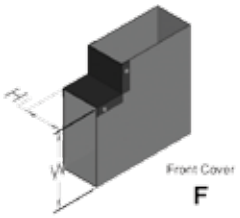
Cable Duct	CD									
Compartment	1	2	3							
Turn Out Type	TO									
Turn Back Type	TB									
Inside Return Flange Snap - on	SI									
Outside Return Flange Snap - on	SO									
Butierfly Cover	B									
Insert Nut Cover	N									
Fixed Screw Cover	F									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	50	100	150	200	300	450	600	900	1000	
Helght (H) mm	50	100	150	200						
Thickness (T) mm	1.0	1.25	1.5	2.0						
Length (L) mm	1000	2440	3000							

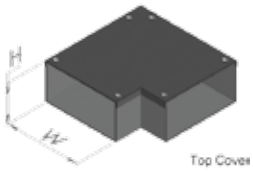
Ordering Code Example : CD2TBBHDG500/100/1.5/2440

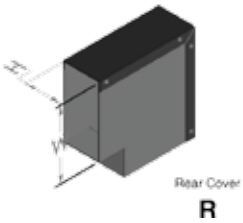
NOTES:

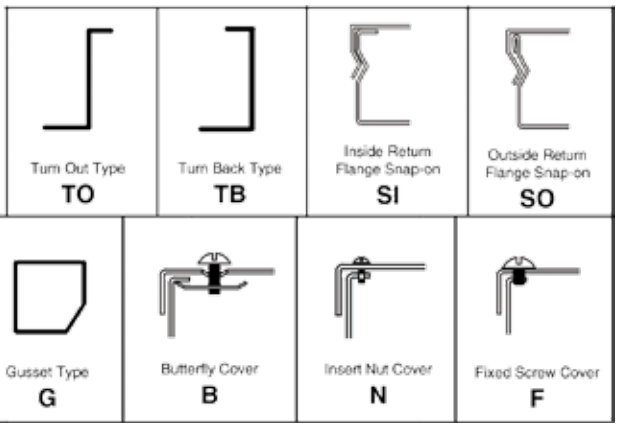
- Waterproof type specifications upon request, turn [illegible] with seal will be used for water proof type duct.
- Special orders can be manufactured upon req[illegible]

Cable Duct Elbows









Ordering Code	CDL										/		/	

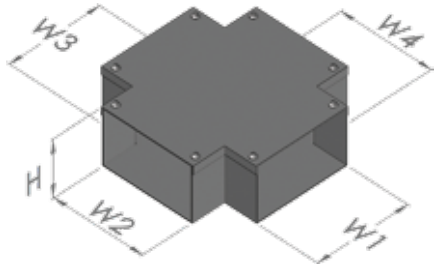
Cable Duct Elbow	CDL									
Compartment	1	2	3							
Top Cover	P									
Front Cover	F									
Rear Cover	R									
Angle	45	60	90							
Corner Type	C									
Round Type	R									
Gusset Type	G									
Radius (R)	120	200								
Turn Out Type	TO									
Turn Back Type	TB									
Inside Return Flange Snap - on	SI									
Outside Return Flange Snap - on	SO									
Butierfly Cover	B									
Insert Nut Cover	N									
Fixed Screw Cover	F									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	100	150	200	300	450	600	750	900	1000	
Helght (H) mm	50	100	150	200						
Thickness (T) mm	1.0	1.25	1.5	2.0						

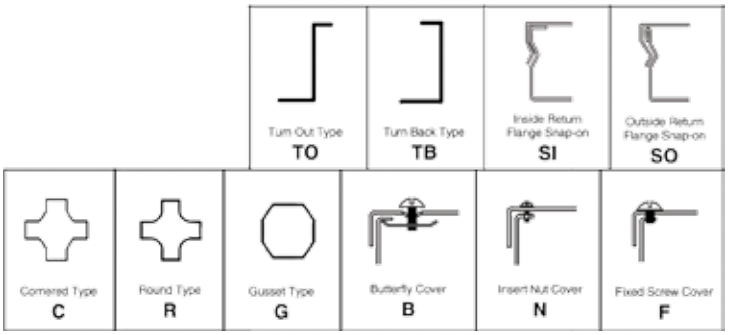
Ordering Code Example : CDL1P90R200TOBHDG200/150/2.0

NOTES:

- Waterproof type specifications upon request, tu[illegible] seal will be used for water proof type duct.
- Special orders can be manufactured upon req[illegible]

Cable Duct Cross





Ordering Code	CDX								/		/		/		/	

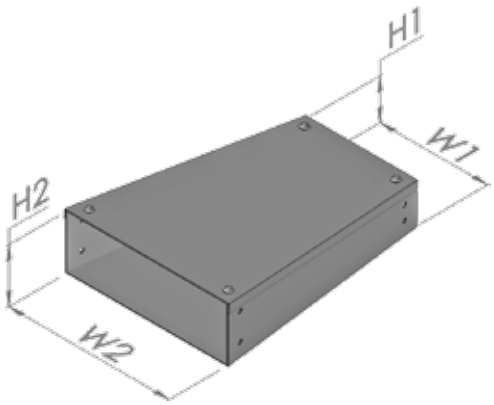
Cable Duct Elbow	CDX								
Compartment	1	2	3						
Corner Type	C								
Round Type	R								
Gusset Type	G								
Radius (R)	150	200							
Turn Out Type	TO								
Turn Back Type	TB								
Inside Return Flange Snap - on	SI								
Outside Return Flange Snap - on	SO								
Butierfly Cover	B								
Insert Nut Cover	N								
Fixed Screw Cover	F								
Black Steel	BST								
Hot Dip Galvanized after Fabrication	HDG								
Hot Dip Galvanized and powder Coating	HDP								
Mill Galvanized	MG								
Steel Epoxy Coating	EPC								
Aluminum	AL								
Stainless Steel 304	S4								
Stainless Steel 316	S6								
Width (W1,W2,W3,W4) mm	100	150	200	300	450	600	750	900	1000
Helght (H) mm	50	100	150	200					
Thickness (T) mm	1.0	1.25	1.5	2.0					

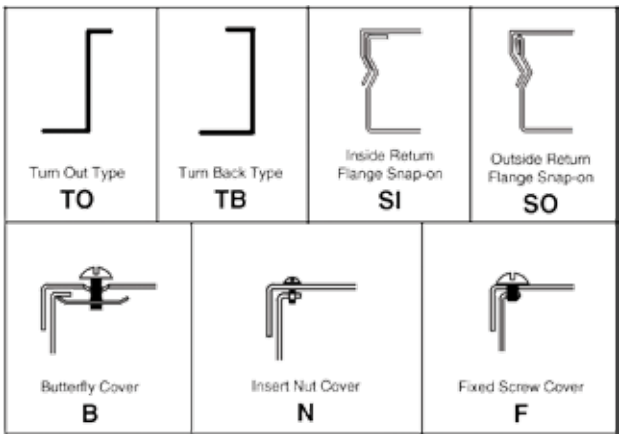
Ordering Code Example : CDX1C150TBNHDG200/300/200/50/1.25

NOTES:

- Waterproof type specifications upon request, turn c[illegible] with seal will be used for water proof type duct.
- Special orders can be manufactured upon req[illegible]

Cable Duct Reducers





Ordering Code	CDY						/		/		/		/	

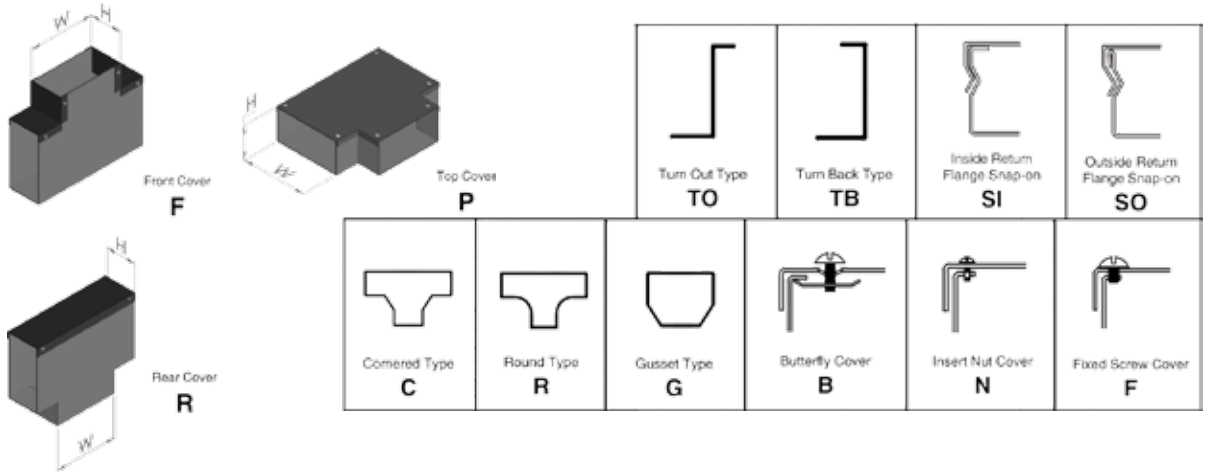
Cable Duct Elbow	CDY									
Compartment	1	2	3							
Turn Out Type	TO									
Turn Back Type	TB									
Inside Return Flange Snap - on	SI									
Outside Return Flange Snap - on	SO									
Butierfly Cover	B									
Insert Nut Cover	N									
Fixed Screw Cover	F									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W1,W2,W3,W4) mm	100	150	200	300	450	600	750	900	1000	
Helght (H) mm	50	100	150	200						
Thickness (T) mm	1.0	1.25	1.5	2.0						

Ordering Code Example : CDY2TONHDG150/300/50/100/1.5

NOTES:

- Waterproof type specifications upon request, turn out type with seal will be used for water proof type duct.
- Special orders can be manufactured upon request.

Cable Duct Tee



Ordering Code	CDT									/		/		/		/	

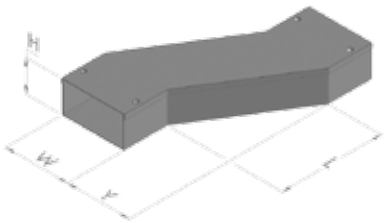
Cable Duct Elbow	CDT									
Compartment	1	2	3							
Top Cover	P									
Front Cover	F									
Rear Cover	R									
Radius (R)	150	200								
Corner Type	C									
Round Type	R									
Gusset Type	G									
Turn Out Type	TO									
Turn Back Type	TB									
Inside Return Flange Snap - on	SI									
Outside Return Flange Snap - on	SO									
Butierfly Cover	B									
Insert Nut Cover	N									
Fixed Screw Cover	F									
Black Steel	BST									
Hot Dip Galvanized after Fabrication	HDG									
Hot Dip Galvanized and powder Coating	HDP									
Mill Galvanized	MG									
Steel Epoxy Coating	EPC									
Aluminum	AL									
Stainless Steel 304	S4									
Stainless Steel 316	S6									
Width (W) mm	100	150	200	300	450	600	750	900	1000	
Helght (H) mm	50	100	150	200						
Thickness (T) mm	1.0	1.25	1.5	2.0						

Ordering Code Example : CDT1P200CTOBHDG150/300/200/50/1.25

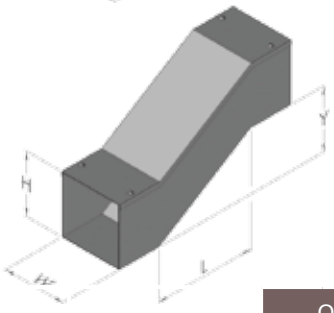
OTES:

- Waterproof type specifications upon request, tu[illegible] seal will be used for water proof type duct.
- Special orders can be manufactured upon req[illegible]st.

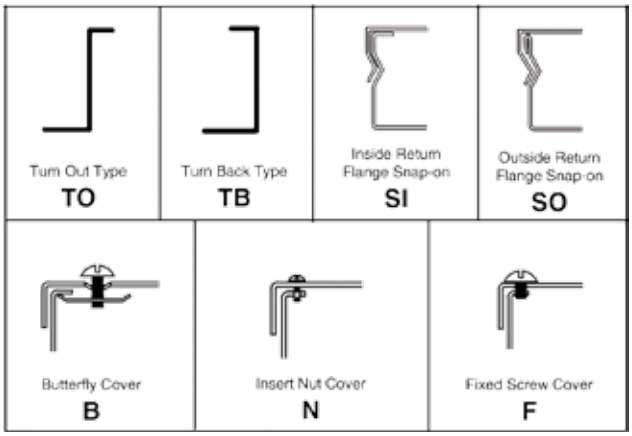
Cable Duct Offset



Horizontal Offset
H



Vertical Offset
V



Ordering Code	CDO							/		/			/	

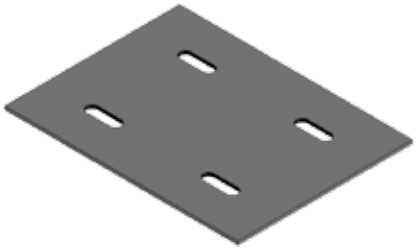
Cable Duct Elbow	CDO								
Vertical Offset	V								
Horizantal Offset	H								
Compartment	1	2	3						
Turn Out Type	TO								
Turn Back Type	TB								
Inside Return Flange Snap - on	SI								
Outside Return Flange Snap - on	SO								
Butierfly Cover	B								
Insert Nut Cover	N								
Fixed Screw Cover	F								
Black Steel	BST								
Hot Dip Galvanized after Fabrication	HDG								
Hot Dip Galvanized and powder Coating	HDP								
Mill Galvanized	MG								
Steel Epoxy Coating	EPC								
Aluminum	AL								
Stainless Steel 304	S4								
Stainless Steel 316	S6								
Width (W) mm	100	150	200	300	450	600	750	900	1000
Offset Distance (Y) mm	(Specity)								
Offset Length (L) mm	(Specity)								
Helght (H) mm	50	100	150	200					
Thickness (T) mm	1.0	1.25	1.5	2.0					

Ordering Code Example : CDOH1TOBHDG300/200/300/100/2.0

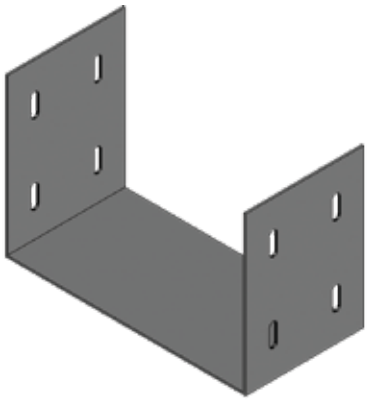
OTES:

- Waterproof type specifications upon request [illegible] with seal will be used for water proof type duct.
- Special orders can be manufactured upon [illegible]

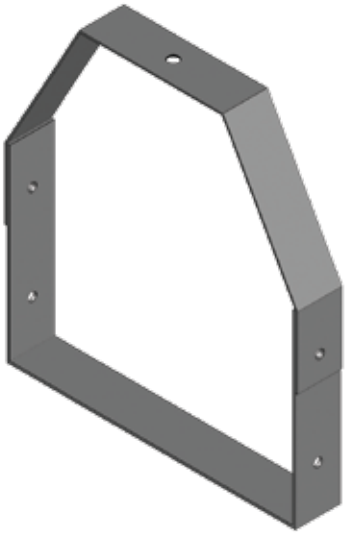
Cable Trunking Accessories



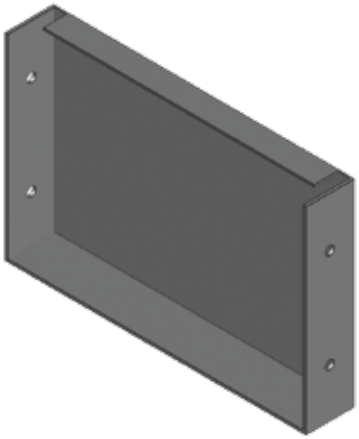
Straight Connector
(CD01)



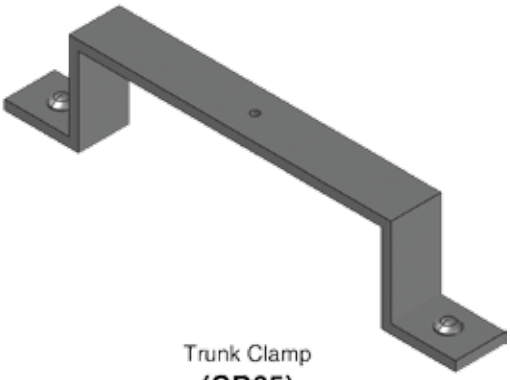
U- Connector
(CD02)



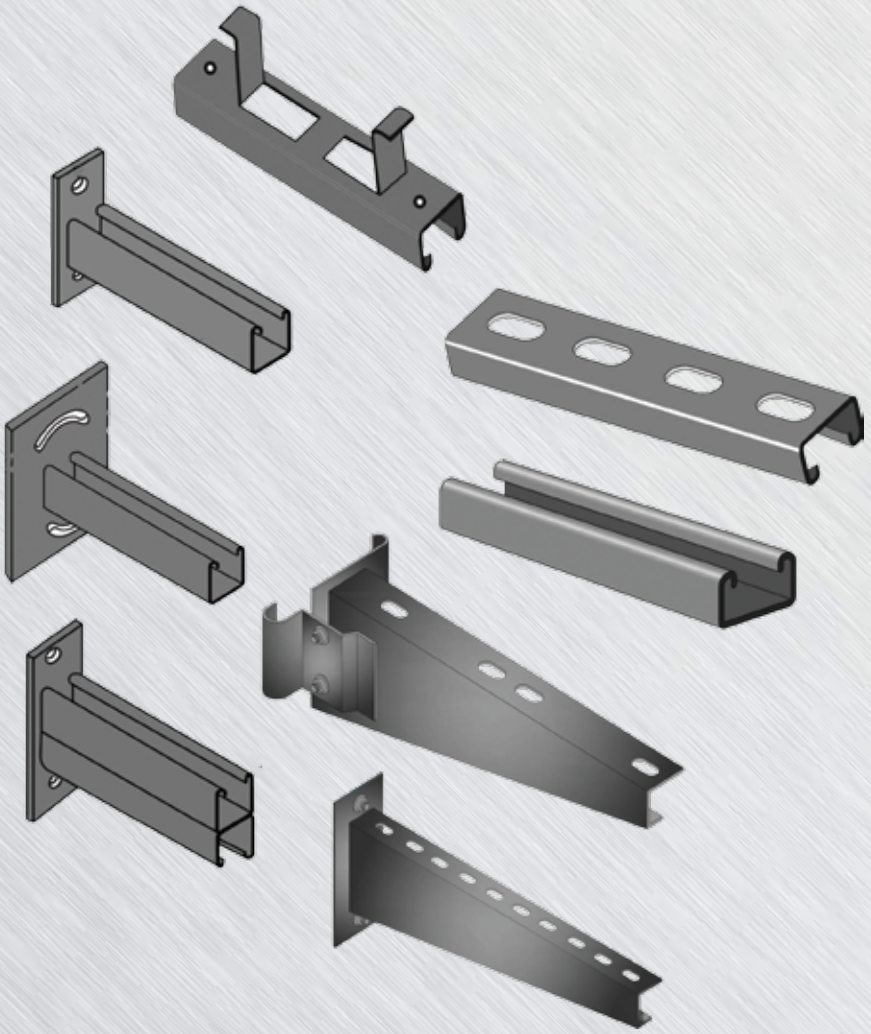
Trunk Hanger Clamp
(CD03)



End Plate
(CD04)



Trunk Clamp
(CD05)

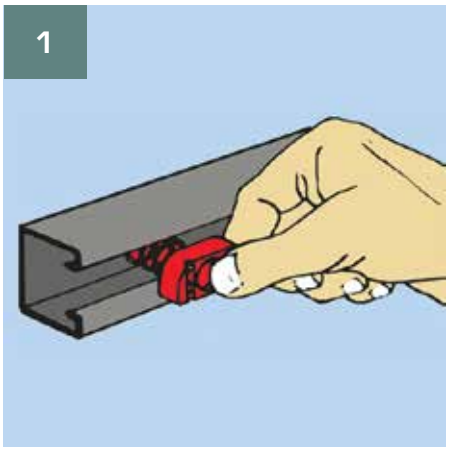


CHANNELS & SUPPORTS SYSTEM

MODERN METAL INDUSTRIES

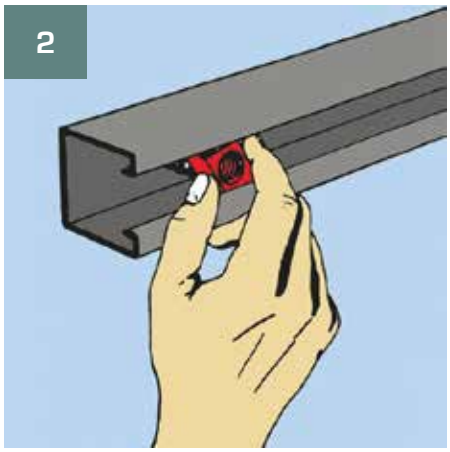
General Technical Information

1



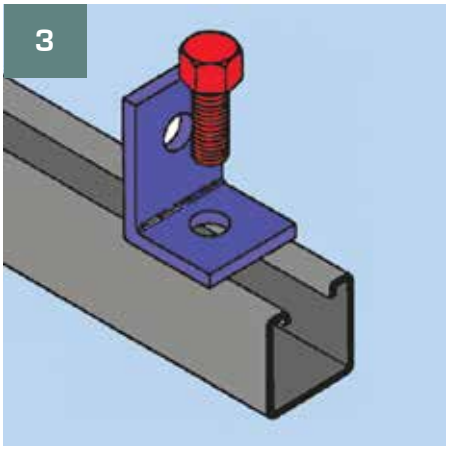
Strong, Fast, Economical and Adjustable Insert the spring unt anywhere along the continuous channel. The rounded nut ends permit easy insertion.

2



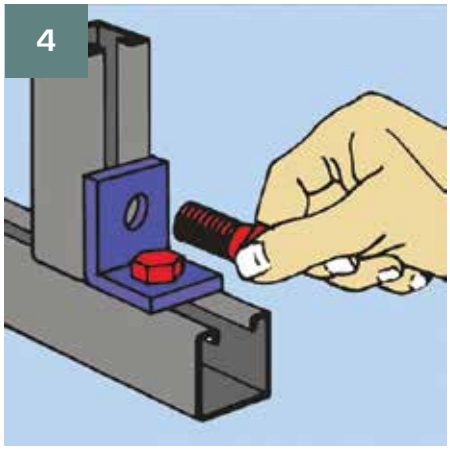
A 90° clockwise turn aligns the grooves in the nut with the inturned edges of the channel.Fittings can be placed anywhere along the channel opening, permitting comlete freedom of adjustment the need for drilling holes is eliminated.

3



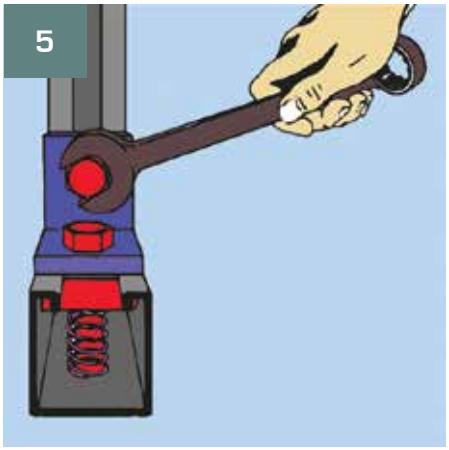
Insert the bolt through the fitting and into the spring nut, (see illustration 5 for end view showing the nut in place)

4



Additional channel sections can now be bolted to the fitting already in place by following procedure described in steps 1-3.

5

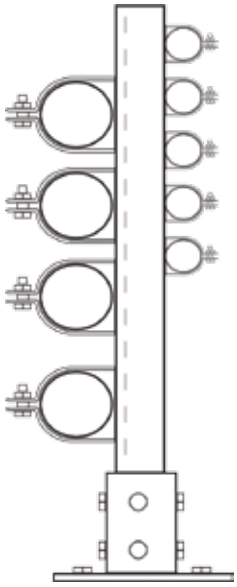


Tightening with a wrench locks the serrated teeth of the nut into the inturned edges of the channel, to complete a strong, vise-like connection.

100%	Adjustable
100%	Reusable
No	Welding
No	Drilling
No	Special Tools

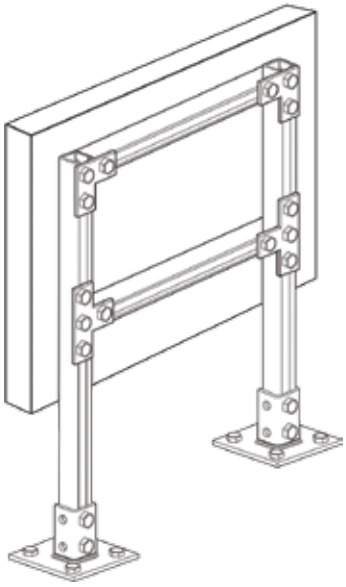
General Technical Information

Utility Stands



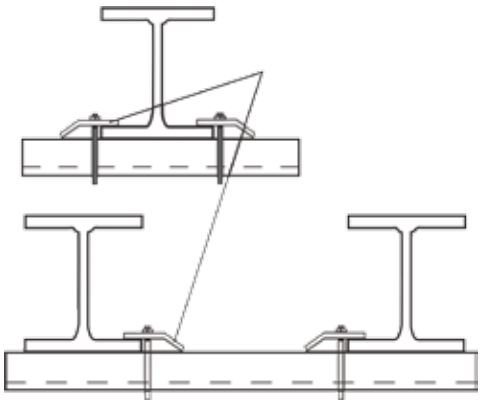
Strut can be configured into functional equipment service stands.

Panel Supports



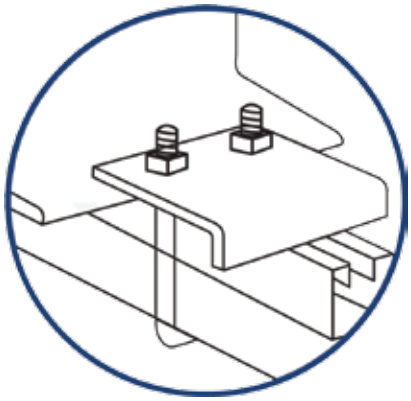
Panel supports can be configured into a variety of support appliceations and attached to existing structurals for stationary or overhead mounting of panel boards, electronics machinery, HVSC, etc.

Suspended Support Configuration



Clamps fit many beam types including; I-Beam, and T-Bar constructions.

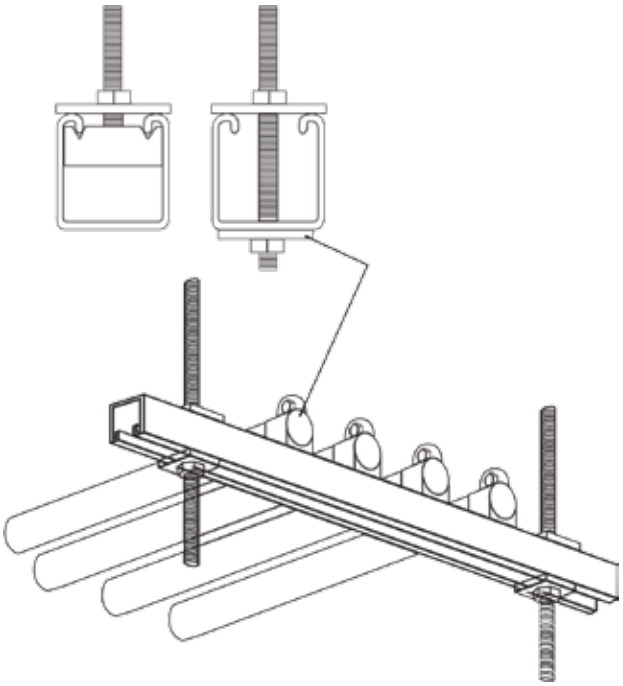
Beam Clamp Support



Beam Clamp Supports can be attached to various sized widths of “I-Beams“ flange thicknesses.

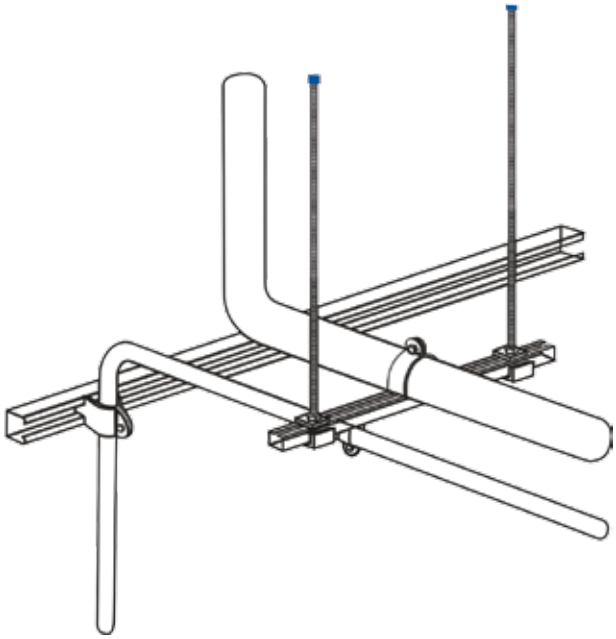
General Technical Information

Hanging



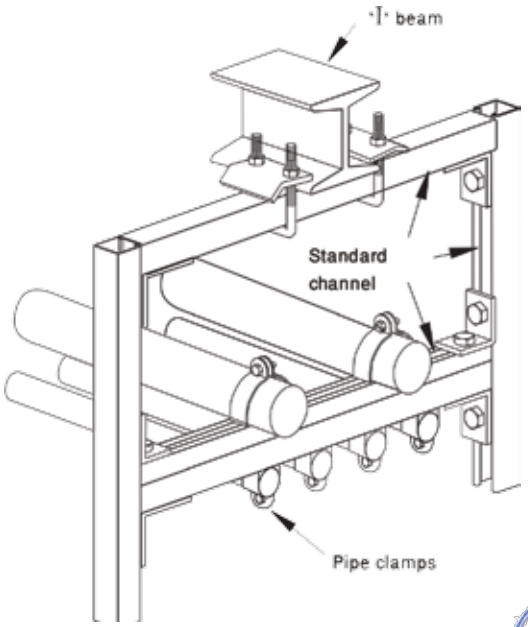
Overhead hanging schemes provide field flexibility

Complex Made Easy

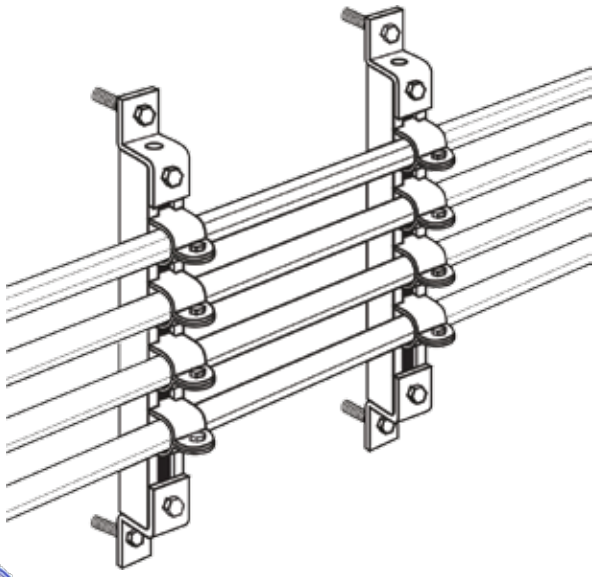


Strut systems make 90° turns a simple adjustment from plane to plane.

Pipe Support



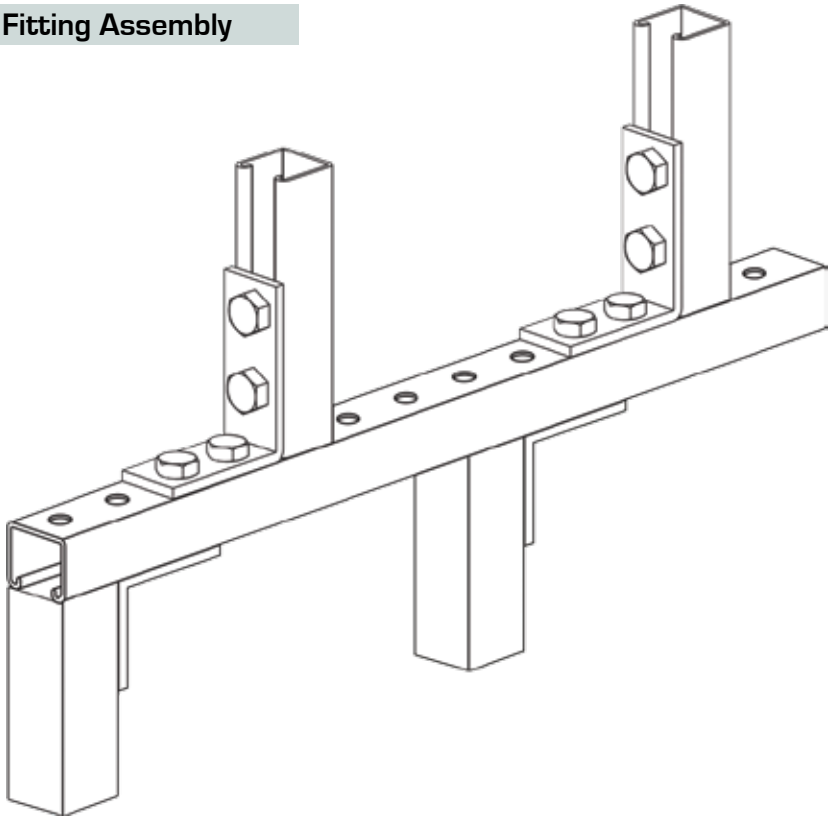
Wall Mount Configuration



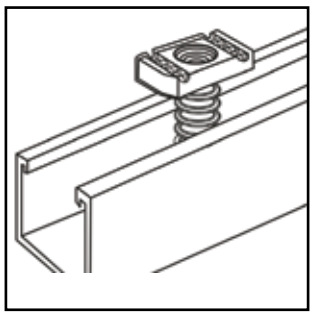
General Technical Information

Standard Channel & Fitting Assembly

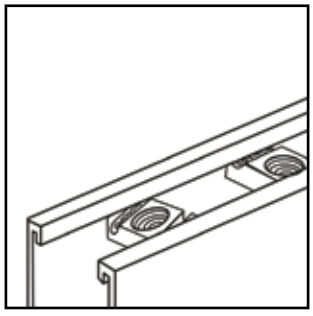
The factory punched holes in channel allow fittings to be attached to either side.



Suspended Support Configuration

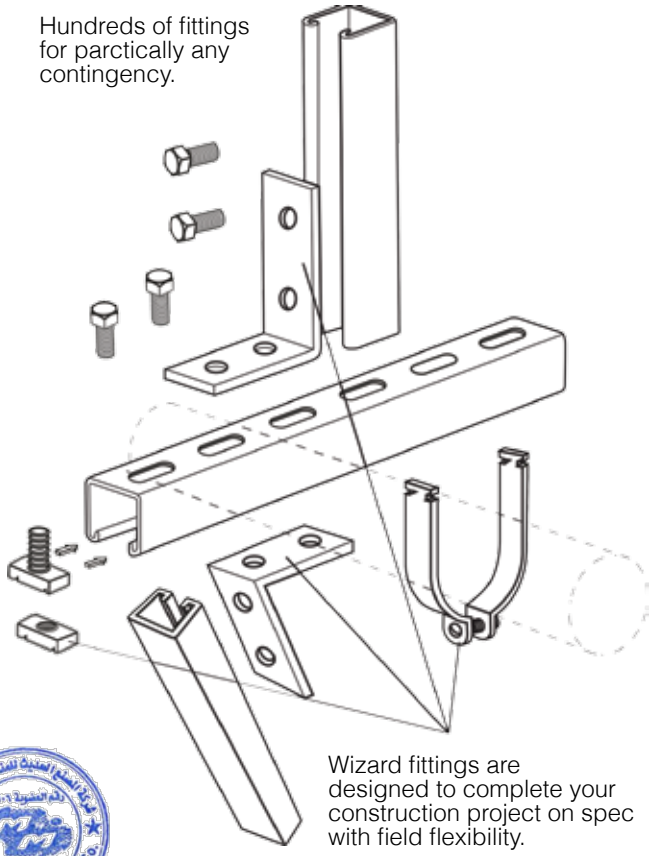


Radiused corner spring nuts insert and turn easily.



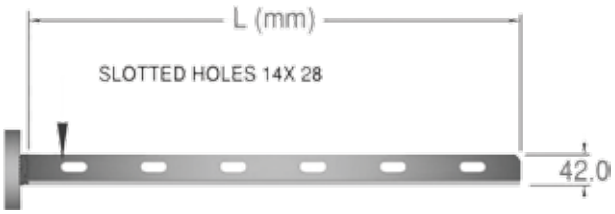
Case hardened nut teeth grip the channel's inner edge.

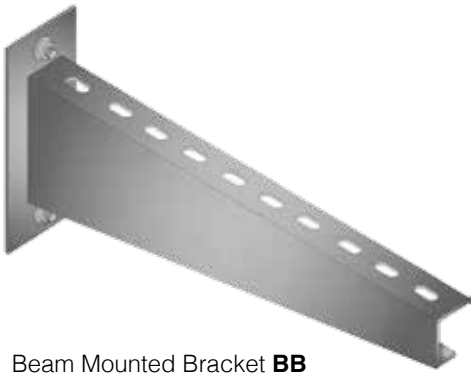
Hundreds of fittings for parctically any contingency.



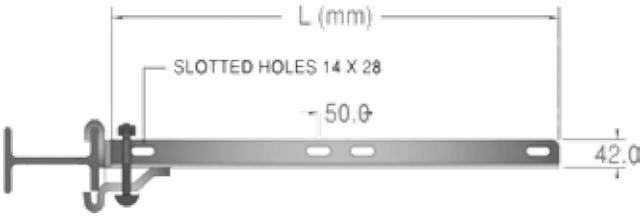
Wizard fittings are designed to complete your construction project on spec with field flexibility.

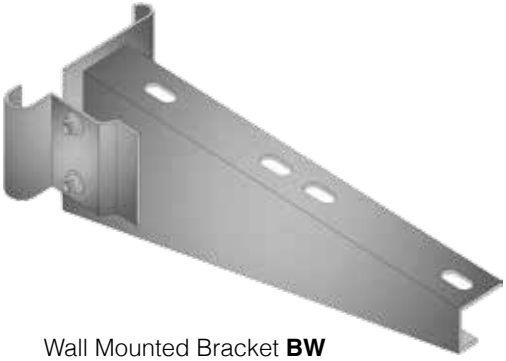
Support System - Brackets





Beam Mounted Bracket **BB**





Wall Mounted Bracket **BW**

Ordering Code				/	

Bracket	BW	BB							
Black Steel	BST								
Hot Dip Galvanized after Fabrication	HDG								
Hot Dip Galvanized and powder Coating	HDP								
Mill Galvanized	MG								
Steel Epoxy Coating	EPC								
Aluminum	AL								
Stainless Steel 304	S4								
Stainless Steel 316	S6								
Thickness (T) mm	1.5 - 6.0								
Width (W) mm	60	110	160	210	310	460	610	910	1010

Ordering Code Example : BBHDG2.5/600

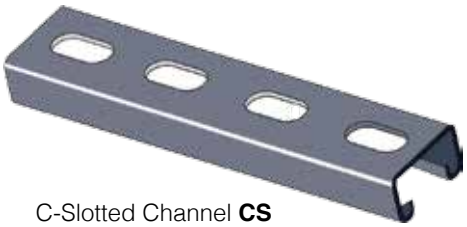
NOTES:

- Support Bracket length = Cable Tray Width + 10 m
- Special orders can be manufactured upon requ

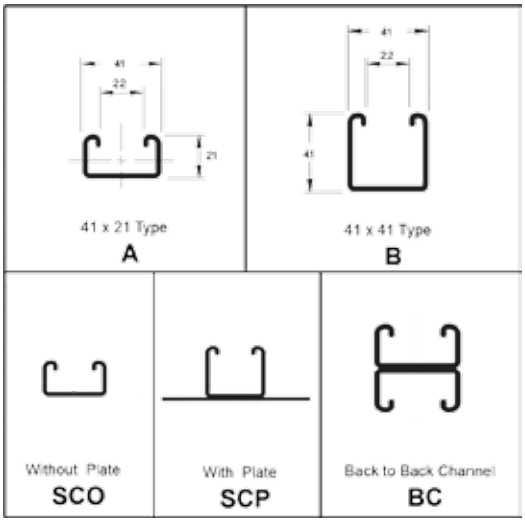
Support System - C-Channels



C-Plain Channel **CP**



C-Slotted Channel **CS**



Ordering Code							/	

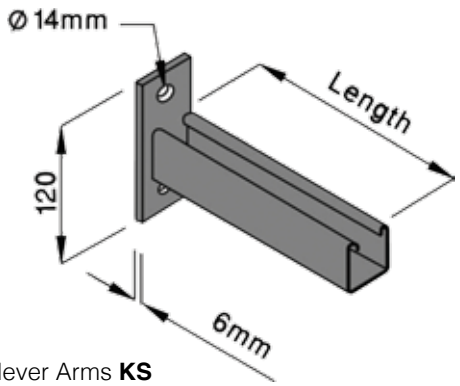
C-Channel	CP	CS
(41x21) Type	A	
(41x21) Type	B	
Single Channel	SCO	SCP
Back to Back Channel	BC	
Black Steel	BST	
Hot Dip Galvanized after Fabrication	HDG	
Hot Dip Galvanized and powder Coating	HDP	
Mill Galvanized	MG	
Steel Epoxy Coating	EPC	
Aluminum	AL	
Stainless Steel 304	S4	
Stainless Steel 316	S6	
Thickness (T) mm	1.5	2.0	2.5
Length (L) mm	200 - 6000	

Ordering Code Example : CPASCOHDG2.0/1500

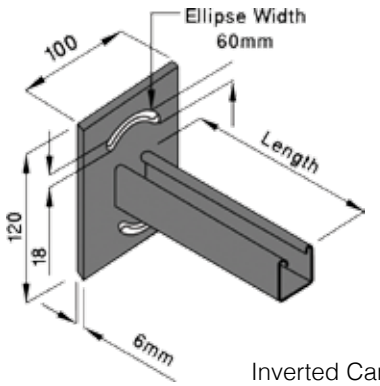
NOTES:

• Special orders can be manufactured upon request

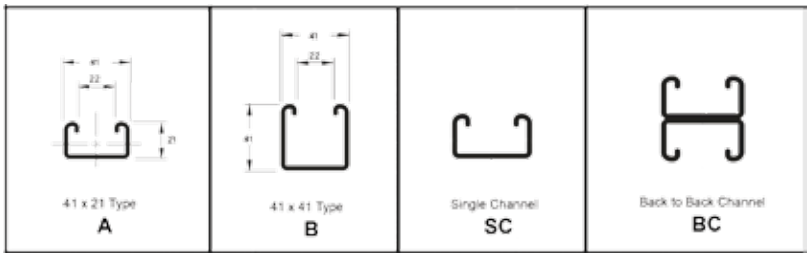
Support System - Cantilever Arms



Single Cantilever Arms **KS**



Inverted Cantilever **KI**



Ordering Code						/	

Cantilever Arms	KS	KI
(41x21) Type	A	
(41x21) Type	B	
Single Channel	SC	
Back to Back Channel	BC	
Black Steel	BST	
Hot Dip Galvanized after Fabrication	HDG	
Hot Dip Galvanized and powder Coating	HDP	
Mill Galvanized	MG	
Steel Epoxy Coating	EPC	
Aluminum	AL	
Stainless Steel 304	S4	
Stainless Steel 316	S6	
Thickness (T) mm	1.5	2.0	2.5
Length (L) mm	69	110	160	210	310	460	610	910	1010

Ordering Code Example : KSABCHDG2.0/1500

NOTES:

- Support Bracket length = Cable Tray Width + 10 m
- Special orders can be manufactured upon requ

Support System - C-Channel Loading Data

41 x 21 C-CHANNEL

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	553	0.45	-	-	3993	1042
500	276	1.81	-	212	3379	995
750	184	4.07	169	94	2198	859
1000	138	7.24	95	53	1353	690
1250	110	1132	61	33	896	548
1500	92	1630	42	23	(634)	(437)
1750	79	22.19	31	-	(471)	(353)
2000	69	28.99	23	-	-	-

41 x 21 C-CHANNEL SLOTTED

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	558	0.50	-	-	-	-
500	254	2.01	-	174	-	-
750	169	4.54	139	77	-	-
1000	127	8.07	78	43	-	-
1250	101	12.61	50	27	-	-
1500	84	18.17	34	-	-	-
1750	72	24.73	25	-	-	-
2000	63	32.30	-	-	-	-

41 x 41 C-CHANNEL

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	1769	0.23	-	-	5869	1693
500	884	0.95	-	-	5379	1650
750	589	2.15	-	571	4479	1554
1000	442	3.82	-	321	3373	1395
1250	535	5.97	-	205	2556	1232
1500	294	8.60	257	142	2020	1093
1750	252	11.70	188	104	1656	976
2000	221	15.29	144	80	1395	879
2250	169	19.35	114	63	1198	797
2500	176	23.89	92	51	1044	725
2750	160	28.91	76	42	(919)	(663)
3000	147	34.40	64	35	(816)	(607)

41 x 41 C-CHANNEL SLOTTED

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	1638	0.27	-	-	5351	1660
500	819	1.05	-	-	5100	1635
750	546	2.35	-	483	4412	1557
1000	409	4.18	-	271	3284	1389
1250	327	6.54	313	174	2436	1211
1500	273	241	217	120	1899	1062
1750	234	12.81	159	88	1546	941
2000	204	16.73	122	67	1297	843
2250	182	21.18	[illegible]	53	1113	761
2500	163	26.15	[illegible]	43	970	691
2750	148	31.64	[illegible]	35	(588)	(631)
3000	136	37.65	[illegible]	30	(761)	(578)

Support System - C-Channel Loading Data

41 x 21 BACK C-CHANNEL

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	1558	0.26	-	-	8228	1834
500	779	1.07	-	-	7740	1809
750	519	2.42	-	446	6865	1756
1000	389	4.30	-	251	5367	1639
1250	311	6.72	289	160	3882	1468
1500	259	9.68	201	111	2838	1289
1750	222	13.18	147	82	2143	1123
2000	194	17.21	113	62	(1668)	(779)
2250	173	21.79	89	49	(1334)	(852)
2500	155	26.90	72	40	(1090)	(745)
2750	141	32.55	59	33	(906)	(655)
3000	129	38.73	50	27	-	-

41 x 41 BACK TO BACK C-CHANNEL

Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	-	-	-	-	119952	2880
500	-	-	-	-	11682	2864
750	1670	1.21	-	-	11186	2833
1000	1253	2.15	-	-	10451	2783
1250	1002	3.36	-	-	9305	2695
1500	835	4.84	-	718	7790	2551
1750	716	6.59	-	527	6293	2367
2000	626	8.61	-	404	5068	2170
2250	556	10.89	-	319	4127	1977
2500	501	13.45	456	258	3410	1796
2750	455	16.27	384	213	2858	1630
3000	417	19.37	323	179	(2427)	(1480)

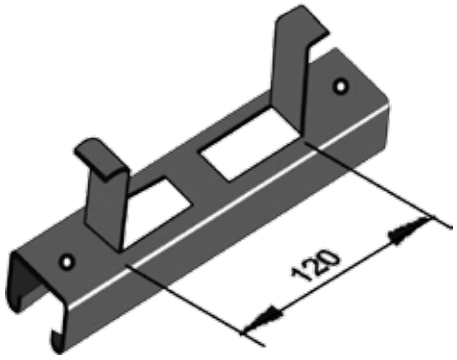
41 x 41 (3x) BACK TO BACK C-CHANNEL

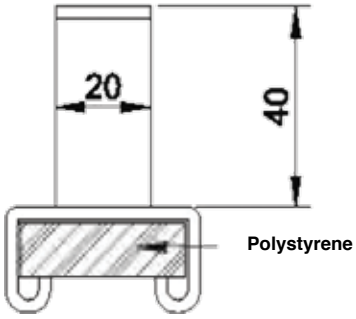
Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	-	-	-	-	16677	4151
500	-	-	-	-	16154	4118
750	-	-	-	-	15093	4045
1000	-	-	-	-	13151	3891
1250	-	-	-	-	10586	3631
1500	-	-	-	-	8175	3297
1750	1586	4.33	-	-	6353	2955
2000	1387	5.66	-	1361	5007	2627
2250	1233	7.16	-	1075	4024	2328
2500	1110	8.84	-	871	3293	2063
2750	1009	10.70	-	720	2744	1833
3000	925	12.74	-	605	(2325)	(1636)

41 x 41 C-CHANNEL WITH PLATE

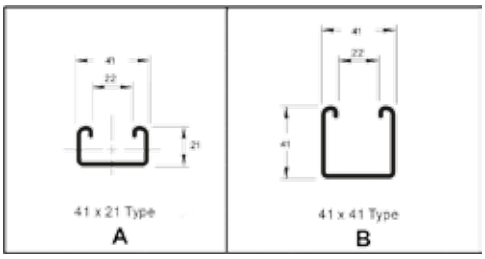
Span or Column Unsupported Height in mm	Uniform Load at 175/ N/mm Stress kg	Deflection in mm at 175/ N/ mm	Uniform Load at Max. Deflec. of 1/1200 th. Span. kg	Uniform Load at Max. Deflec. of 1/360 th. Span. kg	Max. Load of Column loaded @ Centroid kg	Max. Load of Column loaded @ Slot face kg
250	1292	0.17	-	-	-	-
500	1096	0.69	-	-	-	-
750	730	1.55	-	-	-	-
1000	548	2.77	-	-	-	-
1250	438	4.33	-	351	-	-
1500	365	6.23	-	244	-	-
1750	313	84.8	-	179	-	-
2000	274	11.08	247	137	-	-
2250	243	14.03	198	108	-	-
2500	219	17.32	158	87	-	-
2750	199	20.96	131	72	-	-
3000	182	24.94	110	61	-	-

Support System - Concrete Insert C-Channel





C-Channel Concrete Insert **UCI**



Ordering Code					/	

Cantilever Arms	KS	KI	
(41x21) Type	A		
(41x41) Type	B		

Black Steel	BST
Hot Dip Galvanized after Fabrication	HDG
Hot Dip Galvanized and powder Coating	HDP
Mill Galvanized	MG
Steel Epoxy Coating	EPC
Aluminum	AL
Stainless Steel 304	S4
Stainless Steel 316	S6

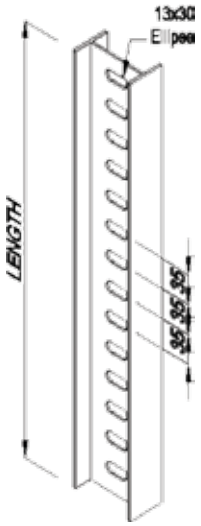
Thickness (T) mm	1.5	2.0	2.5
Length (L) mm	1000 - 6000		

Ordering Code Example : UCIBHDG1.5/1500

NOTES:

- Support Bracket length = Cable Tray Width + 10 m
- Special orders can be manufactured upon requ

Support System - I-Support



I-SUPPORT **IS**

Ordering Code	IS		

I - Supoort	IS

Black Steel	BST
Hot Dip Galvanized after Fabrication	HDG
Hot Dip Galvanized and powder Coating	HDP
Mill Galvanized	MG
Steel Epoxy Coating	EPC
Aluminum	AL
Stainless Steel 304	S4
Stainless Steel 316	S6

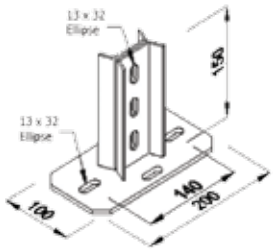
Length (L) mm	1000 - 6000

Ordering Code Example : IS HDG 3000

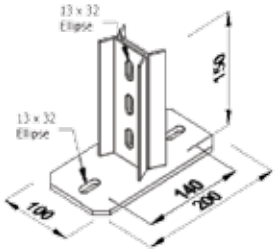
* I-Support thickness as per Standard.

NOTES:

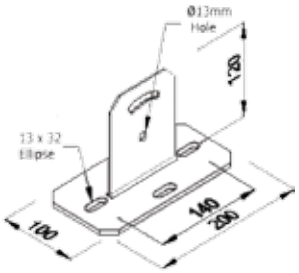
- Support Bracket length = Cable Tray Width + 10 mm.
- Special orders can be manufactured upon request.



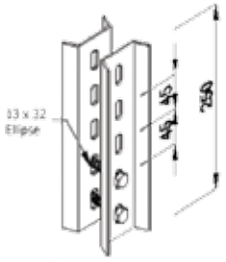
Code: **IS 01**



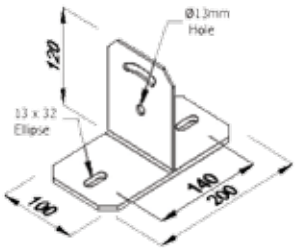
Code: **IS 02**



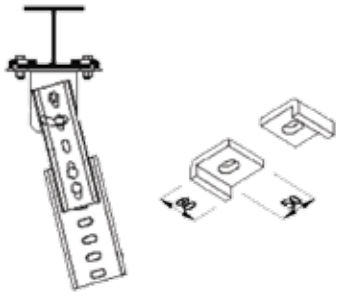
Code: **IS 03**



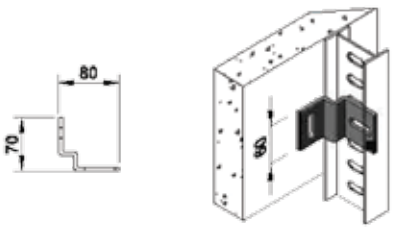
Code: **IS 04**



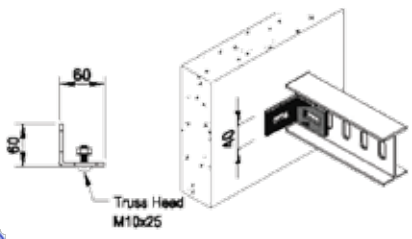
Code: **IS 05**



Code: **IS 06**

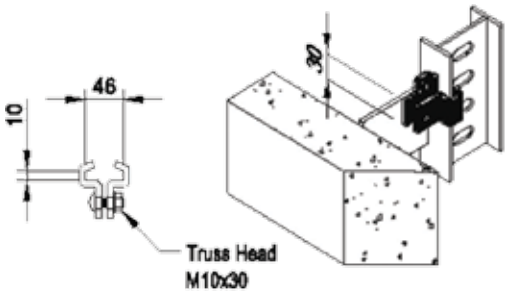


Code: **IS 07**

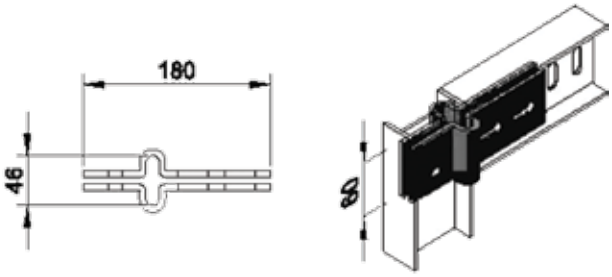


Code: **IS 08**

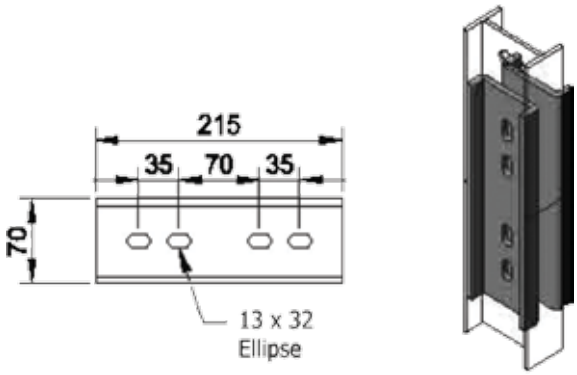
Support System - I-Support



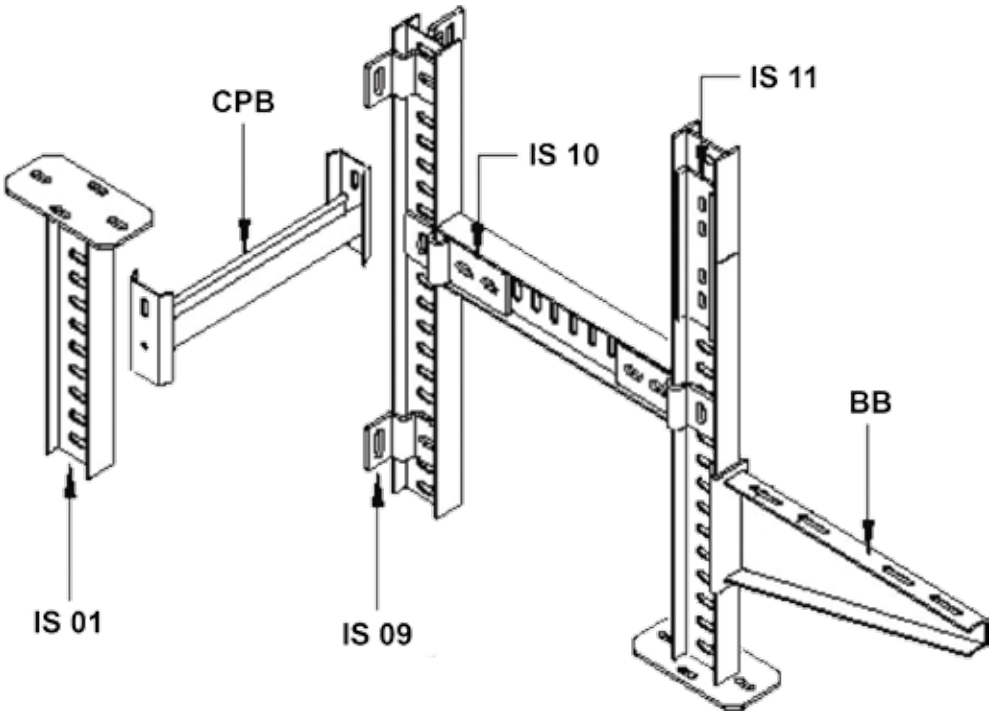
CODE: **IS 09**



CODE: **IS 10**



CODE: **IS 11**

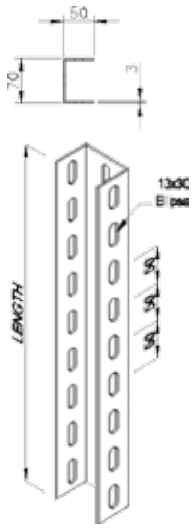


APPLICATION

NOTES:

• Support Bracket length = Cable Tray Width + 1[illegible]

Support System - U-Support



I-SUPPORT **US**

Ordering Code	US		

I - Supoort	US

Black Steel	BST
Hot Dip Galvanized after Fabrication	HDG
Hot Dip Galvanized and powder Coating	HDP
Mill Galvanized	MG
Steel Epoxy Coating	EPC
Aluminum	AL
Stainless Steel 304	S4
Stainless Steel 316	S6

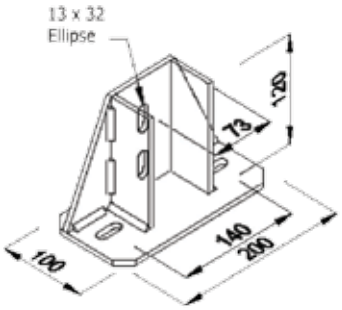
Length (L) mm	1000 - 6000

Ordering Code Example : US HDG 3000

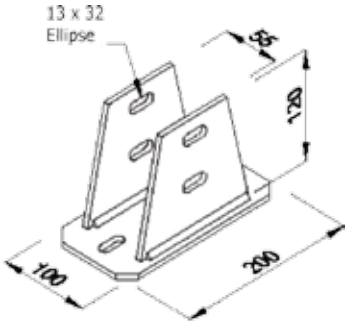
* I-Support thickness as per Standard.

NOTES:

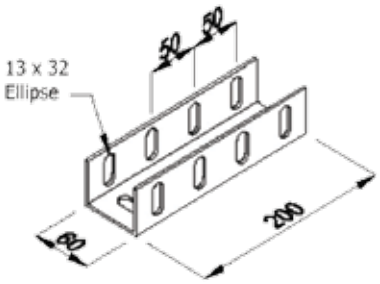
- Special orders can be manufactured upon request.



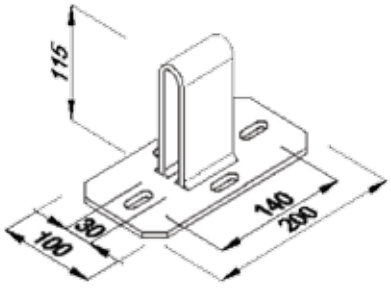
CODE: **US 01**



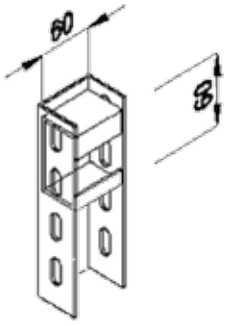
CODE: **US 02**



CODE: **US 03**

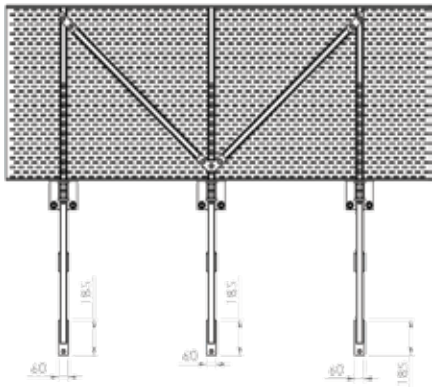


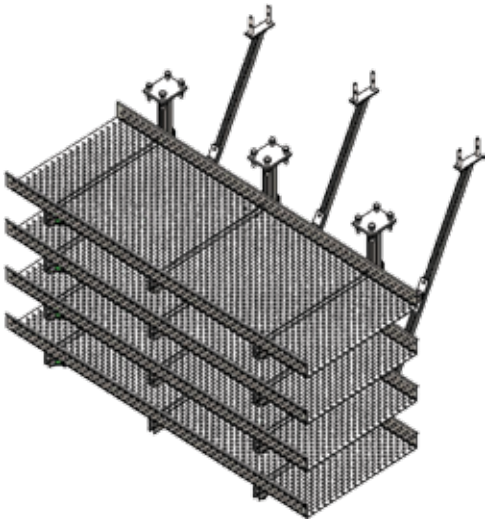
CODE: **US 04**

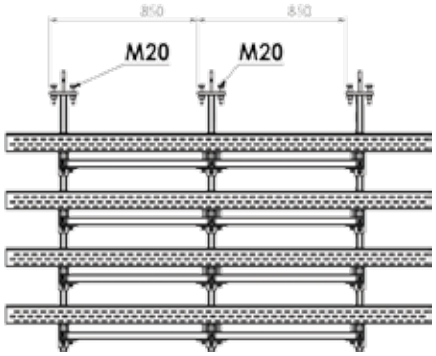


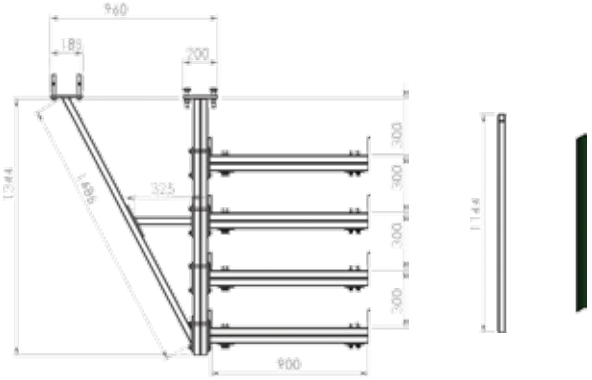
CODE: **US 05**

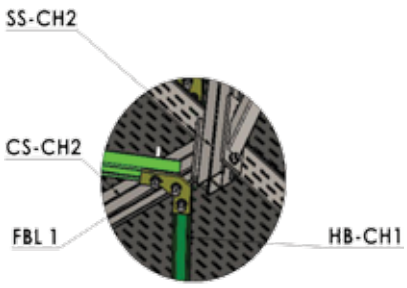
Support System - C-Channel wth Bracket





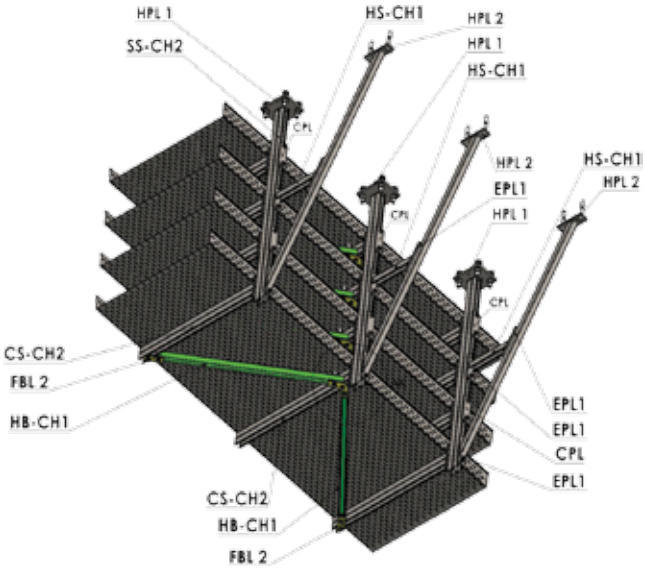






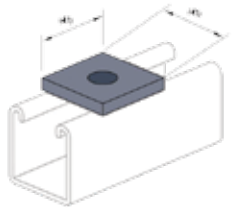
DETAIL AK
SCALE 2 : 15

CODE OF ITEM.	DESCRIPTION
CS-CH2	CANTILEVER SUPPORT 41X41X2.5 _ C-CHANNEL BACK-TO-BACK
SS-CH2	SUSPENDED SUPPORT 41 X41 X2.5 _ C-CHANNEL BACK-TO-BACK
HB-CH1	HORIZONTAL BRACING 41 X 41 X 2.5 _ C- CHANNEL
VB-CH1	VERTICAL BRACING 41 X 41 X2.5 _ C-CHANNEL
HS-CH1	HORIZONTAL SUPPORT 41 X 41 X 2.5
EPL1	END PLATE 185 X 60 X 12
HPL1	HEAD PLATE 200X 175 X 16
HPL2	HEAD PLATE 185 X 60 X 16
CPL	COVER PLATE 185 X60 X12
FPL1	FIXING PLATE 173 X 90 X 6
FPL2	FIXING PLATE 125 X 41 X6

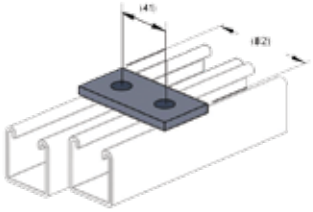


4 LAYER C-CHANNEL W/BR[illegible]OR CABLE TRAY 900

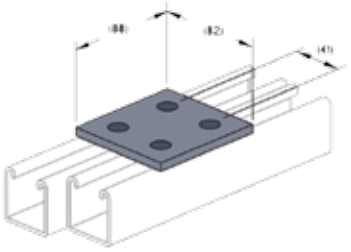
Support System - Accessories



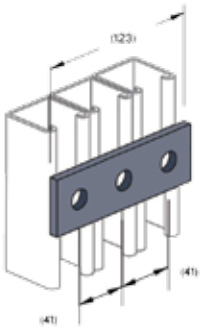
CAC 01



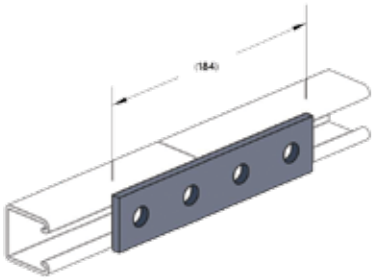
CAC 02



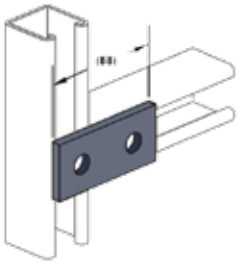
CAC 03



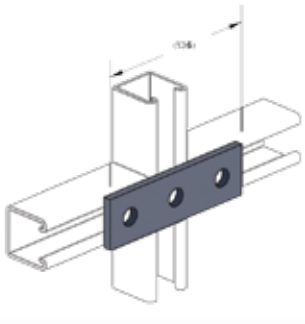
CAC 04



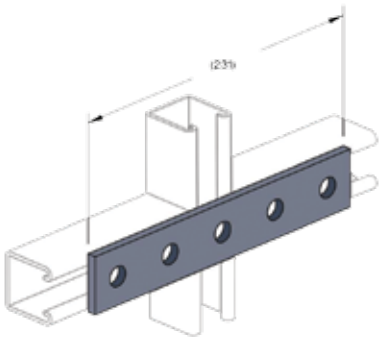
CAC 05



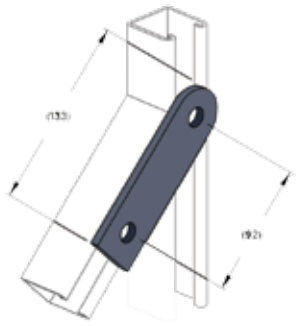
CAC 06



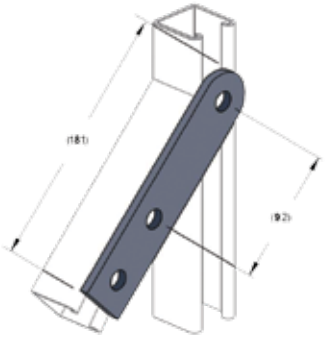
CAC 07



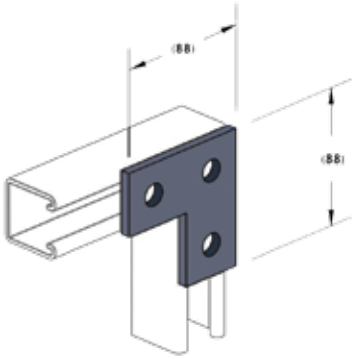
CAC 08

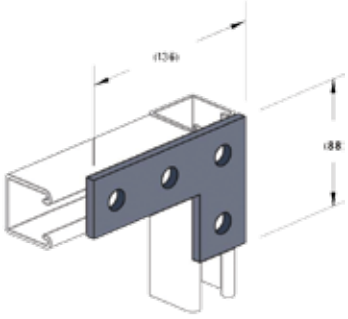


CAC 09



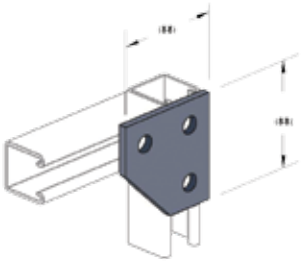
CAC 10



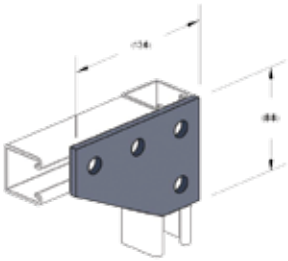


CAC 12

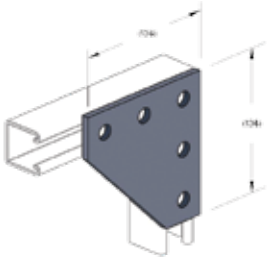
Support System - Accessories



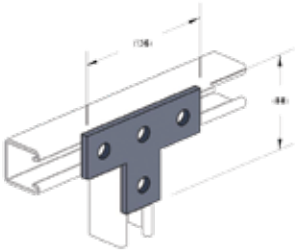
CAC 13



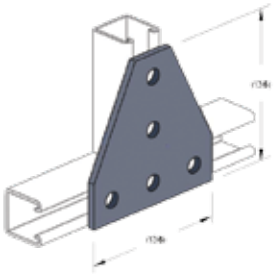
CAC 14



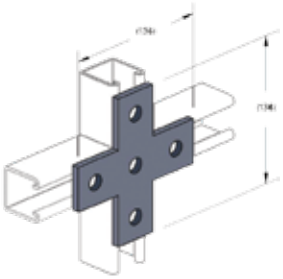
CAC 15



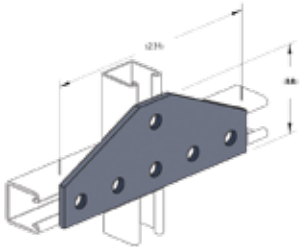
CAC 16



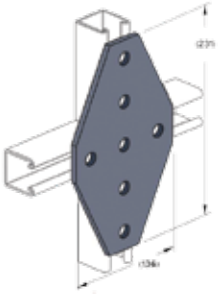
CAC 17



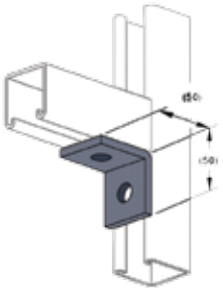
CAC 18



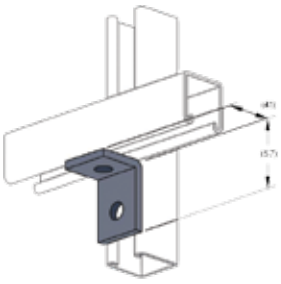
CAC 19



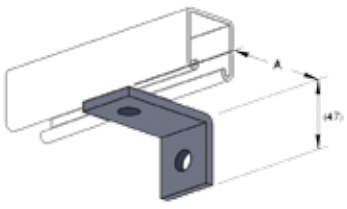
CAC 20

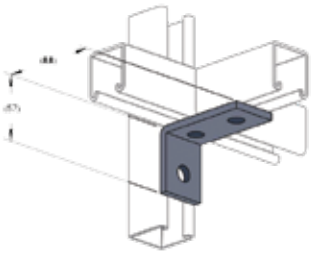


CAC 21



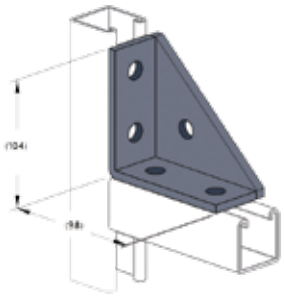
CAC 22



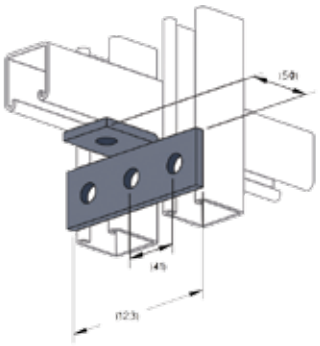


CAC 24

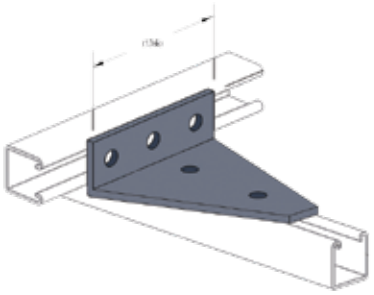
Support System - Accessories



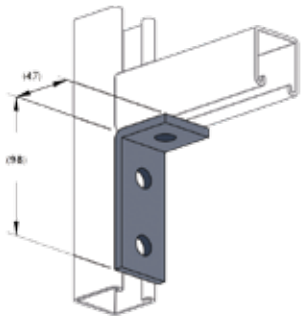
CAC 25



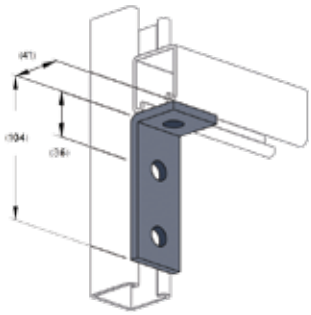
CAC 26



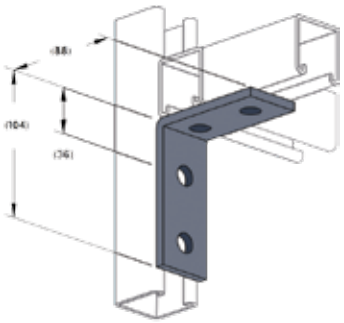
CAC 27



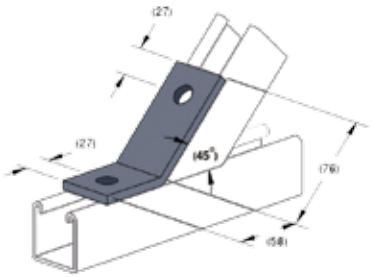
CAC 28



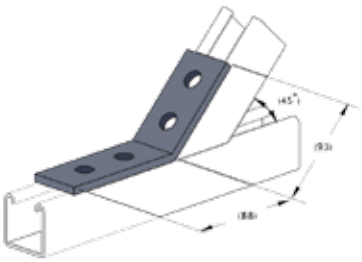
CAC 29



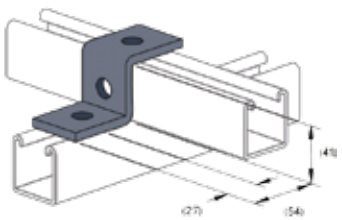
CAC 30



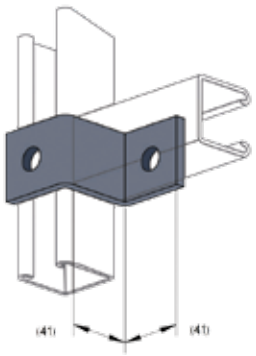
CAC 31



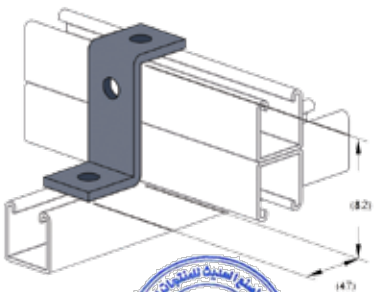
CAC 32



CAC 33

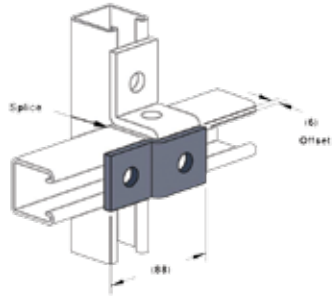


CAC 34

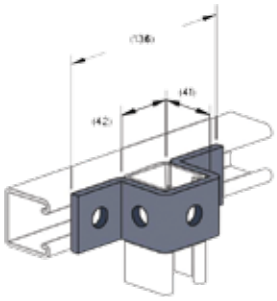


CAC 36

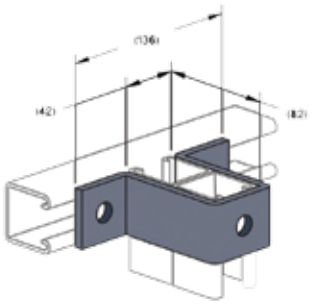
Support System - Accessories



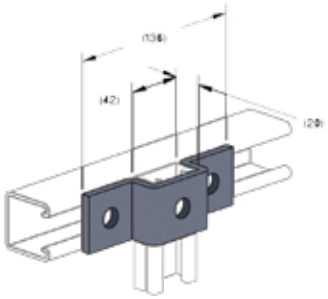
CAC 37



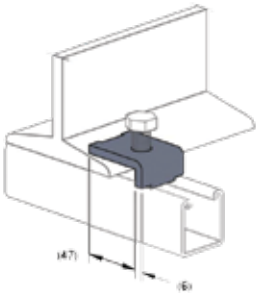
CAC 38



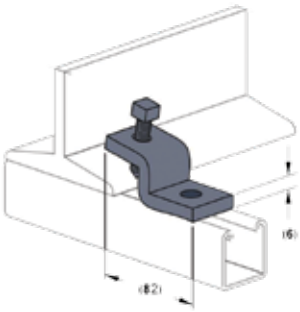
CAC 39



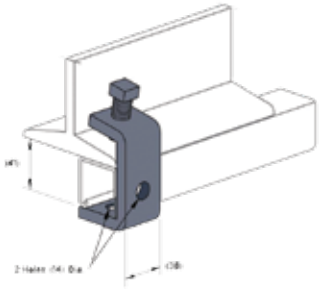
CAC 40



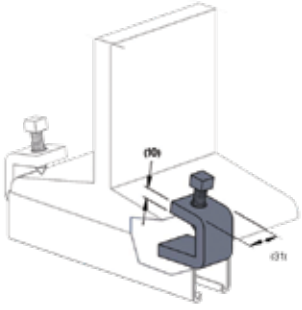
CAC 41



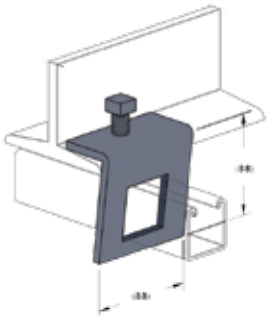
CAC 42



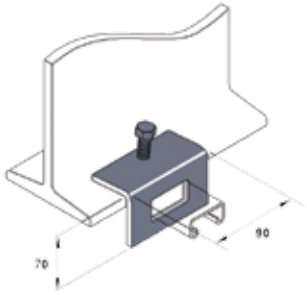
CAC 43



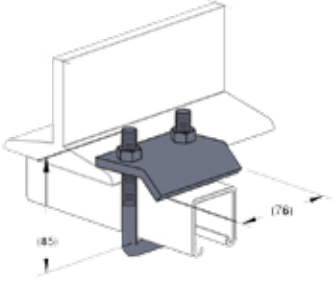
CAC 44



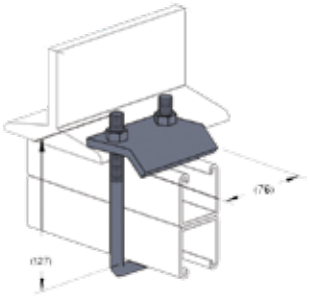
CAC 45



CAC 46

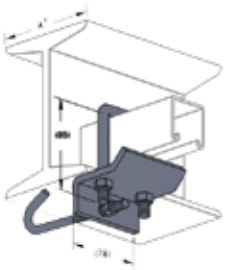


CAC 47

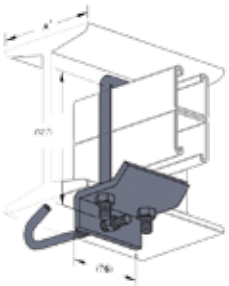


CAC 48

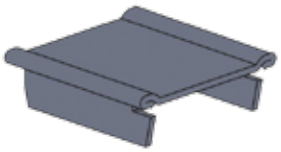
Support System - Accessories



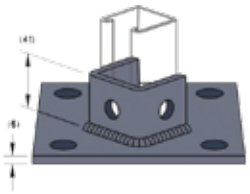
CAC 46

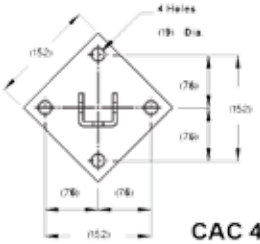


CAC 47

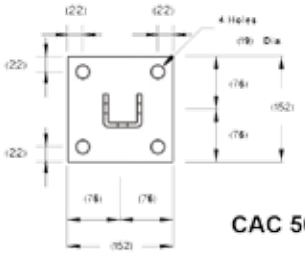


CAC 48

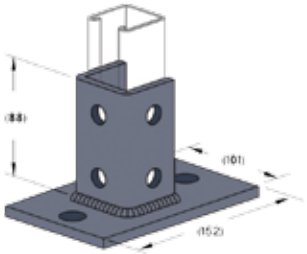


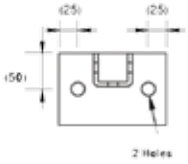


CAC 49

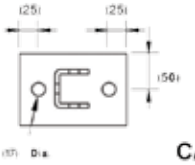


CAC 50

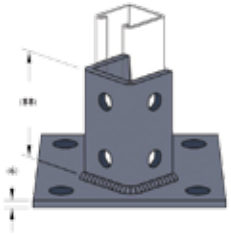


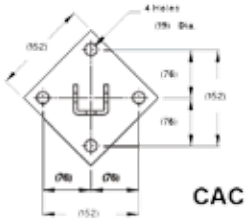


CAC 51

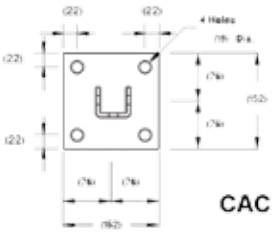


CAC 52

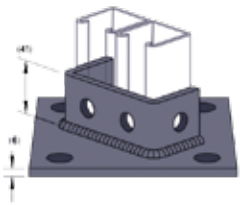


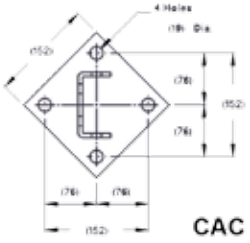


CAC 53

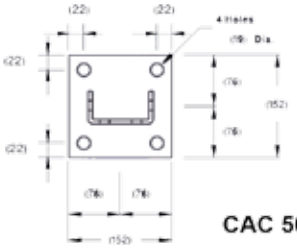


CAC 54

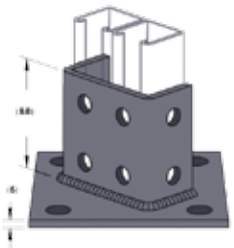


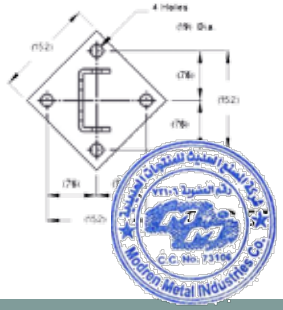


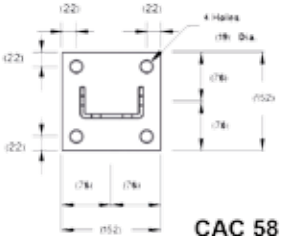
CAC 55



CAC 56



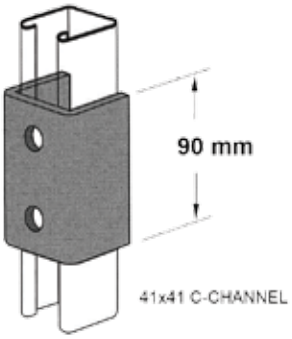




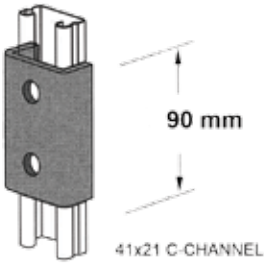
CAC 58

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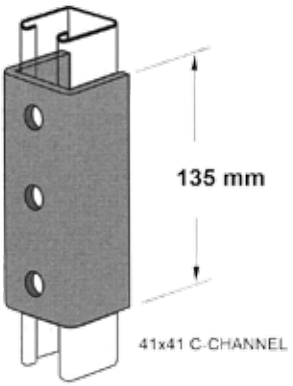
Support System - Accessories



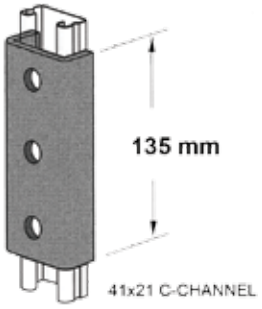
CAC 58



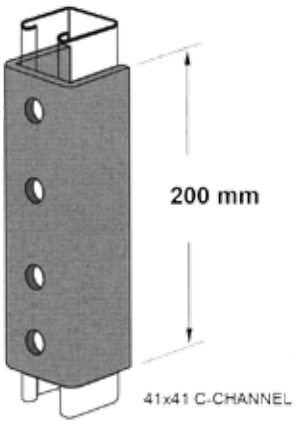
CAC 59



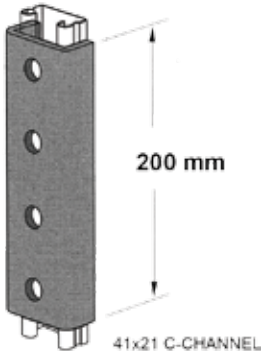
CAC 59



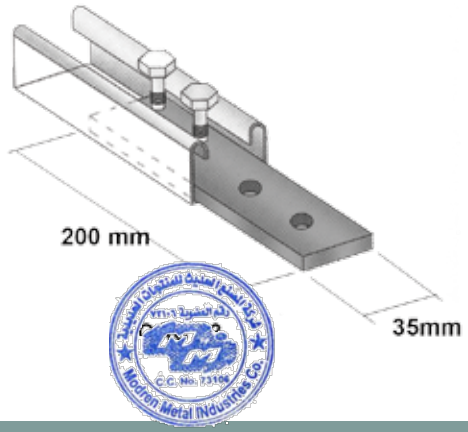
CAC 60



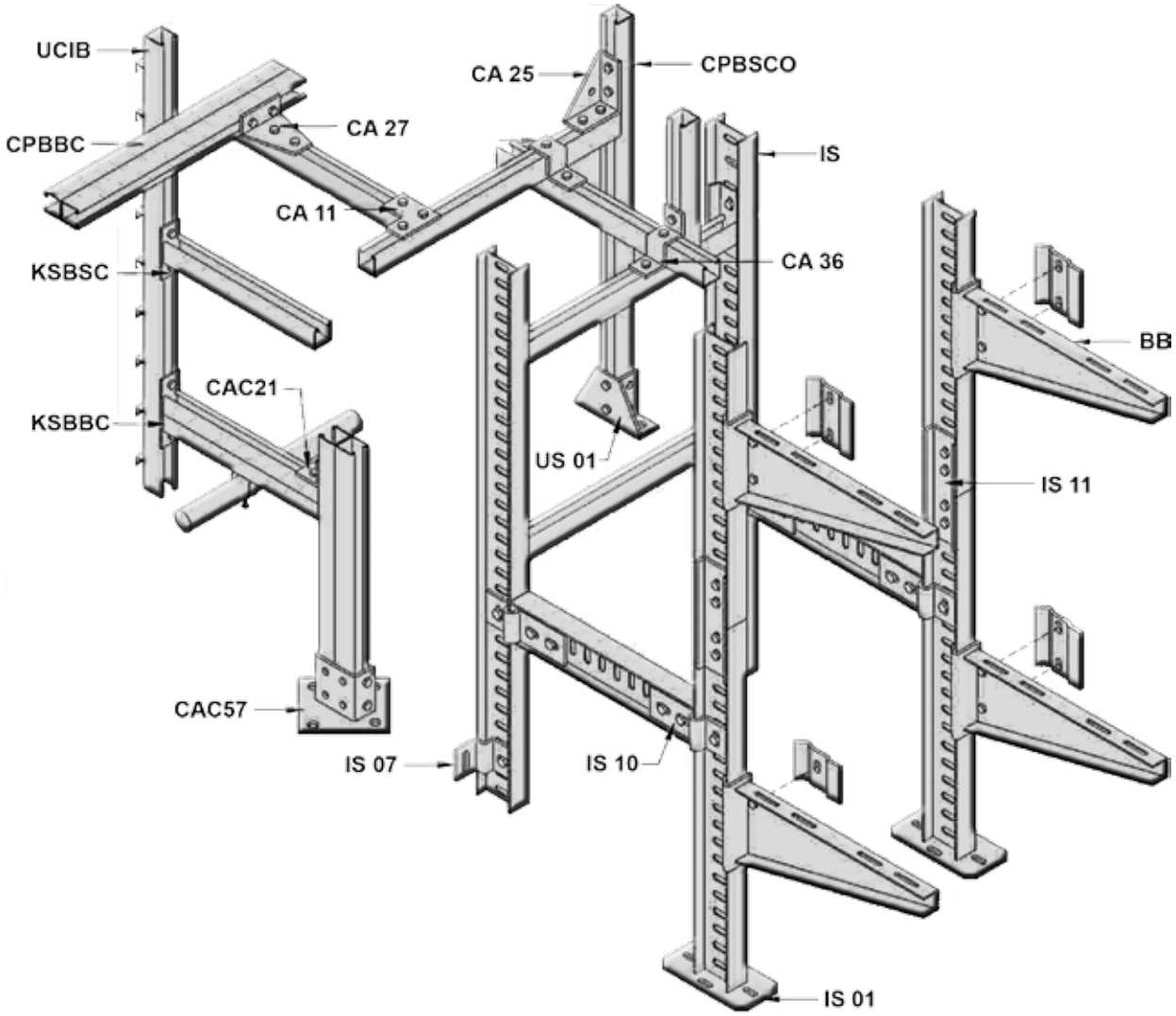
CAC 61



CAC 62



Support System - Channel & Supports Application



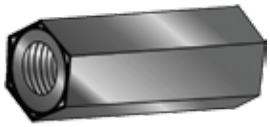


Accessories - Fastening Hardwares



THEADED RODS
(TR)

Ordering Code	TR		
Threaded Rod	TR		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		
Sizes (mm)	6	8	10	12



STUD CONNECTORS
(SC)

Ordering Code	SC		
Stud Connector	SC		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Sizes (mm)	6	8	10	12

BOLTS



HEXAGONAL BOLTS
(B1)



ROUND HEAD BOLTS
(B2)

Ordering Code			
Hexagonal Bolt	BH		
Stud Head Bolt	BR		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		

Sizes (mm)						
	6x16	6x20	6x25	6x30	6x40	6x50
	8x20	8x30	8x40	8x50		
	10x25	10x35	10x50	10x60		
	12x20	12x40	12x50	12x75		

Accessories - Fastening Hardwares



SHORT SPRING NUTS
(SN2)



LONG SPRING NUTS
(SN1)



PLAIN SPRING NUTS
(SN3)

Ordering Code				
Short Spring Nuts	SNS			
Long Spring Nuts	SNL			
Plain Spring Nuts	SNP			
Sizes (mm)	6	8	10	12



HEXAGONAL NUTS
(HN)

Ordering Code	HN		
Hexagonal Nuts	HN		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		
Sizes (mm)	6	8	10	12



SQUARE NUTS
(SN)

Ordering Code	SN		
Square Nuts	SN		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		
Sizes (mm)	6	8	10	12

Accessories - Fastening Hardwares



FLAT WASHERS
(FW)

Ordering Code	FW		
Flat Wahers	FW		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		
Sizes (mm)	6	8	10	12



SPRING WASHERS
(SW)

Ordering Code	SW		
Spring Washers	SW		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Stainless Steel 304	S4		
Stainless Steel 316	S6		
Sizes (mm)	6	8	10	12



UNIFIX
(UFX)

Ordering Code	UFX		
Unifix	UFX		
Elctro - Galvanized	EG		
Hot Dip Galvanized	HDG		
Sizes (mm)	6	8	10	12



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